

“STATISTICAL MODELLING AND MACHINE LEARNING APPROACHES FOR EARLY RISK PREDICTION OF PCOS AMONG YOUNG WOMEN”

Project submitted to the University of Kerala, in partial fulfilment of the requirements for the
Degree of Master of Science in Applied Statistics & Data Analytics

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THIRUVANANTHAPURAM

2024-2026

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I hereby certify that this “***STATISTICAL MODELLING AND MACHINE LEARNING APPROACHES FOR EARLY RISK PREDICTION OF PCOS AMONG YOUNG WOMEN***” is bona fide project work carried out by Ms. ASWATHI BABU RAJAN, Reg.No. 96324615007, M.Sc. Applied Statistics & Data Analytics student in the Department of Statistics, University of Kerala, Kariavattom during 2024-2026 under my supervision and guidance, in partial fulfilment of the requirements for the M.Sc. Degree in Applied Statistics & Data Analytics from the University of Kerala.

Thiruvananthapuram

July 2026

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DECLARATION

I hereby declare that the project report entitled "*Statistical Modelling and Machine Learning Approaches for Early Risk Prediction of PCOS Among Young Women*" bound herewith is the original record of the project work carried out by me under the supervision and guidance of **Dr. Manoj Chacko**, Department of Statistics, University of Kerala, in partial fulfilment of the requirements for the award of the Degree of Master of Science in Applied Statistics and Data Analytics and further that no part thereof has been presented before for any other degree.

All sources of information used in this dissertation have been duly acknowledged through appropriate citations and references.

Thiruvananthapuram

ASWATHI BABU RAJAN

July 2026

ACKNOWLEDGEMENT

I express my heartfelt gratitude to all those who supported and guided me throughout the successful completion of this project, titled "Statistical Modelling and Machine Learning Approaches for Early Risk Prediction of PCOS Among Young Women".

First and foremost, I would like to express my sincere gratitude to my supervisor, Dr. Manoj Chacko, for his valuable guidance, constant encouragement, constructive feedback, and continuous support.

I also extend my sincere thanks to the faculty members of the Department of Statistics, University of Kerala, for providing the academic environment and technical knowledge that formed the foundation of this project.

I am grateful to the hospitals and clinical institutions across Kerala for making the anonymised patient records used in the secondary data analysis available for this study.

I sincerely thank all 303 young women across Kerala who voluntarily participated in the primary survey and shared their time and insights, which formed the foundation of the primary data analysis.

Finally, I would like to thank my family and friends for their constant encouragement and support throughout this academic journey.

Ms. ASWATHI BABU RAJAN

ABSTRACT

Polycystic Ovary Syndrome (PCOS), recently reclassified as Polyendocrine Metabolic Ovarian Syndrome (PMOS) by The Lancet in May 2026, is one of the most common endocrine disorders affecting young women, yet a substantial proportion of affected women in India remain undiagnosed or unaware of their risk. This study, "Statistical Modelling and Machine Learning Approaches for Early Risk Prediction of PCOS Among Young Women," develops and validates statistical and machine learning models for early PCOS risk prediction while quantifying the diagnostic awareness gap among young women in Kerala.

The study draws on two complementary data sources: a secondary clinical dataset of 2,541 anonymised patient records compiled from hospitals across Kerala, and a primary community survey of 303 young women conducted through convenient sampling with snowball recruitment. Following rigorous data cleaning, multicollinearity assessment, and appropriate handling of near-duplicate patient records through patient-grouped train-test splitting to ensure unbiased model evaluation, a range of statistical and machine learning models were developed, including binary logistic regression, LASSO-penalised regression, Decision Tree, Random Forest, K-Nearest Neighbours, Naive Bayes, Support Vector Machine, and Gradient Boosting classifiers, alongside cost-sensitive class weighting to address class imbalance.

Gradient Boosting emerged as the best-performing model (AUC = 0.9997, accuracy = 98.29%, sensitivity = 97.02%, specificity = 98.88%), followed closely by Random Forest (AUC = 0.9887). Permutation feature importance and cross-method validation across logistic regression, LASSO, Random Forest, LDA, and factor analysis consistently identified follicle count, weight gain, LH, hair growth, AMH, and skin darkening as the strongest predictors of PCOS. Analysis of the primary survey revealed that 52.1% of respondents had never been tested for PCOS and 49.8% did not possess full awareness of the condition, while an 8-symptom risk score achieved an AUC of 0.8133 in identifying at-risk individuals from self-reported symptoms alone.

The study demonstrates that machine learning models, particularly ensemble methods, substantially outperform traditional statistical approaches for PCOS risk prediction, while also revealing a significant diagnostic and awareness gap among young women in Kerala. The findings support the development of a low-cost, symptom-based community screening tool and independently corroborate the biological rationale underlying the 2026 PMOS reclassification.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Polycystic Ovary Syndrome (PCOS) — officially renamed Polyendocrine Metabolic Ovarian Syndrome (PMOS) following a landmark global consensus published in *The Lancet* on 12 May 2026 — is one of the most prevalent and clinically complex endocrine disorders affecting women of reproductive age worldwide. Throughout this dissertation, the term PCOS is retained to maintain consistency with the secondary clinical dataset compiled prior to the renaming and with the existing body of published literature. The renaming to PMOS and its clinical implications are discussed in detail in Chapter 2.

PCOS affects an estimated 8 to 13 per cent of women of reproductive age globally, and as many as one in five women in India are estimated to be affected, making it a significant public health concern in the Indian subcontinent (World Health Organization, 2023). Despite its high prevalence, an estimated 70 per cent of women with PCOS remain undiagnosed, primarily due to the heterogeneous and multisystem nature of the condition, which spans endocrine, metabolic, reproductive, dermatological, and psychological domains.

The condition is characterised by a constellation of clinical features including chronic anovulation, hyperandrogenism, polycystic ovarian morphology, insulin resistance, menstrual irregularity, and a spectrum of metabolic complications such as obesity, type 2 diabetes mellitus, dyslipidaemia, and cardiovascular risk. Psychologically, women with PCOS demonstrate significantly elevated rates of depression, anxiety, and diminished quality of life. The multifaceted and heterogeneous nature of PCOS complicates early diagnosis, as no single clinical marker is universally present or definitive across all affected individuals.

In the Indian context, particularly among young women in Kerala, lifestyle factors such as sedentary behaviour, fast food consumption, disrupted sleep patterns, and high levels of academic and occupational stress are increasingly recognised as contributing factors to PCOS risk. The convergence of genetic predisposition and environmental exposures makes early identification of at-risk individuals both clinically challenging and critically important. This study is situated within this context, responding to the urgent need for data-driven early risk prediction tools tailored to young Indian women.

1.2 Significance of Early Risk Prediction

Early detection and timely intervention in PCOS can substantially reduce the long-term burden of the condition. Women diagnosed at an early stage can benefit from lifestyle modification, hormonal management, and metabolic monitoring, thereby reducing the risk of progression to type 2 diabetes, infertility, endometrial cancer, and cardiovascular disease. However, the current diagnostic pathway for PCOS in India is predominantly reactive — most women seek clinical attention only after experiencing significant symptoms — rather than proactive and preventive.

A robust and clinically validated statistical prediction model capable of identifying PCOS risk in young women using readily observable clinical, hormonal, demographic, and lifestyle variables could serve as a cost-effective early screening tool, particularly in resource-limited healthcare settings. Such a model, if integrated into routine gynaecological or general health screening programmes, has the potential to significantly reduce the burden of delayed diagnosis.

The primary community survey conducted as part of this study further underscores this need. Of 303 young women surveyed across Kerala, 52.1 per cent had never been tested for PCOS, and 49.8 per cent did not possess full awareness of the condition. These findings confirm that the diagnostic gap is real, widespread, and uniform across age groups, educational levels, and residential settings — and that it cannot be addressed through targeted interventions alone.

1.3 Statement of the Problem

Despite the high global and national prevalence of PCOS, a substantial proportion of affected women — particularly young women in India — remain undiagnosed for years following the onset of symptoms. The absence of a validated, accessible, and statistically sound early risk prediction tool tailored to this population represents a critical gap in preventive healthcare. Existing clinical diagnostic criteria, such as the Rotterdam criteria (2003), require laboratory investigations and ultrasound imaging, which are not always accessible or affordable in community-level screening contexts.

Furthermore, the existing literature on PCOS risk prediction models predominantly focuses on Western populations, with limited representation of South Asian women, who exhibit distinct clinical and metabolic risk profiles. The role of lifestyle factors, dietary patterns, stress levels,

sleep behaviour, and family history — variables particularly relevant in the Indian socio-cultural context — has not been comprehensively integrated into published prediction models.

This study addresses these gaps by developing and validating a comprehensive statistical and machine learning-based risk prediction framework for PCOS, using a large combined clinical dataset from Kerala, supplemented by primary survey data collected from 303 young women across rural, semi-urban, and urban areas of Kerala. The study focuses on early detection using clinically meaningful and practically accessible predictor variables, with the goal of informing community-level PCOS screening protocols.

1.4 Rationale for the Study

The rationale for this study is grounded in three interconnected imperatives — clinical, statistical, and societal.

Clinically, early identification of PCOS risk enables timely intervention, reducing the long-term endocrine, metabolic, and psychological burden. The development of a prediction model that does not rely exclusively on laboratory or imaging investigations broadens the applicability of early screening to community-level healthcare settings. The present study demonstrates that an eight-symptom self-reported risk score achieves an AUC of 0.8133 — good clinical discrimination — without any blood tests or ultrasound, supporting the feasibility of community-level screening.

Statistically, the richness of the combined clinical dataset — comprising 2,541 patient records with 43 predictor variables spanning hormonal, anthropometric, symptomatic, and lifestyle domains — provides an ideal foundation for applying and comparing a comprehensive range of statistical and machine learning methodologies, including logistic regression, regularisation techniques, ensemble classifiers, dimensionality reduction, factor analysis, discriminant analysis, cluster analysis, and model validation frameworks. This study applies 33 distinct statistical and machine learning methods — one of the most comprehensive analytical frameworks applied to PCOS prediction in the published literature.

Societally, given that an estimated 70 per cent of women with PCOS in India remain undiagnosed (WHO, 2023), and that PCOS is associated with significant impairment of quality of life, fertility, and long-term metabolic health, there is an urgent public health imperative to develop validated, data-driven tools that support early clinical decision-making. The concurrent renaming of PCOS to PMOS (Teede et al., 2026) further underscores the need for

research that captures the full multisystem nature of the condition — a perspective this study's multivariate analytical approach inherently supports.

1.5 Data Sources and Description

This study employs a two-source data strategy combining secondary clinical data and primary survey data. The secondary dataset comprises 2,541 patient records with 45 analytical variables, compiled from clinical records across multiple hospitals in Kerala, capturing hormonal assays, anthropometric measurements, clinical symptoms, and PCOS diagnosis. A structured primary survey — the ‘PCOS Early Risk Assessment Survey’ — was administered digitally to young women across Kerala via purposive and snowball sampling, comprising 31 questions across seven sections covering demographic, menstrual health, clinical symptom, lifestyle, family history, medical history, and awareness domains. A total of 303 complete responses were received. Full details of data collection, cleaning, and encoding for both sources are provided in Chapter 3 (Methodology).

1.6 Objectives of the Study

The following fourteen specific objectives have been formulated to guide the analysis:

Secondary Data Objectives:

Objective 1: To summarise and compare clinical, hormonal, demographic, and lifestyle characteristics between PCOS and non-PCOS groups using descriptive statistics, Mann-Whitney U test, independent samples t-test, and Chi-squared test.

Objective 2: To explore multivariate relationships among predictor variables using Pearson correlation analysis and to apply Principal Component Analysis (PCA) for dimensionality reduction and visual separation of PCOS risk groups.

Objective 3: To detect and assess multicollinearity among predictor variables using Variance Inflation Factor (VIF) and correlation matrix analysis, ensuring stability and interpretability of the regression model.

Objective 4: To develop a binary logistic regression model for early PCOS risk prediction and to interpret the odds ratios with 95% confidence intervals of significant clinical predictors.

Objective 5: To apply LASSO penalised logistic regression for automatic feature selection and identify the most clinically meaningful and statistically significant subset of predictors for PCOS risk prediction.

Objective 6: To develop and compare multiple supervised machine learning classification models — including Decision Tree, Random Forest, K-Nearest Neighbours, Naive Bayes, Support Vector Machine, and Gradient Boosting — for early PCOS prediction based on accuracy, sensitivity, specificity, and AUC-ROC.

Objective 7: To address the observed class imbalance using cost-sensitive class weighting and evaluate its impact on model sensitivity and minority class detection performance.

Objective 8: To validate all developed prediction models using stratified k-fold cross-validation and ROC curve analysis, ensuring model reliability and generalisability.

Objective 9: To interpret the best performing prediction model using permutation importance analysis and feature importance rankings, and discuss the clinical relevance of identified predictors for early PCOS screening.

Objective 10: To apply Factor Analysis, Linear Discriminant Analysis, and K-Means Cluster Analysis to identify underlying clinical factor structures, discriminant functions, and patient subgroups.

Primary Data Objectives:

Objective 11: To describe the demographic profile, symptom prevalence, lifestyle patterns, and health-seeking behaviour of young women in Kerala using primary survey data.

Objective 12: To quantify PCOS awareness levels and identify significant associations between awareness, health-seeking behaviour, and demographic characteristics using Chi-squared tests.

Objective 13: To construct and validate a PCOS symptom risk score, assess its discriminatory ability using ROC curve analysis, and evaluate associations with lifestyle factors using ANOVA, Kruskal-Wallis test, Spearman rank correlation, and logistic regression.

Objective 14: To validate symptom prevalence consistency between primary and secondary datasets using two-proportion Z-tests, Chi-squared goodness of fit, and Spearman rank correlation, and to quantify the community-level PCOS diagnostic gap

1.7 Scope and Limitations

The scope of this study encompasses the development and evaluation of statistical and machine learning models for binary PCOS risk prediction using a combined secondary clinical dataset and primary survey data from Kerala, India. The analytical scope covers all fourteen objectives across descriptive analysis, multivariate exploration, statistical modelling, machine learning comparison, class imbalance correction, model validation, model interpretation, and cross-dataset comparison.

The following limitations are acknowledged:

- This study relies primarily on secondary clinical data collected in hospital settings. The study population may not be fully representative of the broader population of young women in Kerala, particularly those who have not sought clinical consultation. The secondary dataset is drawn from clinical hospital records across Kerala. Findings may not generalise directly to women from other geographical regions or ethnicities without further external validation.
- The primary survey data is based on self-reported symptoms and was not validated against clinical diagnosis for all respondents. It is used for descriptive and awareness analysis rather than predictive modelling.
- The study is cross-sectional in nature. Causal relationships between predictor variables and PCOS diagnosis cannot be established; all associations are correlational.
- Certain clinically relevant variables — including androgen levels, insulin levels, and detailed dietary intake data — are not available in the dataset, which may limit the completeness of the predictor set.
- The prediction models developed are intended as decision-support tools and are not designed to replace clinical diagnosis by qualified medical professionals.
- The primary survey sample, while diverse in residential setting, is predominantly composed of postgraduate students aged 21 to 25 years, which may limit its representativeness of older or less educated women.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

This chapter presents a comprehensive review of existing literature on Polycystic Ovary Syndrome (PCOS) — recently renamed Polyendocrine Metabolic Ovarian Syndrome (PMOS) — encompassing its epidemiology, clinical characteristics, diagnostic criteria, hormonal and metabolic pathophysiology, and the application of statistical and machine learning methods for risk prediction. The review is organised into eleven thematic sections, each corresponding to a specific dimension of the research problem addressed in this study. The literature review draws from peer-reviewed publications in clinical endocrinology, reproductive medicine, biostatistics, and applied machine learning, with particular emphasis on studies conducted on Indian and South Asian populations. Studies published between 2017 and 2026 are prioritised, with foundational earlier works cited where relevant. The chapter culminates with a discussion of the landmark renaming of PCOS to PMOS published concurrently with this study and identifies the specific research gaps this study seeks to address.

2.2 Epidemiology of PCOS

2.2.1 Global Prevalence

PCOS is one of the most prevalent endocrine disorders affecting women of reproductive age worldwide. The global prevalence is estimated at 8 to 13 per cent when applying the Rotterdam criteria, and as high as 21 per cent under the Androgen Excess Society criteria, reflecting considerable variation depending on the diagnostic framework employed. The condition affects an estimated 170 million women globally — approximately one in eight women of reproductive age — making it a significant public health burden with implications extending beyond reproductive health to encompass metabolic, cardiovascular, and psychological domains (Teede et al., 2026).

Despite its high prevalence, diagnostic delays are widespread. Studies indicate that up to 70 per cent of women with PCOS remain undiagnosed at the community level, with many experiencing symptoms for several years before receiving a formal clinical diagnosis. This

diagnostic gap is attributable to the heterogeneous clinical presentation of the condition, the requirement for specialist investigations, and persistent misconceptions among both patients and healthcare providers regarding the nature of the disorder. The primary community survey conducted as part of this study found that 52.1 per cent of 303 young women surveyed in Kerala had never been tested for PCOS — directly confirming the existence of this diagnostic gap at the community level in South India.

2.2.2 Prevalence in India

India represents a particularly concerning epidemiological context for PCOS. A nationwide multicentric cross-sectional study involving 9,824 women aged 18 to 40 years conducted across five geographical zones of India reported a weighted national prevalence of 19.6 per cent under the Rotterdam 2003 criteria, with PCOS phenotype C — characterised by hyperandrogenism and polycystic ovarian morphology without anovulation — being the most prevalent phenotype, accounting for 40.8 per cent of cases. Among women with PCOS in this study, 43.2 per cent had obesity, 91.9 per cent had dyslipidaemia, and 24.9 per cent had metabolic syndrome.

A 2024 cross-sectional survey conducted among 1,164 college-going females aged 18 to 25 years in Delhi-NCR reported a PCOS prevalence of 17.40 per cent in this age group — significantly higher than the pooled prevalence of 8.41 per cent reported in similar studies across India from 2010 to 2024 (Gill et al., 2025). A systematic review and meta-analysis of PCOS prevalence across India reported a pooled prevalence of 11.33%, with estimates varying by diagnostic criteria used (Bharali et al., 2022). This finding is consistent with the primary survey findings of the present study, which identified a self-reported PCOS prevalence of 17.5 per cent among 303 young women in Kerala — closely matching the national college-student estimate. PCOS is widely recognised as the most common endocrine disorder in Indian women of reproductive age, yet remains severely underdiagnosed and undertreated due to inadequate awareness, limited screening infrastructure, and fragmented care pathways.

2.3 Clinical Characteristics and Diagnostic Criteria

2.3.1 Diagnostic Criteria

The Rotterdam Consensus criteria (Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group, 2004) remain the most widely adopted diagnostic framework for PCOS and

require the presence of at least two of the following three features: oligo-anovulation or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovarian morphology on ultrasound defined as the presence of 12 or more follicles measuring 2 to 9 mm in diameter or an ovarian volume exceeding 10 mL in at least one ovary. The Rotterdam criteria recognise four distinct PCOS phenotypes, reflecting the heterogeneous clinical presentation of the condition.

The National Institutes of Health (NIH) 1990 criteria are more restrictive, requiring both anovulation and hyperandrogenism. The Androgen Excess Society criteria require androgen excess as a mandatory feature. Current international clinical guidelines recommend the Rotterdam criteria as the standard for clinical diagnosis, emphasising the importance of anti-Müllerian hormone (AMH) as a biomarker for polycystic ovarian morphology. In the present study, the clinical dataset employed Rotterdam-based diagnostic criteria, with follicle count and AMH emerging as the two strongest predictors of PCOS status — consistent with these diagnostic guidelines.

2.3.2 Clinical Features

The clinical manifestations of PCOS are heterogeneous and span multiple organ systems. Reproductive features include menstrual irregularity, anovulatory infertility, and recurrent pregnancy loss. Dermatological features arising from androgen excess include hirsutism, acne, androgenic alopecia, and acanthosis nigricans. Metabolic features include insulin resistance, central obesity, dyslipidaemia, and elevated risk of type 2 diabetes and cardiovascular disease. Psychological features include elevated rates of depression, anxiety, and diminished health-related quality of life.

In the secondary clinical dataset employed by this study, the variables most strongly differentiating PCOS from non-PCOS patients in bivariate analysis were follicle count (Mann-Whitney $p < 0.0001$), AMH ($p < 0.0001$), weight gain (Chi-squared $p < 0.0001$), irregular cycle ($p < 0.0001$), and skin darkening ($p < 0.0001$). In the primary community survey, the most prevalent symptoms among confirmed PCOS respondents were mood swings (70%), skin darkening (70%), abdominal obesity (77%), weight gain (68%), and hair growth (55%) — broadly consistent with the clinical literature.

2.4 Statistical Modelling Approaches for PCOS Prediction

2.4.1 Binary Logistic Regression

Logistic regression is the most widely applied statistical method for PCOS risk prediction due to its interpretability and robustness with binary outcomes, typically achieving AUC values between 0.75 and 0.90 on clinical datasets while retaining interpretability advantages essential for clinical decision support. The application of logistic regression in the present study, including odds ratios and threshold optimisation, is presented in Chapter 5.

2.4.2 LASSO Regularisation for Feature Selection

LASSO regularisation (Tibshirani, 1996) applies an L1 penalty to the logistic regression objective function, forcing the coefficients of uninformative predictors to exactly zero, making it particularly valuable for feature selection in high-dimensional medical datasets. The application of LASSO-based feature selection in the present study is presented in Section 5.4.

2.4.3 Multicollinearity Assessment

Multicollinearity among predictors is a well-recognised challenge in clinical regression modelling. Variance Inflation Factor (VIF) assessment is recommended prior to logistic regression, with VIF above 10 indicating severe multicollinearity requiring remediation (O'Brien, 2007). The VIF assessment conducted for the present study's predictor set is presented in Section 5.2.

2.4.4 Factor Analysis and Discriminant Analysis

Factor analysis is a multivariate technique that identifies underlying latent constructs explaining the covariance structure among observed variables, and can reveal clinically meaningful dimensions that individual variable analysis may miss. Linear Discriminant Analysis (LDA) identifies the linear combination of features that best separates predefined groups. The application of both methods to the present study's clinical predictor set is presented in Chapter 4.

2.5 Machine Learning Approaches for PCOS Prediction

2.5.1 Overview

The application of machine learning to clinical prediction and diagnosis has expanded substantially over the past decade, with ensemble methods consistently outperforming individual classifiers and statistical models in PCOS detection. A machine learning model developed on a large electronic health records dataset of over 30,000 women demonstrated that ensemble and regularised methods meaningfully improved PCOS risk prediction over single-method approaches (Zad et al., 2024).

2.5.2 Decision Trees and Random Forest

Decision trees provide interpretable classification rules by recursively partitioning the feature space. Random Forest (Breiman, 2001) addresses the overfitting limitation of individual trees by averaging predictions across many independently trained trees. Results for both methods in the present study are presented in Sections 6.2 and 6.3.

2.5.3 Support Vector Machine

SVM classifies observations by identifying the maximum-margin hyperplane separating classes, with the Radial Basis Function kernel allowing it to capture non-linear decision boundaries. Results for SVM in the present study are presented in Section 6.6.

2.5.4 K-Nearest Neighbours and Naive Bayes

KNN classifies new observations based on the majority class among their k nearest training examples, while Naive Bayes applies Bayes' theorem under a conditional independence assumption that is often violated by intercorrelated clinical predictors. Results for both methods in the present study are presented in Sections 6.4 and 6.5.

2.5.5 XGBoost and Gradient Boosting

XGBoost implements gradient boosted decision trees (Friedman, 2001) with regularisation and has consistently achieved state-of-the-art performance on structured tabular datasets, including PCOS prediction using clinical and ultrasound features. Results for Gradient Boosting in the present study are presented in Section 6.7.

2.6 Class Imbalance and Resampling Techniques

Class imbalance — where one class substantially outnumbers the other — is a pervasive challenge in medical classification tasks. The Synthetic Minority Oversampling Technique (SMOTE), introduced by Chawla et al. (2002), generates synthetic minority class samples through linear interpolation between existing minority observations. SMOTE has been widely adopted in medical machine learning research. In the present study, the 69:31 class imbalance was addressed through cost-sensitive class weighting applied during model training. All seven models used `class_weight = 'balanced'` or an equivalent mechanism, which proved effective — Gradient Boosting and KNN both achieved sensitivity of 97.02% and 95.24% respectively on the patient-grouped evaluation without requiring synthetic oversampling.

2.7 Model Validation and Calibration

Cross-validation is a standard technique for assessing model generalisability; stratified k-fold cross-validation preserves class proportions in each fold and provides robust performance estimates. ROC curve analysis and Youden's J statistic for optimal threshold selection are recommended for clinical screening models. The cross-validation and ROC analysis conducted for the present study's models are presented in Sections 6.9 and 6.10.

2.8 Model Interpretability and Permutation Importance Analysis

A critical challenge in clinical machine learning is the trade-off between predictive accuracy and model interpretability. SHAP (SHapley Additive exPlanations) decomposes individual model predictions into additive feature contributions grounded in cooperative game theory, and has been applied to gradient boosting models for PCOS prediction to identify influential clinical features, including follicle count, AMH, weight gain, and irregular cycle. The permutation importance analysis conducted for the present study's best-performing model is presented in Section 6.12.

2.9 Community-Level Screening and Awareness Studies

2.9.1 PCOS Awareness Among Young Women

Studies on PCOS awareness among young women in India consistently report inadequate knowledge, particularly at the community level. Similarly, a study among adolescent and young

girls in South India found that while awareness of PCOS existed, detailed knowledge of symptoms and risk factors remained limited (Jabeen et al., 2022). The present study found that 50.2% of surveyed young women in Kerala were fully aware of PCOS, 40.6% had partial awareness, and 9.3% had minimal or no knowledge. Critically, Chi-squared analysis revealed that awareness did not vary significantly by education level ($p = 0.227$), residence ($p = 0.430$), or age group ($p = 0.222$) — suggesting the awareness gap is uniform across all demographic groups and cannot be addressed by targeted interventions alone.

2.9.2 Self-Reported Symptom-Based Screening

The development of symptom-based screening tools for community-level PCOS risk stratification has received increasing research attention. In the present study, a simple eight-symptom risk score, incorporating weight gain, hair growth, skin darkening, hair loss, pimples, mood swings, abdominal obesity, and irregular menstrual cycle, was developed as a practical community screening tool. Using a cut-off of four or more symptoms, this risk score achieved an AUC of 0.8133 with an optimal sensitivity of 79.2%, demonstrating good discriminatory ability without requiring blood tests or ultrasound examinations. To further improve predictive performance, a logistic regression model was developed using the same self-reported symptom variables. Unlike the simple risk score, logistic regression assigns different statistical weights to each symptom based on its contribution to PCOS prediction. The analysis identified irregular menstrual cycle ($OR = 13.51$, $p < 0.001$) and mood swings ($OR = 5.77$, $p = 0.002$) as the strongest independent predictors of confirmed PCOS diagnosis, with the model achieving a higher AUC of 0.8898. These findings indicate that while the symptom-based risk score provides a simple and practical approach for community-level screening, the logistic regression model offers improved predictive accuracy by accounting for the relative importance of individual symptoms.

2.9.3 Lifestyle Factors and PCOS Risk

Multiple studies have documented the association between lifestyle factors and PCOS risk. Dietary quality has been associated with PCOS risk in prior research, with processed food consumption linked to elevated androgen levels. In the present study, Spearman rank correlation confirmed significant positive associations between PCOS risk score and stress level ($\rho = 0.30$, $p < 0.001$) and fast food consumption ($\rho = 0.18$, $p = 0.002$), and a significant negative association with sleep duration ($\rho = -0.14$, $p = 0.014$). One-way ANOVA revealed no significant difference in mean risk score across age groups ($F = 0.11$, $p = 0.895$), and

Kruskal-Wallis test confirmed no significant variation across residential settings ($H = 0.01$, $p = 0.995$) — establishing that PCOS symptom burden is uniformly distributed and that lifestyle factors, particularly stress and diet, are modifiable determinants of PCOS risk regardless of demographic background.

2.10 The Renaming of PCOS to PMOS — A Landmark Development

This study was conducted concurrently with this landmark development.

On 12 May 2026 — during the course of this research — a landmark global consensus study was published in *The Lancet* officially renaming Polycystic Ovary Syndrome (PCOS) to Polyendocrine Metabolic Ovarian Syndrome (PMOS). The process, led by Professor Helena Teede at Monash University, involved 56 organisations and drew responses from over 14,360 individuals over more than a decade (Teede et al., 2026).

The primary impetus for the renaming was the recognised inaccuracy of the existing terminology. The term 'polycystic ovary syndrome' implied the presence of pathological ovarian cysts — which are not a feature of the condition. The structures visualised on ultrasound are arrested antral follicles rather than true cysts, and there is no increase in abnormal ovarian cysts in the condition. This misnomer contributed to delayed diagnosis, fragmented care, and social stigma.

The new name PMOS — Polyendocrine Metabolic Ovarian Syndrome — was constructed to reflect the multisystem nature of the condition. Polyendocrine recognises multiple interacting hormonal systems including insulin, androgens, and neuroendocrine hormones. Metabolic acknowledges insulin resistance, obesity, and cardiometabolic risks. Ovarian retains the connection to ovulatory disturbances and infertility. The renaming is expected to prompt updates to clinical guidelines, medical education, and international disease classification systems including ICD and DSM.

The relevance of this development to the present study is profound and multifaceted. The findings of this research independently validate the multisystem conceptualisation encapsulated by PMOS through several converging lines of evidence. First, factor analysis identified four clinical factor structures corresponding precisely to the polyendocrine (gonadotropin factor: LH loading 0.86, FSH loading 0.48), metabolic (metabolic-lifestyle

factor: weight gain 0.67, fast food 0.60), and ovarian (androgenic-ovarian factor: skin darkening 0.49, AMH 0.40, follicle count 0.36) dimensions of PMOS. Second, permutation importance analysis confirmed that the dominant predictors span endocrine (follicle count, AMH), metabolic (weight gain), and dermatological-androgenic (hair growth, skin darkening) domains. Third, individual symptom-level comparisons across both datasets (Chapter 4) indicate that these multisystem predictors are consistently observed across both clinical and community populations. This study may therefore be considered among the first applied statistical and machine learning analyses explicitly aligned with the PMOS conceptual framework.

2.11 Research Gaps and Justification for the Present Study

The review of existing literature identifies the following specific gaps that the present study addresses:

- The majority of published PCOS prediction models have been developed on Western or East Asian populations. Studies focusing specifically on South Indian women with distinct hormonal profiles and lifestyle risk factors are limited.
- Existing studies typically apply one or two machine learning algorithms in isolation. Comprehensive comparisons of multiple statistical and machine learning models on the same dataset with consistent evaluation metrics are uncommon.
- Few studies simultaneously apply VIF-based multicollinearity assessment, LASSO feature selection, cost-sensitive class weighting, factor analysis, discriminant analysis, cluster analysis, and permutation importance interpretability within a single integrated analytical framework.
- The application of Youden's J statistic for optimal threshold selection — rather than the default 0.50 threshold — is rarely reported in PCOS prediction literature despite its clinical importance for maximising screening sensitivity.
- Studies incorporating both clinical laboratory variables and self-reported lifestyle and symptom variables within the same prediction model are scarce, limiting the development of community-level screening tools
- Cross-dataset validation comparing community survey findings with clinical dataset findings using formal statistical tests — including two-proportion Z-tests and Spearman rank correlation — has not been reported in PCOS prediction literature.

CHAPTER 3

DATA AND METHODOLOGY

3.1 Introduction

This chapter provides a detailed and comprehensive account of the data sources, data preparation procedures, variable encoding, and the complete set of statistical and machine learning methodologies applied in this study. The chapter is organised to reflect the sequential analytical pipeline followed — from data collection and quality assessment, through cleaning and preprocessing, to exploratory analysis, statistical modelling, machine learning implementation, model validation, and cross-dataset comparison. All analyses were conducted using Python version 3.12 in the Google Colaboratory cloud-based environment. A total of 33 distinct statistical and machine learning methods were applied across the two datasets, making this one of the most comprehensive analytical frameworks applied to PCOS risk prediction in the published literature.

3.2 Research Design

This study adopts a quantitative, cross-sectional research design combining secondary data analysis and primary survey-based data collection. The secondary clinical dataset provides the foundation for all statistical modelling and machine learning analyses. The primary survey data contributes supplementary community-level evidence on self-reported symptoms, lifestyle factors, and PCOS awareness among young women in Kerala.

The study is exploratory-predictive in nature. The exploratory component identifies significant predictors through descriptive and inferential analysis. The predictive component develops, compares, and validates statistical and machine learning risk prediction models. The cross-dataset validation component formally tests the consistency of findings between the two independent data sources using inferential statistical tests. The study does not establish causal relationships, as it is cross-sectional and based on observational data.

3.3 Data Sources

3.3.1 Secondary Clinical Dataset

The secondary clinical dataset comprises anonymised patient records collected from multiple hospitals across Kerala, India. The dataset contains patient-level data on hormonal assays, anthropometric measurements, clinical symptoms, and PCOS diagnosis. Two independently compiled subsets of Kerala hospital clinical data — accessed as open-access datasets for academic research purposes — were merged to form a single analytical file containing 2,541 patient records and 45 variables, referred to throughout this study as the Combined Dataset. This dataset has been utilised in multiple published studies on PCOS prediction, including research published in Scientific Reports (2025) and the International Journal of Electrical and Computer Engineering (2024), all describing it as clinical data collected from hospitals across Kerala, India..

The binary outcome variable is PCOS (Y/N), encoded as 1 for PCOS-positive and 0 for PCOS-negative. The class distribution was 1,756 Non-PCOS (69.1%) and 785 PCOS (30.9%), representing a moderate class imbalance with a ratio of approximately 2.2:1.

The dataset was accessed as a secondary data source for academic research purposes. Ethical considerations relevant to secondary data use — including patient anonymisation and data confidentiality — were ensured by the original data collectors. No personally identifiable information was used in this analysis.

3.3.2 Primary Survey Dataset

A structured primary survey instrument was developed and administered to young women in Kerala to complement the secondary clinical data with community-level evidence. The survey — titled 'PCOS Early Risk Assessment Survey' — was designed under the supervision of Dr. Manoj Chacko, Department of Statistics, University of Kerala, Kariavattom, and comprised 31 structured questions across seven thematic sections:

- **Section A** — Demographic Information (Questions 1 to 6): age group, weight range, height range, blood group, residence type, and education level
- **Section B** — Menstrual Health (Questions 7 to 10): cycle regularity, cycle length, bleeding duration, and menstrual pain

- **Section C** — Physical and Clinical Symptoms (Questions 11 to 15): weight gain, hair growth, skin darkening, hair loss, and pimples
- **Section D** — Lifestyle and Mental Wellbeing (Questions 16 to 24): fast food consumption, exercise frequency, occupation, stress level, sleep hours, sleep time, wake time, mood swings, and abdominal obesity
- **Section E** — Family History (Questions 25 to 26): family history of PCOS and family history of diabetes
- **Section F** — Medical History (Questions 27 to 30): PCOS diagnosis, gynaecologist consultation, thyroid diagnosis, and diabetes diagnosis
- **Section G** — PCOS Awareness (Question 31): self-assessed PCOS awareness level

The survey was administered digitally via Google Forms through convenient sampling with snowball recruitment to young women across rural, semi-urban, and urban areas of Kerala. A total of 303 complete responses were received. The sample comprised young women aged 15 to 35 years, with 84.5 per cent aged 21 to 25 years, 9.6 per cent below 20 years, and 5.9 per cent above 25 years.

For analytical purposes, the original five age categories were consolidated into three clinically meaningful bands consistent with PCOS epidemiological literature:

- Below 20 years (n = 29, 9.6%) — combining 15–17 and 18–20 age groups
- 21 to 25 years (n = 256, 84.5%)
- Above 25 years (n = 18, 5.9%) — combining 26–30 and 31–35 age groups

Question 27 — "Have you ever received a diagnosis of PCOS from a qualified medical professional?" — serves as the primary outcome variable for the survey dataset. Responses were classified as confirmed PCOS (n = 53), confirmed Non-PCOS (n = 71), consulted but no diagnosis (n = 21), and never tested (n = 158).

3.4 Data Quality Assessment and Cleaning — Secondary Dataset

3.4.1 Initial Data Audit

Prior to statistical analysis and model development, a comprehensive data quality assessment was conducted on the combined secondary dataset to identify data inconsistencies, missing values, coding errors, and potential anomalies. The audit revealed a small number of data quality issues requiring correction before analysis. Specifically, non-numeric entries were identified in the AMH (ng/mL) and II beta-HCG (mIU/mL) variables, preventing these variables from being recognised as numeric data types. A small number of invalid category codes were also detected in the Cycle (R/I) variable, where values inconsistent with the data dictionary were present. Additionally, five missing values were identified across the Marriage Status (Yrs) and Fast Food (Y/N) variables. Extreme values observed in the FSH and LH variables were determined to be data entry errors, most likely resulting from misplaced decimal points rather than genuine clinical measurements. No exact duplicate rows were detected in the dataset. However, measures were taken during model development to minimise potential information leakage arising from near-duplicate patient records. All identified issues were addressed during the data preprocessing stage to ensure data integrity and suitability for subsequent statistical analyses and machine learning modelling.

3.4.2 Cleaning Procedures

The following sequential cleaning procedures were applied:

- **AMH column:** The four 'a' entries were converted to NaN using `pd.to_numeric` with `errors='coerce'`. Resulting missing values were imputed with the column median (3.90 ng/mL), as AMH values are right-skewed and median imputation is more robust than mean imputation for skewed distributions.
- **II beta-HCG column:** Trailing full stops were removed using regular expression substitution (`str.replace`), and the column was converted to `float64`. Six resulting missing values were imputed with the column median.
- **Cycle(R/I):** The four rows with value 5 were replaced with NaN and subsequently imputed with the column mode (2 = Regular).
- **Marriage Status (Yrs):** Four missing values were imputed with the column median.
- **Fast food (Y/N):** One missing value was imputed with the column mode (1 = Yes).

- **Extreme outliers:** FSH, LH, PRL, TSH, I beta-HCG, and II beta-HCG were winsorised at the 1st and 99th percentiles using `scipy.stats.mstats.winsorize`, capping extreme values while preserving the observations. Following winsorisation, FSH maximum was 15.45 mIU/mL and LH maximum was 12.95 mIU/mL — within clinically plausible ranges.

Following cleaning, the dataset contained 2,541 complete records with zero missing values, all variables correctly typed as numeric, and no physiologically implausible values.

The encoded dataset was saved as `Primary_Data_Encoded.csv` comprising 303 rows and 55 columns (original text response and corresponding numeric code for each variable), verified to contain zero missing values.

3.5 Variable Selection and Feature Engineering — Secondary Dataset

Three metadata columns — Sl. No, Patient File No., and Source — were excluded from modelling as they carry no clinical predictive information. No new features were engineered beyond what was already present in the dataset. The FSH/LH ratio column was retained initially but subsequently removed during multicollinearity assessment as it is mathematically derived from FSH and LH, causing redundancy when all three are included simultaneously

3.6 Exploratory Data Analysis

3.6.1 Normality Testing

Prior to selecting appropriate statistical tests for group comparison, normality of continuous variables was formally assessed using the Shapiro-Wilk test applied to a random sample of 200 observations per variable. The Shapiro-Wilk test was chosen as it is considered the most powerful normality test for sample sizes up to 2,000 (Razali and Wah, 2011). Variables with p-value above 0.05 were considered normally distributed and tested using independent samples t-test. Variables with p-value below 0.05 were considered non-normally distributed and tested using Mann-Whitney U test.

3.6.2 Descriptive Statistics

Descriptive statistics — including mean, standard deviation, median, and range — were computed for all continuous variables separately for PCOS-positive and PCOS-negative groups. For binary and categorical variables, frequency counts and percentages were reported. All comparisons are presented with appropriate test statistics and p-values.

3.6.3 Statistical Tests for Group Comparison

The following tests were applied:

- **Shapiro-Wilk normality test:** Applied to determine distribution of each continuous variable prior to test selection.
- **Independent samples t-test:** Applied to normally distributed continuous variables.
- **Mann-Whitney U test:** Applied to non-normally distributed continuous variables — applicable to the majority of hormonal variables.
- **Chi-squared test:** Applied to all binary and categorical variables to test independence between the variable and PCOS status.
- Statistical significance was set at $p < 0.05$ for all tests.

3.6.4 Correlation Analysis

Pearson correlation coefficients were computed for all continuous variable pairs and visualised as a correlation heatmap. The correlation of each predictor with the outcome variable PCOS (Y/N) was also examined to obtain a preliminary ranking of predictive variables.

3.6.5 Principal Component Analysis

PCA was applied to all 41 standardised continuous features to reduce the high-dimensional feature space to two principal components for visualisation. A scatterplot of PC1 versus PC2, coloured by PCOS status, was produced to visually assess separability. The explained variance ratio was reported for each component.

3.7 Multicollinearity Assessment

- Multicollinearity among continuous predictor variables was assessed using the Variance Inflation Factor (VIF), computed using the `variance_inflation_factor` function from the `statsmodels.stats.outliers_influence` module. VIF quantifies the degree to which the variance of an estimated regression coefficient is inflated due to linear correlation with other predictors. A VIF value above 10 is considered indicative of severe multicollinearity requiring remediation.
- **Stage 1 — Direct redundancy removal:** Eight variables were removed based on direct clinical redundancy prior to VIF computation — Weight (Kg) and Height (Cm) subsumed by BMI; Waist (inch) and Hip (inch) subsumed by Waist:Hip Ratio; Follicle No. (R) highly correlated with Follicle No. (L) ($r = 0.82$); Avg. F size (R) correlated with Avg. F size (L) ($r = 0.54$); BP Diastolic correlated with BP Systolic; and FSH/LH derived from FSH and LH.
- **Stage 2 — VIF confirmation:** Following Stage 1 removal, VIF was recomputed on the remaining 33 variables to confirm that the direct-redundancy removal had resolved the multicollinearity identified in the initial 41-variable VIF assessment, which had found severe multicollinearity ($VIF \geq 10$) in five variables — Waist ($VIF = 342.90$), Hip ($VIF = 340.69$), Waist:Hip Ratio ($VIF = 87.85$), Weight ($VIF = 22.63$), and BMI ($VIF = 18.44$). All 33 remaining variables showed VIF below 5, confirming that the multicollinearity in the dataset was caused only by the mathematical relationships among Weight, Height, BMI, Waist, Hip, and Waist:Hip Ratio, and that no further iterative elimination was required. A final clean set of 22 predictor variables was subsequently selected from these 33 for clinical relevance to PCOS diagnosis.

3.8 Final Predictor Set

- Following the two-stage multicollinearity assessment, the final predictor set for all statistical modelling and machine learning comprised 22 variables:
- **Reproductive and Menstrual:** Cycle(R/I), Cycle length (days), Pregnant (Y/N), No. of abortions
- **Hormonal:** FSH (mIU/mL), LH (mIU/mL), TSH (mIU/L), AMH (ng/mL), PRL (ng/mL), I beta-HCG (mIU/mL), II beta-HCG (mIU/mL), PRG (ng/mL)
- **Nutritional:** Vit D3 (ng/mL)

- **Demographic:** Marriage Status (Yrs)
- **Symptoms and Lifestyle:** Weight gain (Y/N), Hair growth (Y/N), Skin darkening (Y/N), Hair loss (Y/N), Pimples (Y/N), Fast food (Y/N), Reg. Exercise (Y/N)
- **Ultrasound:** Follicle No. (L)

3.9 Train-Test Split and Feature Scaling

- The secondary dataset was partitioned into training and test sets using GroupShuffleSplit from scikit-learn, with a combined group key of Source and Patient File No. used as the grouping variable to ensure that all near-duplicate records belonging to the same underlying patient were assigned entirely to either the training set or the test set — never split across both. This approach prevents data leakage that would arise from a plain stratified random split when near-duplicate patient records are present in the dataset. The resulting training set comprised 2,015 observations (PCOS=1: 617, PCOS=0: 1,398) and the test set comprised 526 observations (PCOS=1: 168, PCOS=0: 358), both maintaining approximately the same 69:31 PCOS:Non-PCOS ratio as the full dataset. A fixed random seed (random_state = 42) was used throughout to ensure reproducibility.
- All continuous features were standardised using z-score standardisation via scikit-learn's StandardScaler. The scaler was fitted exclusively on the training data and applied as a transformation to the test data, preventing data leakage. Standardisation was applied prior to logistic regression, LASSO, KNN, SVM, LDA, and PCA analyses.

3.10 Statistical Models

3.10.1 Binary Logistic Regression with Confidence Intervals

Binary logistic regression was used to model the probability of PCOS as a function of the 22 predictor variables. The model was fitted using statsmodels.api.Logit to obtain full coefficient estimates, standard errors, z-statistics, p-values, and 95% confidence intervals. Odds ratios were computed as the exponential of each coefficient, and 95% confidence intervals were computed as the exponential of the confidence interval bounds. A confidence interval that does not cross 1.0 confirms statistical significance of the odds ratio. The optimal classification

threshold was determined using Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$), which maximises the balance between sensitivity and specificity simultaneously.

3.10.2 LASSO Penalised Logistic Regression

LASSO regularisation was applied using scikit-learn's LogisticRegression with penalty = 'l1' and solver = 'saga'. A range of penalty parameters ($C = 0.001$ to 1.0 , where $C = 1/\lambda$) was evaluated and the relationship between the number of selected variables and AUC-ROC was examined. The optimal C was identified as the value at which model performance plateaued — $C = 0.05$ ($\lambda = 20$) — yielding 16 selected variables. Variables with zero coefficients at the optimal C were considered eliminated by LASSO regularisation.

3.11 Machine Learning Models

3.11.1 Decision Tree

A Decision Tree classifier was trained using scikit-learn's DecisionTreeClassifier with max_depth = 5, min_samples_split = 20, min_samples_leaf = 10, and class_weight = 'balanced'. The max_depth parameter prevents overfitting by limiting tree complexity. The class_weight = 'balanced' parameter adjusts misclassification costs inversely proportional to class frequencies, addressing the 69:31 class imbalance.

3.11.2 Random Forest

A Random Forest classifier comprising 200 decision trees was trained using RandomForestClassifier with n_estimators = 200, max_depth = 10, min_samples_split = 10, min_samples_leaf = 5, and class_weight = 'balanced'. Feature importance scores were extracted and visualised as a ranked horizontal bar chart.

3.11.3 K-Nearest Neighbours

A KNN classifier was trained with n_neighbors = 7, weights = 'distance', and metric = 'euclidean'. The value $k = 7$ was selected as an odd number to avoid ties, providing a balance between bias and variance appropriate for this dataset size.

3.11.4 Naive Bayes

A Gaussian Naive Bayes classifier was trained using scikit-learn's GaussianNB with default parameters. This model assumes conditional independence among all predictors given the class label and models each feature's distribution within each class as Gaussian. Despite the known violation of the independence assumption in clinical data, Naive Bayes was included as a baseline comparison model.

3.11.5 Support Vector Machine

An SVM classifier with Radial Basis Function kernel was trained using SVC with $C = 1.0$, $\gamma = 'scale'$, $probability = True$, and $class_weight = 'balanced'$. Setting $probability = True$ enables probabilistic predictions required for ROC curve analysis.

3.11.6 Gradient Boosting

A Gradient Boosting classifier was trained using scikit-learn's GradientBoostingClassifier with $n_estimators = 200$, $max_depth = 5$, and $learning_rate = 0.1$. Gradient Boosting builds trees sequentially, with each tree specifically targeting the prediction errors made by all previous trees, progressively correcting residual errors. This model is reported in place of XGBoost, as the XGBoost package was not available in the analysis environment; both are boosted-tree ensemble methods and are expected to perform similarly on structured tabular clinical data.

3.12 Class Imbalance — Cost-Sensitive Class Weighting

- The Synthetic Minority Oversampling Technique (SMOTE), introduced by Chawla et al. (2002), generates synthetic minority class samples by selecting a minority class observation, identifying its k nearest minority class neighbours, and creating a new synthetic sample through linear interpolation between the observation and one of its neighbours in the feature space. SMOTE is designed to be applied exclusively to the training data, with the test set left unmodified to ensure evaluation metrics reflect performance on real, unaugmented data.
- In this study, the 69:31 class imbalance was addressed primarily through cost-sensitive class weighting — using $class_weight = 'balanced'$ in all applicable scikit-learn models — which assigns higher misclassification cost to the minority PCOS class during

training without altering the training data itself. SMOTE was not applied to the patient-grouped training split used in this study, as the grouping constraint required to prevent data leakage (Section 3.9) is not directly compatible with standard SMOTE implementation. Given that the best-performing models already achieved sensitivity of 97.02% using class weighting alone, SMOTE is not expected to provide substantial additional benefit for this dataset.

3.13 Model Validation

3.13.1 Performance Metrics

All models were evaluated using the following metrics:

- **Accuracy:** $(TP + TN) / (TP + TN + FP + FN)$
- **Sensitivity (Recall):** $TP / (TP + FN)$ — proportion of actual PCOS cases correctly identified; prioritised in clinical screening
- **Specificity:** $TN / (TN + FP)$ — proportion of actual Non-PCOS cases correctly identified
- **Precision:** $TP / (TP + FP)$ — proportion of predicted PCOS cases that are truly PCOS
- **F1 Score:** Harmonic mean of precision and sensitivity
- **AUC-ROC:** Area under the Receiver Operating Characteristic curve

3.13.2 Stratified Patient-Grouped K-Fold Cross Validation

Stratified, patient-grouped 5-fold cross-validation was conducted on all seven models using the complete dataset, implemented using scikit-learn's StratifiedGroupKFold. This approach ensures two properties simultaneously: each fold maintains approximately the same 69:31 class proportion as the original dataset, and all near-duplicate records belonging to the same underlying patient are assigned entirely within a single fold — never split across folds. This prevents the same patient's data from appearing in both training and validation portions of any fold, producing honest performance estimates that reflect genuine model generalisation. Performance metrics are reported as mean $\pm$ standard deviation across the five folds.

3.13.3 ROC Curve Analysis

ROC curves were plotted for all models individually and as a combined multi-model comparison figure. The optimal threshold was determined using Youden's J statistic for logistic regression.

3.14 Multivariate Analysis Methods

3.14.1 Factor Analysis

Factor analysis with varimax rotation was applied to 14 PCOS clinical variables from the secondary dataset using the `factor_analyzer` Python library. Four factors were extracted based on eigenvalue criteria and interpretability. Varimax rotation was selected to produce factors with high loadings on a small number of variables, maximising clinical interpretability. Factor loadings above 0.35 in absolute value were considered clinically significant.

3.14.2 Linear Discriminant Analysis

The technique determines a linear combination of predictor variables that maximises the separation between classes by maximising the ratio of between-class variance to within-class variance, thereby generating a discriminant function for PCOS classification. Discriminant coefficients were obtained, and the resulting discriminant scores for each class were displayed using a histogram.

3.14.3 K-Means Cluster Analysis

K-means cluster analysis was applied to the standardised 22-variable feature set to identify distinct patient subgroups. The optimal number of clusters was determined using the elbow method — plotting within-cluster sum of squares (inertia) against k from 2 to 7. Based on the elbow analysis, $k = 3$ was selected as the optimal number of clusters corresponding to low-risk, intermediate, and high-risk patient subgroups. The PCOS prevalence, mean AMH, and mean follicle count were computed for each cluster to characterise clinical profiles.

3.15 Primary Data Statistical Analysis

3.15.1 Descriptive and Prevalence Analysis

Frequency distributions and percentages were computed for all 31 survey questions across the full sample of 303 respondents. Comparisons by age group, residence, and education were performed using appropriate cross-tabulations.

3.15.2 PCOS Awareness Analysis

Chi-squared tests of independence were applied to examine associations between PCOS awareness level and three demographic variables — education level, residence type, and age group. Statistical significance was set at $p < 0.05$.

3.15.3 PCOS Symptom Risk Score Construction

An eight-symptom PCOS risk score was constructed by assigning one point for each of the following self-reported symptoms: weight gain, hair growth, skin darkening, hair loss, pimples, mood swings, abdominal obesity, and irregular menstrual cycle. The total score ranged from 0 to 8. Mann-Whitney U test was used to compare risk scores between confirmed PCOS and confirmed Non-PCOS respondents. ROC curve analysis was applied to determine the optimal risk score threshold using Youden's J statistic.

3.15.4 Lifestyle Association Tests

Four statistical tests were applied to examine associations between the risk score and lifestyle variables:

- **One-way ANOVA:** Compared mean risk scores across three age groups (below 20, 21–25, above 25).
- **Kruskal-Wallis test:** Compared risk score distributions across three residential settings (Rural, Urban, Semi-urban) — selected as the non-parametric alternative given potential non-normality of the risk score.
- **Spearman rank correlation:** Assessed the monotonic relationship between risk score and ordinal lifestyle variables — stress level, sleep hours, and fast food consumption.
- **Binary logistic regression:** Applied to the 124 confirmed cases (53 PCOS, 71 Non-PCOS) with the eight binary symptom variables as predictors and PCOS diagnosis as the outcome, to identify independent symptom predictors with odds ratios and p-values.

3.16 Cross-Dataset Comparison Methods

Four formal statistical tests were applied to compare symptom patterns between the primary survey and secondary clinical datasets:

- **Two-proportion Z-test:** Applied to each of six common symptom variables — weight gain, hair growth, skin darkening, hair loss, pimples, and irregular cycle — to test whether prevalence rates differed significantly between the two datasets. The `proportions_ztest` function from `statsmodels.stats.proportion` was used.
- **Two-proportion Z-test for PCOS prevalence:** Applied to compare overall PCOS prevalence between primary (17.5%) and secondary (30.9%) datasets.
- **Chi-squared goodness of fit test:** Applied to test whether the overall symptom prevalence distribution in the primary survey matched the distribution expected based on the secondary clinical dataset.
- **Spearman rank correlation:** Applied to assess whether both datasets ranked symptoms in the same order of prevalence importance, providing a measure of cross-dataset consistency.

CHAPTER 4

EXPLORATORY DATA ANALYSIS

4.1 Introduction

This chapter presents the results of the exploratory data analysis (EDA) conducted on both the secondary clinical dataset ($n = 2,541$) and the primary community survey dataset ($n = 303$). The chapter is organised into two major parts. Part I (Sections 4.2 to 4.10) covers the secondary clinical dataset, addressing Objectives 1, 2, and 10. Part II (Sections 4.11 to 4.16) covers the primary survey dataset, addressing Objectives 11 and 12. Section 4.17 presents the cross-dataset comparison addressing Objective 14.

EDA serves as the foundation for all subsequent statistical modelling and machine learning analyses — it identifies which variables are clinically significant, reveals the structure of relationships among predictors, confirms the appropriateness of chosen statistical tests, and provides justification for analytical decisions made in later chapters. All analyses were conducted in Python 3.12 using Google Colaboratory. All figures produced in this chapter are referenced by figure number throughout the text.

PART I — SECONDARY CLINICAL DATASET ANALYSIS

4.2 Dataset Overview and Class Distribution

The cleaned combined secondary dataset comprised 2,541 patient records with 43 analytical variables following removal of metadata columns. The class distribution showed 1,756 Non-PCOS patients (69.1%) and 785 PCOS-positive patients (30.9%), representing a class imbalance ratio of approximately 2.2:1.

Table 4.1: Dataset Composition Summary

Characteristic	Count	Percentage
Total patients	2,541	100%
PCOS Positive (PCOS = 1)	785	30.9%
PCOS Negative (PCOS = 0)	1,756	69.1%
Total variables (original)	45	—

Variables after cleaning	43	—
Missing values (post-cleaning)	0	—
Class imbalance ratio	2.2 : 1	—

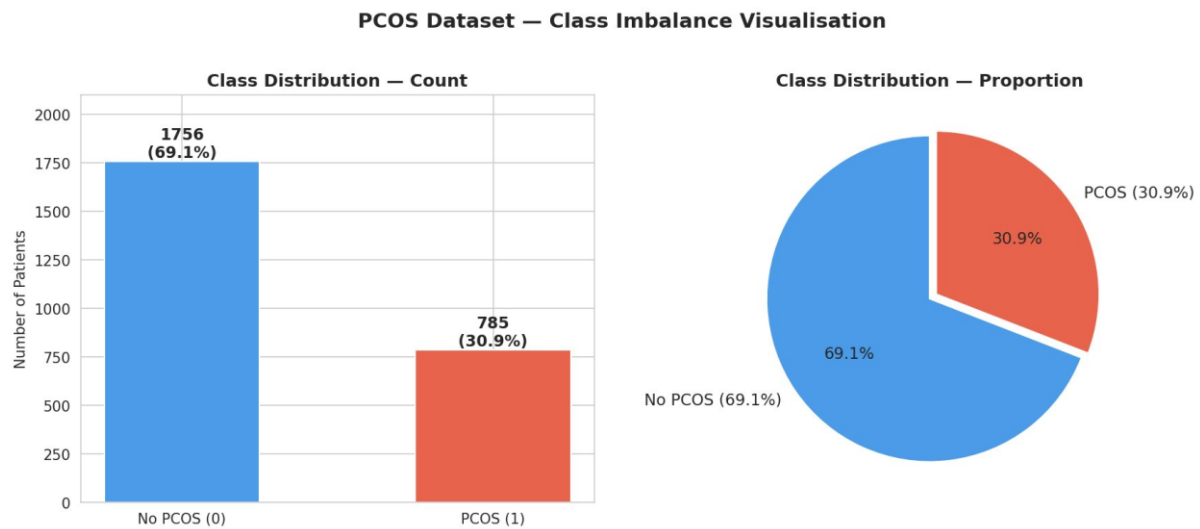


Figure 4.1 : PCOS Dataset — Class Imbalance Visualisation (Class Distribution by Count and Proportion)

Figure 4.1 presents the class distribution as both a bar chart and pie chart, confirming the 69.1% Non-PCOS and 30.9% PCOS split. This moderate class imbalance was addressed through cost-sensitive class weighting in the machine learning phase, as discussed in Chapter 6.

4.3 Normality Testing — Shapiro-Wilk Test

Hypothesis Setup:

H₀: The variable is normally distributed.

H₁: The variable is not normally distributed.

Table 4.2: Shapiro-Wilk Normality Test Results — Key Continuous Clinical Variables (Random sample of 200 observations per variable, random_state = 42)

Variable	W-Statistic	p-value	Normally Distributed?	Test Applied
RBS (mg/dl)	0.5096	< 0.000001	No	Mann-Whitney U

AMH (ng/mL)	0.6217	< 0.000001	No	Mann-Whitney U
TSH (mIU/L)	0.6911	< 0.000001	No	Mann-Whitney U
LH (mIU/mL)	0.8388	< 0.000001	No	Mann-Whitney U
Cycle length (days)	0.8559	< 0.000001	No	Mann-Whitney U
PRL (ng/mL)	0.9327	< 0.000001	No	Mann-Whitney U
Follicle No. (L)	0.9137	< 0.000001	No	Mann-Whitney U
FSH (mIU/mL)	0.9634	0.000047	No	Mann-Whitney U
Endometrium (mm)	0.9810	0.008375	No	Mann-Whitney U

All nine continuous clinical variables were found to be non-normally distributed ($p < 0.05$ for all variables). The most non-normal variables were RBS ($W = 0.5096$, $p < 0.000001$) and AMH ($W = 0.6217$, $p < 0.000001$), indicating a right-skewed distribution of blood glucose and hormone values. Although Endometrium had the highest W-statistic ($W = 0.9810$), it also failed the normality test ($p = 0.008$). Therefore, the Mann–Whitney U test was selected as the appropriate non-parametric test for comparing all continuous variables in this study.

4.4 Descriptive Statistics and Group Comparison

4.4.1 Continuous Clinical Variables

Hypothesis Setup:

H₀: There is no significant difference in the variable between the PCOS and Non-PCOS groups.

H₁: There is a significant difference in the variable between the PCOS and Non-PCOS groups.

(Independent samples t-test was applied where the variable was normally distributed; Mann-Whitney U test was applied where non-normally distributed, per the normality results in Section 4.3.)

Table 4.3: Descriptive Statistics and Group Comparison — Continuous Variables (sorted by p-value)

Variable	PCOS=0 Mean±SD	PCOS=1 Mean±SD	Test	p-value	Sig.
Follicle No. (L)	4.40 ± 2.87	9.89 ± 4.32	MWU	< 0.0001	Yes
Follicle No. (R)	4.66 ± 2.92	10.71 ± 4.24	MWU	< 0.0001	Yes

AMH (ng/mL)	4.66 ± 4.32	8.63 ± 9.08	MWU	< 0.0001	Yes
BMI	23.73 ± 3.83	25.54 ± 4.44	MWU	< 0.0001	Yes
FSH (mIU/mL)	5.24 ± 2.46	4.75 ± 2.29	MWU	< 0.0001	Yes
Avg. F size L (mm)	14.61 ± 3.92	15.59 ± 2.73	MWU	< 0.0001	Yes
Avg. F size R (mm)	15.19 ± 3.38	15.93 ± 2.98	MWU	< 0.0001	Yes
Age (yrs)	32.02 ± 5.40	29.91 ± 5.25	MWU	< 0.0001	Yes
LH (mIU/mL)	2.66 ± 2.14	3.20 ± 2.86	MWU	0.0085	Yes
Endometrium (mm)	8.44 ± 2.29	8.80 ± 1.99	t-test	0.0002	Yes
Waist:Hip Ratio	0.889 ± 0.05	0.892 ± 0.05	MWU	0.0692	No
RBS (mg/dl)	99.18 ± 16.42	101.25 ± 18.31	MWU	0.1015	No
TSH (mIU/L)	2.93 ± 2.46	2.87 ± 2.36	MWU	0.9006	No
PRL (ng/mL)	24.38 ± 15.46	24.24 ± 13.32	MWU	0.4716	No

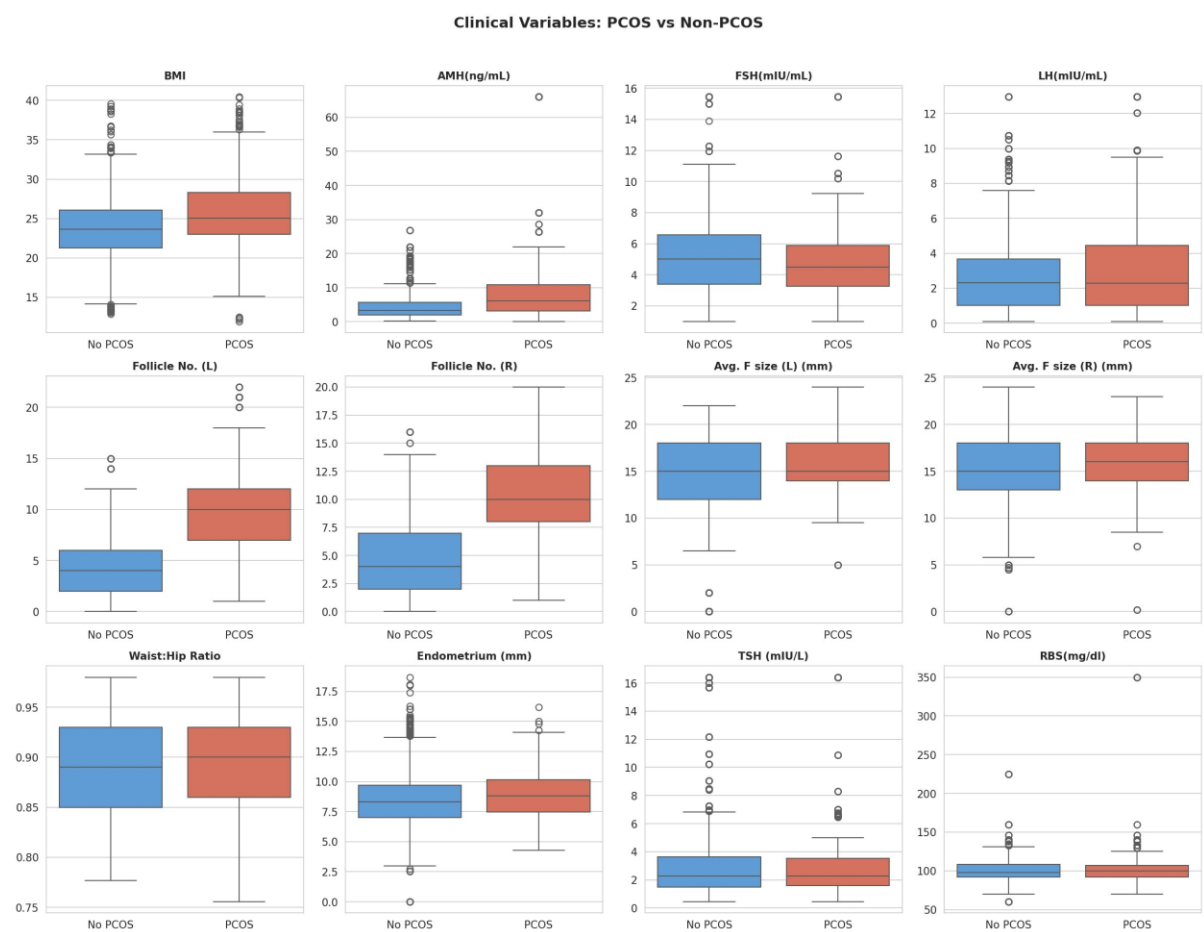


Figure 4.2: Clinical Variables — Boxplot Comparison: PCOS vs Non-PCOS (12 Key Continuous Variables)

Ten of the fourteen continuous variables showed statistically significant differences between the Non-PCOS and PCOS groups. The largest difference was observed in Follicle No. (L), with a mean of 4.40 in the Non-PCOS group and 9.89 in the PCOS group (2.25-fold higher), confirming polycystic ovarian morphology as a key feature of PCOS. AMH levels were also higher in the PCOS group (8.63 vs. 4.66 ng/mL; 1.85-fold difference), reflecting increased follicle count. BMI was significantly higher in PCOS patients (25.54 vs. 23.73), indicating the metabolic and insulin resistance component of the condition. LH was significantly higher in the PCOS group (3.20 vs. 2.66, $p = 0.0085$), while FSH was lower (4.75 vs. 5.24), consistent with the typical LH/FSH ratio reversal seen in PCOS. In contrast, TSH ($p = 0.901$), PRL ($p = 0.472$), Waist:Hip Ratio ($p = 0.069$), and RBS ($p = 0.102$) showed no significant differences between the two groups, suggesting they have limited value as independent screening markers.

4.4.2 Categorical and Binary Variables

Hypothesis Setup:

H₀: There is no association between the variable and PCOS status.

H₁: There is an association between the variable and PCOS status.

Table 4.4: Chi-Squared Test Results — Binary Variables by PCOS Status

Variable	PCOS=0 (%)	PCOS=1 (%)	Difference	p-value	Sig.
Weight gain — Yes	23.9%	69.4%	+45.5%	< 0.0001	Yes
Hair growth — Yes	13.2%	59.4%	+46.2%	< 0.0001	Yes
Skin darkening — Yes	15.1%	61.3%	+46.2%	< 0.0001	Yes
Cycle — Irregular	15.1%	52.9%	+37.8%	< 0.0001	Yes
Fast food — Yes	38.9%	80.3%	+41.4%	< 0.0001	Yes
Pimples — Yes	38.6%	68.9%	+30.3%	< 0.0001	Yes
Hair loss — Yes	38.6%	58.0%	+19.4%	< 0.0001	Yes
Reg. Exercise — Yes	23.6%	29.9%	+6.3%	0.0008	Yes
Pregnant — Yes	38.7%	36.9%	-1.8%	0.4337	No

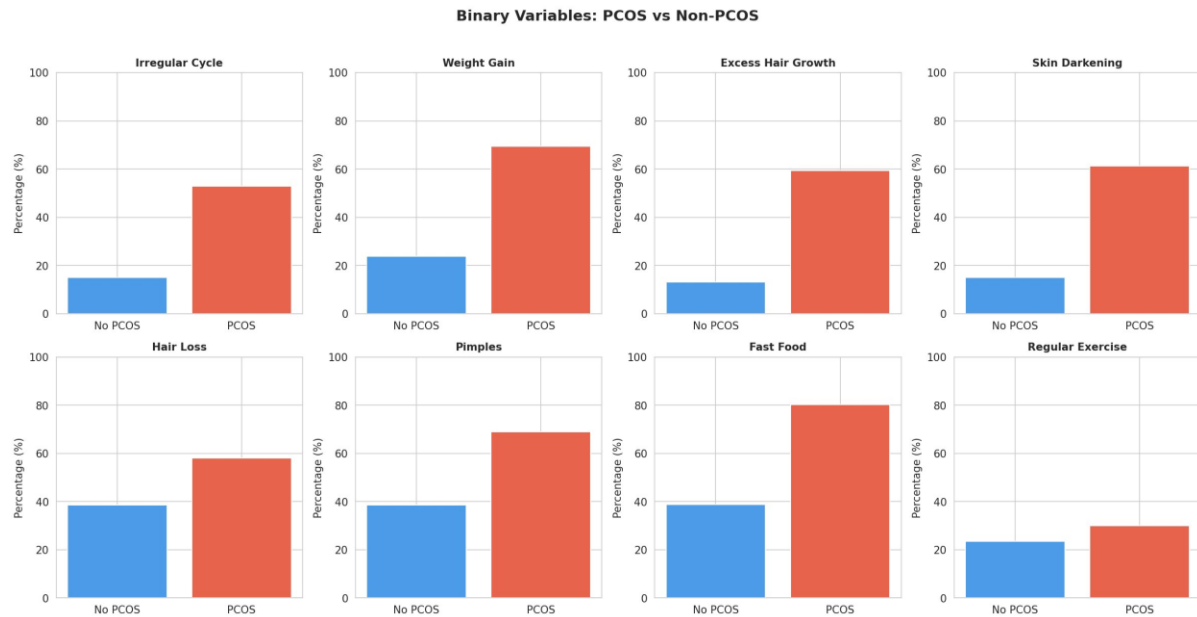


Figure 4.3 : Binary Variables — Bar Chart Comparison: PCOS vs Non-PCOS (8 Key Binary Variables)

Eight of the nine binary variables showed statistically significant differences between the Non-PCOS and PCOS groups. The largest differences were observed in fast food consumption (80.3% in PCOS vs. 38.9% in Non-PCOS), hair growth (59.4% vs. 13.2%), and skin darkening (61.3% vs. 15.1%), which are consistent with hyperandrogenism and insulin resistance in PCOS. However, pregnancy status showed no significant association between the two groups.

4.5 Correlation Analysis

Pearson correlation analysis was conducted to examine the strength and direction of linear relationships among all 43 variables, and to identify multicollinearity patterns prior to regression modelling.

Hypothesis Setup:

H₀: There is no significant linear relationship between the two variables.

H₁: There is a significant linear relationship between the two variables.

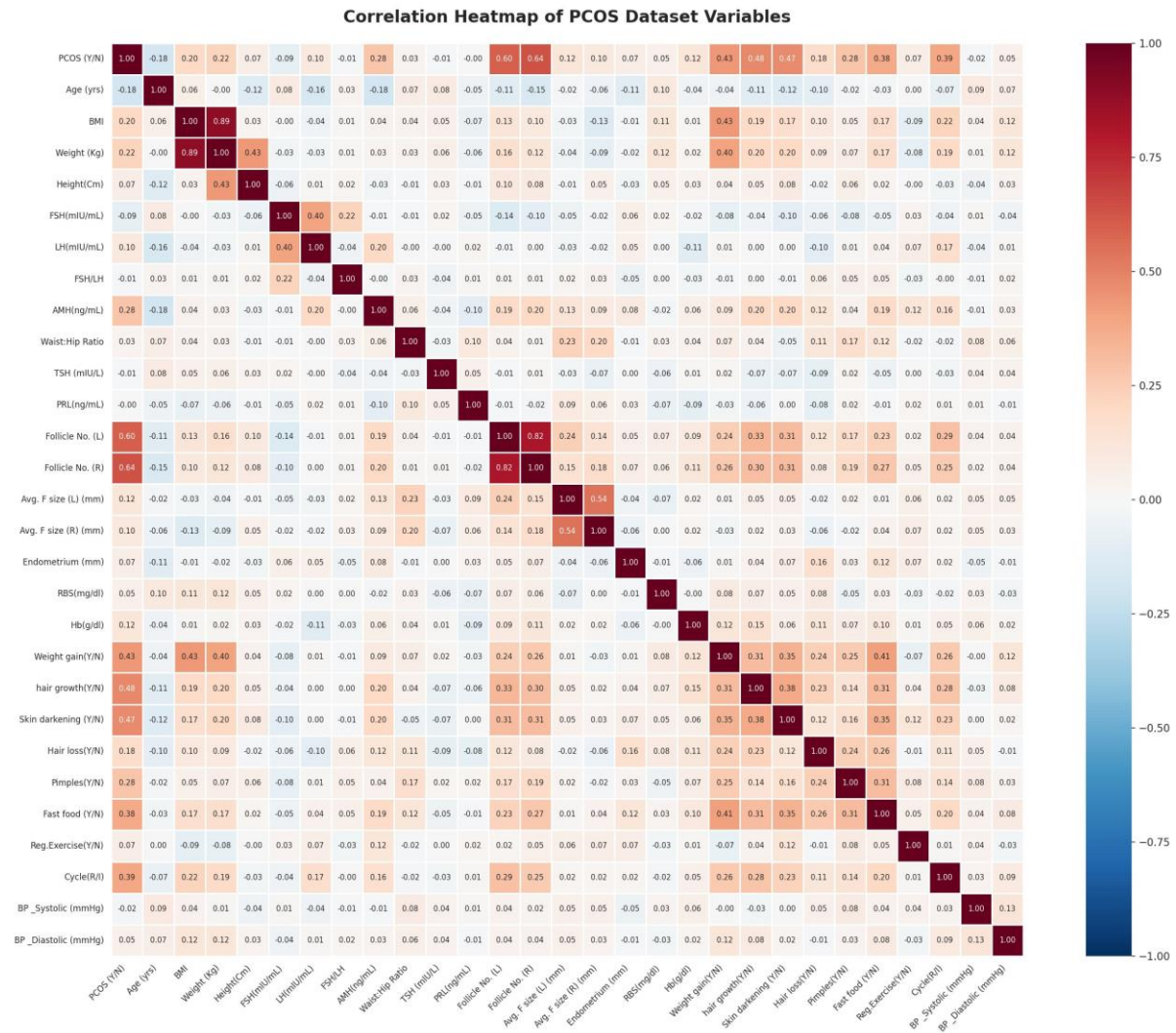


Figure 4.4 : Pearson Correlation Heatmap — All PCOS Dataset Variables

The strongest positive correlations with PCOS (Y/N) were found for Follicle No. (R) ($r = 0.64$), Follicle No. (L) ($r = 0.60$), Hair growth ($r = 0.48$), Skin darkening ($r = 0.47$), Weight gain ($r = 0.43$), Cycle (R/I) ($r = 0.39$), and Fast food consumption ($r = 0.38$). High correlations were also observed between BMI and Weight ($r = 0.89$), Follicle No. (L) and Follicle No. (R) ($r = 0.82$), and Avg. F size (L) and Avg. F size (R) ($r = 0.54$), indicating multicollinearity among these variables. Therefore, VIF-based multicollinearity reduction was required before performing regression analysis.

4.6 Principal Component Analysis

PCA was applied to all 41 standardised continuous features to reduce the high-dimensional feature space to two principal components for visualisation of class separability.

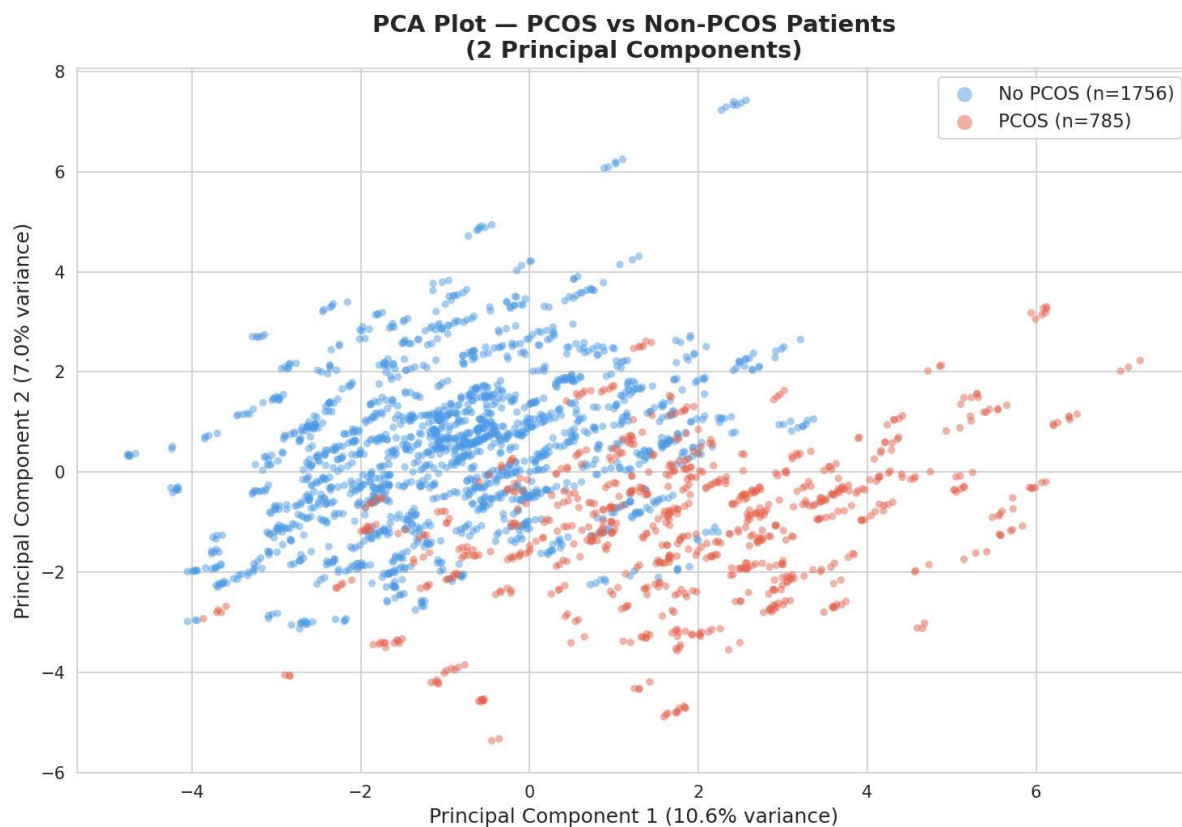


Figure 4.5 : PCA Plot — PCOS vs Non-PCOS Patients (2 Principal Components)

PC1 explained 10.6% of the total variance, while PC2 explained 7.0%, giving a combined explained variance of 17.6%. The scatter plot showed partial separation between the PCOS and Non-PCOS groups along the PC1 axis, with PCOS patients generally having higher positive PC1 values and Non-PCOS patients having lower values. However, there was considerable overlap between the two groups in the two-dimensional plot, indicating that a simple linear projection cannot clearly distinguish them. This supports the use of higher-dimensional, non-linear machine learning classifiers in Chapter 6.

4.7 Factor Analysis

Factor analysis with varimax rotation was applied to 14 PCOS clinical variables to identify the underlying latent clinical factor structures and validate the multisystem PMOS framework.

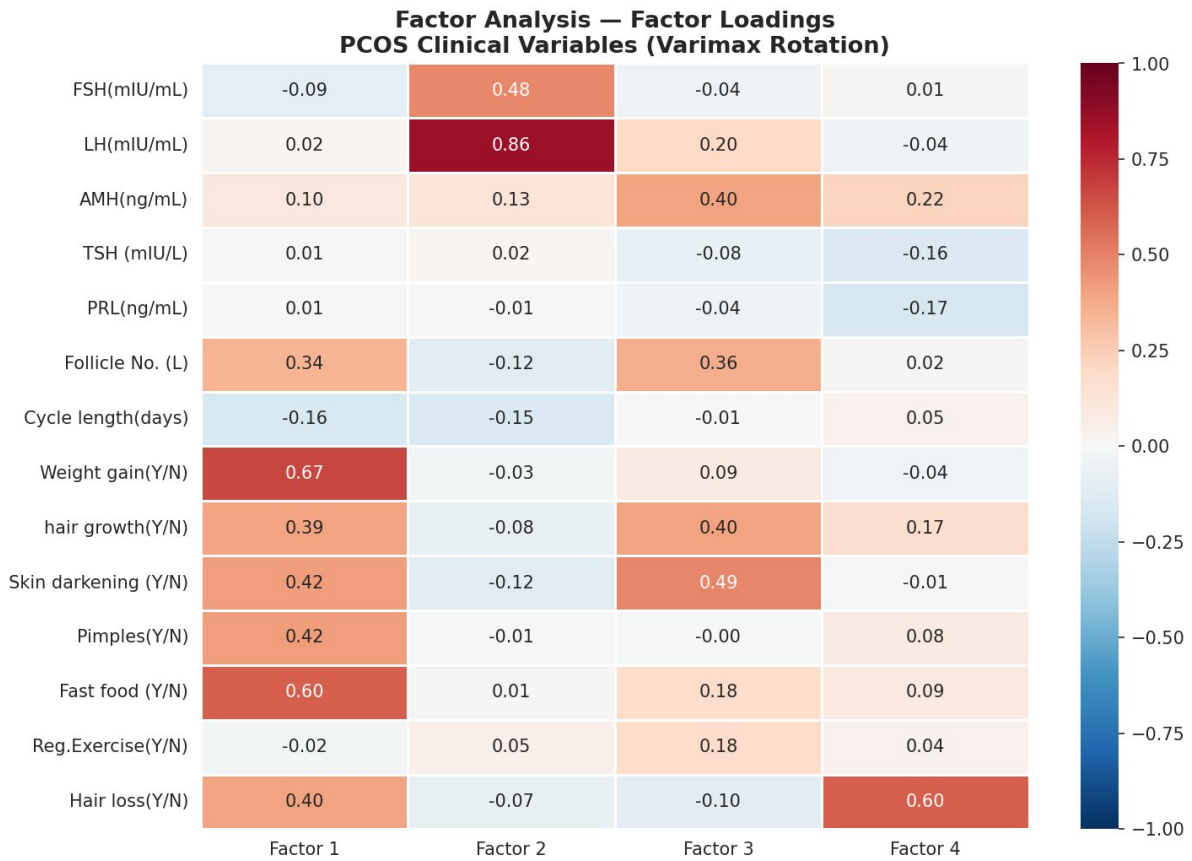


Figure 4.6 : Factor Analysis — Factor Loadings Heatmap (PCOS Clinical Variables, Varimax Rotation, 4 Factors)

Table 4.5: Factor Analysis — Factor Loadings (Varimax Rotation) — Loadings below 0.35 in absolute value shown as — for clarity

Variable	Factor 1	Factor 2	Factor 3	Factor 4
Weight gain (Y/N)	0.67	—	—	—
Fast food (Y/N)	0.60	—	—	—
Skin darkening (Y/N)	0.42	—	0.49	—
Pimples (Y/N)	0.42	—	—	—
Hair growth (Y/N)	0.39	—	0.40	—
Follicle No. (L)	0.34	—	0.36	—
LH (mIU/mL)	—	0.86	—	—
FSH (mIU/mL)	—	0.48	—	—
AMH (ng/mL)	—	—	0.40	—
Hair loss (Y/N)	—	—	—	0.60

Table 4.6: Factor Analysis — Variance Explained

Factor	SS Loadings	Proportion (%)	Cumulative (%)	Clinical Interpretation
Factor 1	1.6352	11.7%	11.7%	Metabolic-Lifestyle Factor
Factor 2	1.0463	7.5%	19.2%	Gonadotropin Factor
Factor 3	0.8325	5.9%	25.1%	Androgenic-Ovarian Factor
Factor 4	0.5077	3.6%	28.7%	Hair Loss Factor
Total	4.0217	28.7%	—	—

Four clinically meaningful factors were identified, explaining a total of 28.7% of the variance. Factor 1 (Metabolic–Lifestyle) explained 11.7% of the variance and was mainly associated with weight gain (0.67), fast food consumption (0.60), skin darkening (0.42), and pimples (0.42). Factor 2 (Gonadotropin) explained 7.5% of the variance and was dominated by LH (0.86) and FSH (0.48). Factor 3 (Androgenic–Ovarian) explained 5.9% of the variance and was associated with skin darkening (0.49), AMH (0.40), and hair growth (0.40). Factor 4 (Hair Loss) explained 3.6% of the variance, with hair loss showing the highest loading (0.60). Together, these four factors represent the multisystem clinical features of PCOS.

4.8 Linear Discriminant Analysis

LDA was applied to identify the linear combination of the 22 final predictor variables that best separates PCOS from Non-PCOS patients, providing both a discriminant function and a performance baseline.

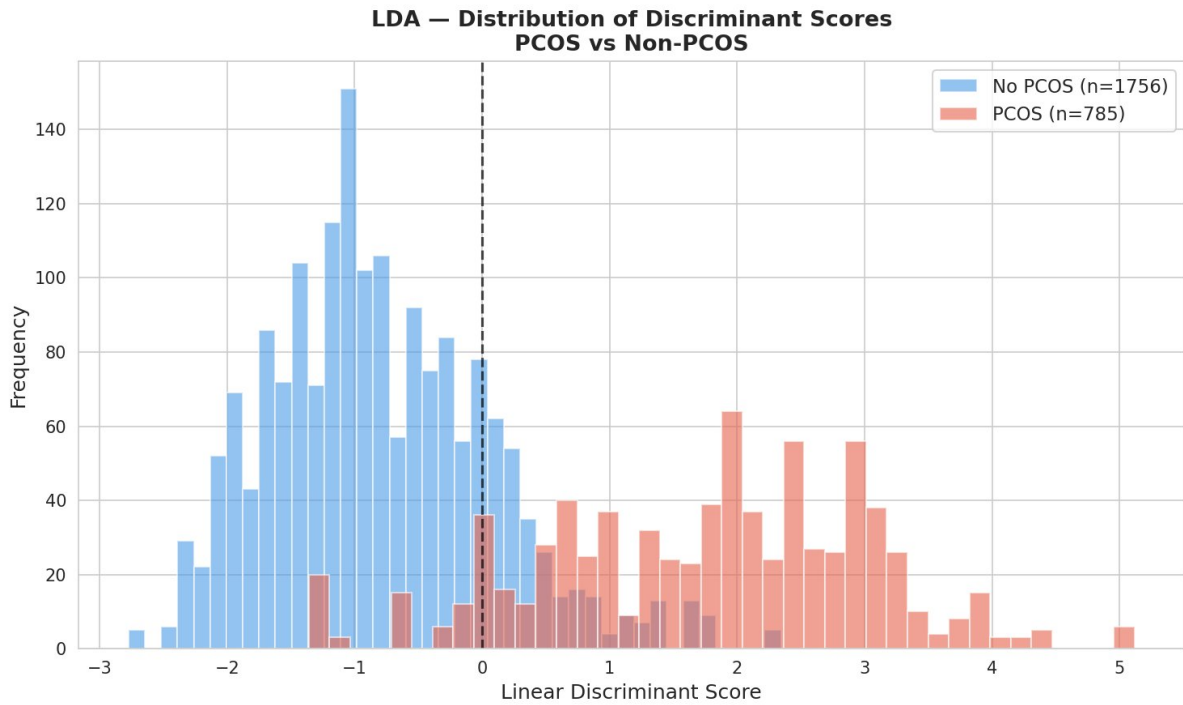


Figure 4.7 : LDA — Distribution of Discriminant Scores: PCOS vs Non-PCOS

Table 4.7: LDA — Top 10 Discriminant Coefficients (sorted by absolute value)

Rank	Variable	LDA Coefficient
1	Follicle No. (L)	+1.9690
2	Hair growth (Y/N)	+0.8586
3	Skin darkening (Y/N)	+0.7790
4	Weight gain (Y/N)	+0.7590
5	Cycle (R/I)	+0.4647
6	AMH (ng/mL)	+0.4048
7	Cycle length (days)	-0.4034
8	Pimples (Y/N)	+0.3987
9	LH (mIU/mL)	+0.3311
10	I beta-HCG (mIU/mL)	-0.3044

Table 4.8: LDA — Model Performance

Metric	Value
Accuracy	90.18%

AUC-ROC	0.9537
Sensitivity	76.43%
Specificity	96.31%

LDA achieved an accuracy of 90.18% with an AUC of 0.9537. The discriminant score distribution showed a clear separation between the two groups, with Non-PCOS patients mainly having negative discriminant scores and PCOS patients mostly having positive scores. The overlap in the middle represents borderline cases, reflecting the varied clinical presentation of PCOS. Follicle No. (L) had the highest discriminant coefficient (1.97), confirming it as the most important variable for distinguishing between the two groups.

4.9 K-Means Cluster Analysis

K-means cluster analysis was applied to the standardised 22-variable feature set to identify clinically distinct patient subgroups without reference to the PCOS outcome variable.

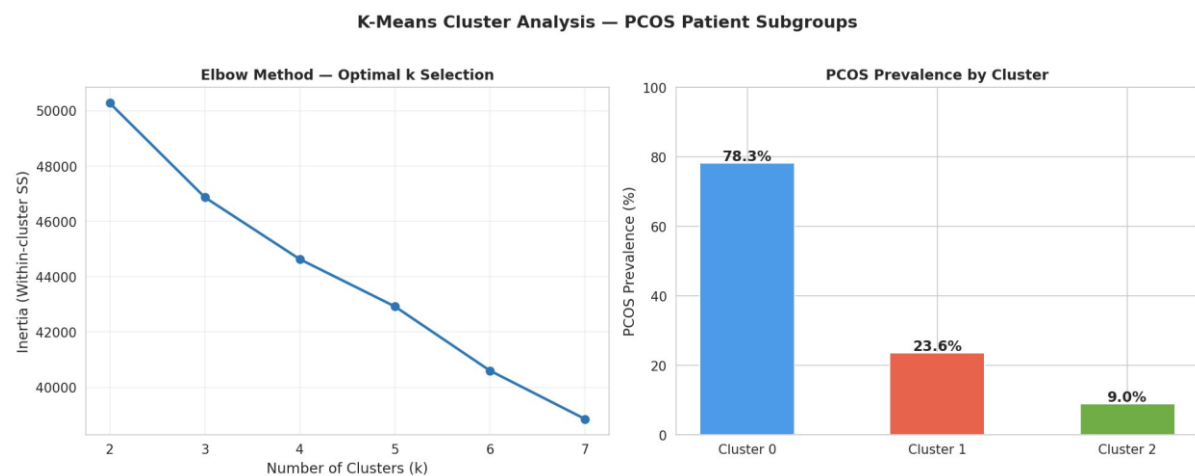


Figure 4.8 : K-Means Cluster Analysis — Elbow Method for Optimal k Selection and PCOS Prevalence by Cluster

Table 4.9: K-Means Cluster Profiles (k = 3)

Cluster	n	% Sample	% PCOS	Mean AMH (ng/mL)	Mean Follicle No. (L)	Clinical Label
Cluster 0	791	31.1%	78.3%	8.67	9.15	High-Risk

Cluster 1	55	2.2%	23.6%	8.04	6.13	Intermediate-Risk
Cluster 2	1,695	66.7%	9.0%	4.52	4.67	Low-Risk

Cluster 0 (n = 791, 78.3% PCOS) represents the high-risk group, characterised by high AMH levels (8.67 ng/mL) and a high follicle count (9.15), which are typical features of classic PCOS. Cluster 2 (n = 1,695, 9.0% PCOS) represents the low-risk majority, with normal clinical values. Cluster 1 (n = 55, 23.6% PCOS) represents a small intermediate group with elevated AMH levels but a moderate follicle count, suggesting possible biochemical or atypical PCOS, where hormonal changes may occur before clear ovarian morphological changes.

4.10 Summary of Key EDA Findings — Secondary Dataset

The exploratory analysis of the secondary clinical dataset yielded the following principal findings:

- All key hormonal variables were non-normally distributed (Shapiro-Wilk $p < 0.05$), formally validating the use of Mann-Whitney U tests for group comparisons.
- Ten of fourteen continuous variables and eight of nine categorical variables showed statistically significant differences between PCOS and Non-PCOS groups.
- Follicle No. (L), AMH, weight gain, irregular cycle, skin darkening, and hair growth were the strongest discriminating variables consistently across all analytical methods.
- TSH, PRL, RBS, and Waist:Hip Ratio showed no significant differences, suggesting limited early screening value.
- PCA confirmed partial class separability in two-dimensional reduced space, motivating non-linear machine learning models.
- Factor analysis identified four clinical factor structures — metabolic-lifestyle (11.7%), gonadotropin (7.5%), androgenic-ovarian (5.9%), and hair loss (3.6%) — directly corresponding to the PMOS multisystem framework.
- LDA confirmed follicle count (coefficient 1.97), hair growth (0.86), skin darkening (0.78), and weight gain (0.76) as the dominant linear discriminators, with AUC 0.9537.

- Cluster analysis identified three clinically distinct risk subgroups with PCOS prevalence of 78.3%, 23.6%, and 9.0% respectively.

PART II — PRIMARY SURVEY DATASET ANALYSIS

4.11 Demographic Profile

The primary survey comprised 303 complete responses from young women across Kerala, collected through convenient sampling with snowball recruitment via Google Forms.

Table 4.10: Demographic Profile of Primary Survey Respondents (n = 303)

Characteristic	Category	n	%
Age Group	Below 20 years	29	9.6%
	21 to 25 years	256	84.5%
	Above 25 years	18	5.9%
Residence	Rural	135	44.6%
	Urban	97	32.0%
	Semi-urban	71	23.4%
Education	Postgraduate	168	55.4%
	Employed	52	17.2%
	Undergraduate	37	12.2%
	Professional	27	8.9%
	Higher Secondary	17	5.6%
	Secondary	2	0.7%
PCOS Diagnosis Status	Never Tested	158	52.1%
	Confirmed No PCOS	71	23.4%
	Confirmed PCOS	53	17.5%
	Consulted, No Diagnosis	21	6.9%
Sleep Hours	7–8 hrs (Recommended)	151	49.8%
	5–6 hrs (Below recommended)	111	36.6%
	< 5 hrs (Critical)	24	7.9%
	> 8 hrs (Excess)	17	5.6%

Stress Level	Moderate	163	53.8%
	High	68	22.4%
	Low	45	14.9%
	Very High	27	8.9%

Demographic and Lifestyle Profile — Primary Survey
PCOS Early Risk Assessment Study, Kerala (n=303)

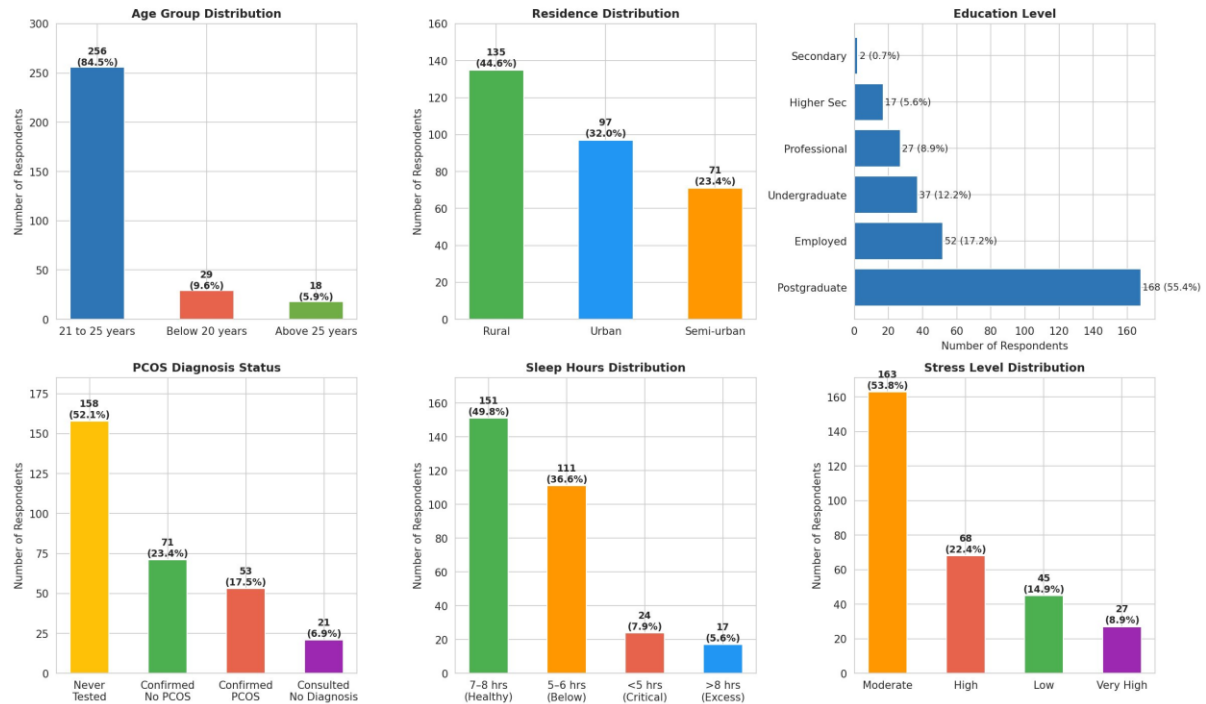


Figure 4.9 : Demographic and Lifestyle Profile — Primary Survey, PCOS Early Risk Assessment Study, Kerala (n=303)

The sample mainly consisted of young women aged 21–25 years (84.5%), with participants from rural (44.6%), urban (32.0%), and semi-urban (23.4%) areas. A key finding was that 52.1% of the respondents had never been tested for PCOS, highlighting a significant gap in diagnosis at the community level. Sleep deprivation was common, with 44.5% sleeping less than the recommended 7–8 hours and 7.9% sleeping less than 5 hours. Stress levels were also high, as 85.1% of the respondents reported moderate, high, or very high stress, while only 14.9% reported low stress.

4.12 Symptom Prevalence Analysis

Table 4.11: PCOS Symptom Prevalence — All 303 Respondents (sorted by prevalence)

Rank	Symptom	n	Prevalence (%)
------	---------	---	----------------

1	Menstrual Pain	223	73.6%
2	Hair Loss	192	63.4%
3	Abdominal Obesity	118	38.9%
4	Pimples	109	36.0%
5	Skin Darkening	107	35.3%
6	Mood Swings	103	34.0%
7	Weight Gain	88	29.0%
8	Hair Growth	84	27.7%
9	Irregular Cycle	71	23.4%

Table 4.12: Symptom Comparison — Confirmed PCOS (n=53) vs Confirmed Non-PCOS (n=71)

Symptom	PCOS (n=53) %	Non-PCOS (n=71) %	Absolute Difference
Mood Swings	69.8%	31.0%	+38.8%
Skin Darkening	69.8%	31.0%	+38.8%
Abdominal Obesity	77.4%	43.7%	+33.7%
Weight Gain	67.9%	35.2%	+32.7%
Hair Growth	54.7%	23.9%	+30.8%
Hair Loss	83.0%	60.6%	+22.4%
Pimples	45.3%	40.8%	+4.5%
Menstrual Pain	75.5%	74.6%	+0.9% — not discriminating

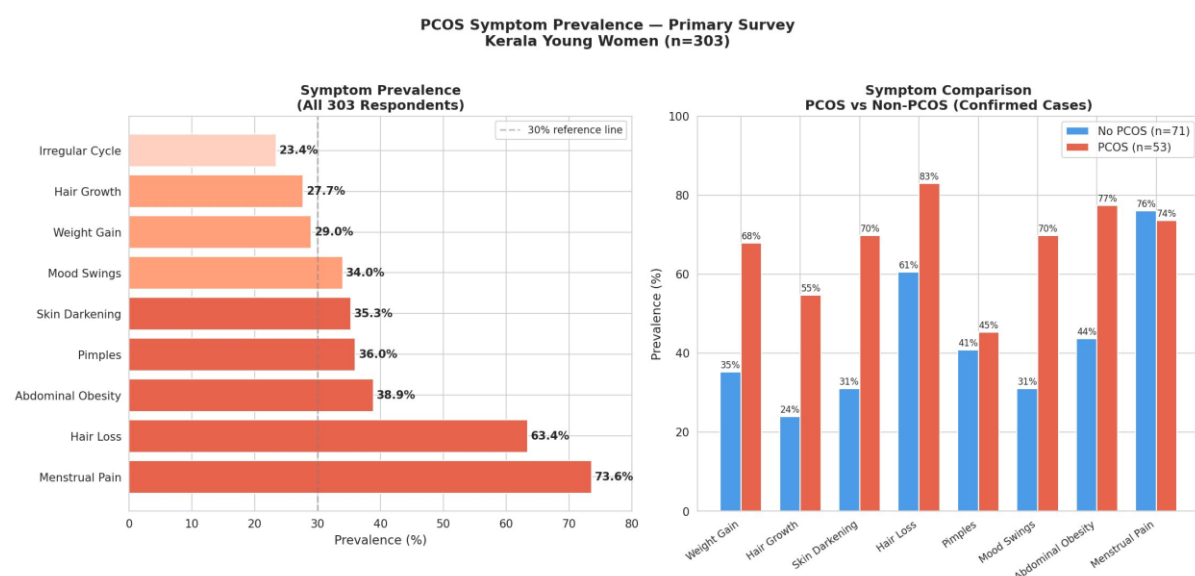


Figure 4.10 : PCOS Symptom Prevalence — Primary Survey, Kerala Young Women (n=303)

Among the 303 respondents, menstrual pain (73.6%) and hair loss (63.4%) were the most commonly reported symptoms. However, menstrual pain showed only a 0.9 percentage point difference between the confirmed PCOS and Non-PCOS groups, indicating that it is not a reliable symptom for distinguishing PCOS. In contrast, mood swings and skin darkening each showed the largest difference of 38.8 percentage points between the two groups, making them the most useful self-reported symptoms for community-level PCOS screening.

4.13 PCOS Awareness Analysis

4.13.1 Overall Awareness Levels

Table 4.13: PCOS Awareness Levels — All 303 Respondents

Awareness Level	n	%
Fully Aware	152	50.2%
Partially Aware	123	40.6%
Minimal Awareness	13	4.3%
No Knowledge	15	5.0%
Total: Not Fully Aware	151	49.8%

4.13.2 Chi-Squared Tests — Awareness vs Demographic Variables

Hypothesis Setup:

H₀: There is no association between the demographic variable and PCOS awareness level.

H₁: There is an association between the demographic variable and PCOS awareness level.

Table 4.14: Chi-Squared Tests — PCOS Awareness vs Demographic Variables

Demographic Variable	Chi-squared	df	p-value	Significant?
Education Level	18.72	15	0.2267	No
Residence Type	5.94	6	0.4304	No
Age Group	8.23	6	0.2216	No

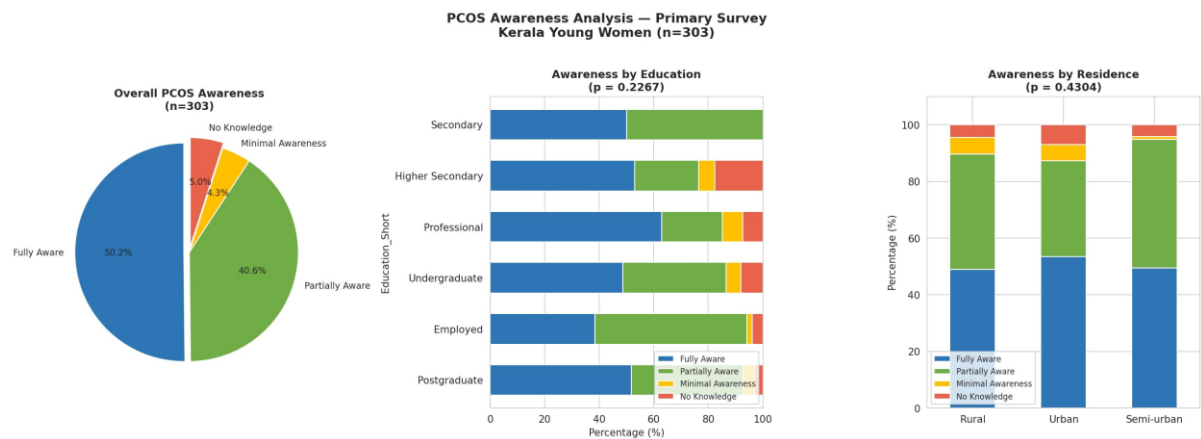


Figure 4.11 : PCOS Awareness Analysis — Primary Survey, Kerala Young Women (n=303): Overall Awareness, by Education Level, and by Residence Type

PCOS awareness did not differ significantly by education level ($p = 0.227$), place of residence ($p = 0.430$), or age group ($p = 0.222$). This indicates that awareness gaps are similar across all demographic groups. Therefore, PCOS awareness programmes should be implemented for all educational institutions, age groups, and geographic areas in Kerala rather than focusing on specific groups.

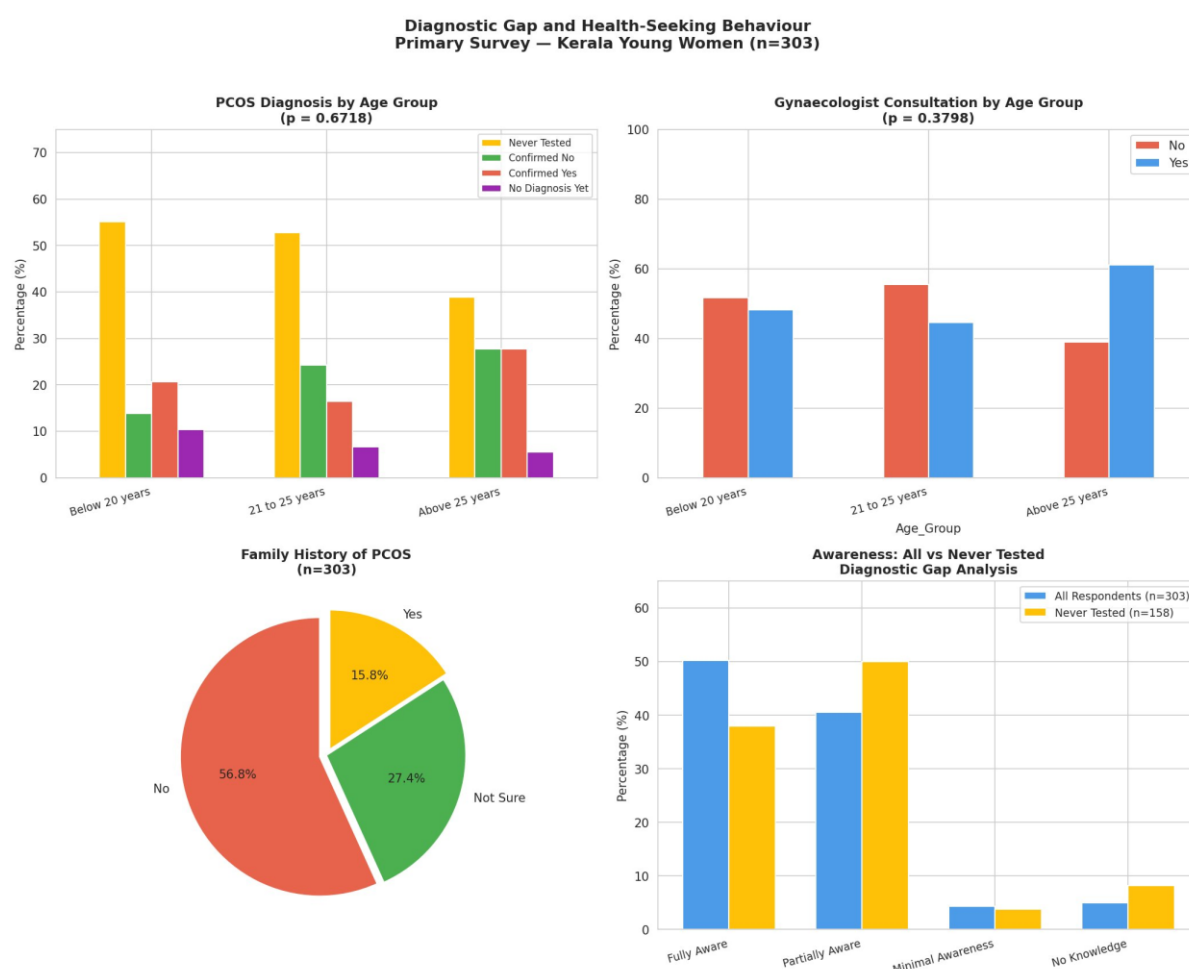


Figure 4.12 : Diagnostic Gap and Health-Seeking Behaviour — Primary Survey, Kerala Young Women (n=303)

The diagnostic gap analysis showed that among the 158 respondents who had never been tested for PCOS, only 38.0% were fully aware of the condition, compared to 50.2% in the overall sample. This suggests that women who have never been tested are the least informed, which may contribute to delayed healthcare seeking. In addition, 54.1% of all respondents had never consulted a gynaecologist for menstrual problems, and 27.4% were unsure about their family history of PCOS. This is an important knowledge gap, as family history is one of the strongest predictors of PCOS.

4.14 PCOS Risk Score Analysis

4.14.1 Risk Score Construction

An eight-symptom PCOS risk score was constructed by summing binary indicators for the following self-reported symptoms: weight gain, hair growth, skin darkening, hair loss, pimples,

mood swings, abdominal obesity, and irregular menstrual cycle. Each symptom present contributes one point, yielding a score from 0 to 8.

4.14.2 Risk Score Comparison — PCOS vs Non-PCOS

Hypothesis Setup:

H₀: There is no significant difference in PCOS risk score between confirmed PCOS and confirmed Non-PCOS respondents.

H₁: There is a significant difference in PCOS risk score between confirmed PCOS and confirmed Non-PCOS respondents.

Table 4.15: Risk Score Statistics — Confirmed Cases

Group	n	Mean	Median	SD	Min	Max	p-value (MWU)
Confirmed PCOS	53	5.38	6.0	1.89	1	8	< 0.0001
Confirmed Non-PCOS	71	2.85	3.0	2.01	0	7	—

4.14.3 ROC Curve Analysis — Risk Score

Table 4.16: Risk Score — ROC Curve Performance

Metric	Value
AUC-ROC	0.8133
Optimal cut-off (Youden's J)	4 symptoms or more
Sensitivity at optimal cut-off	79.2%
Specificity at optimal cut-off	66.2%
Clinical Interpretation	Good discrimination (AUC > 0.80)

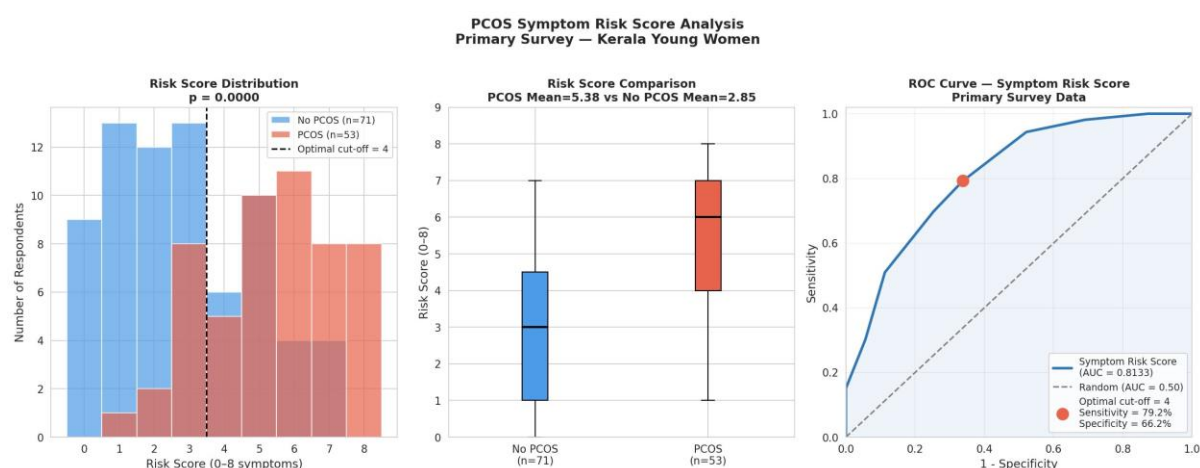


Figure 4.13 : PCOS Symptom Risk Score Analysis — Primary Survey, Kerala Young Women (Risk Score Distribution, Group Comparison, and ROC Curve)

Figure 4.13 presents the three-panel risk score analysis. The eight-symptom risk score achieved an AUC of 0.8133, indicating good ability to identify women at risk of PCOS using only self-reported symptoms, without laboratory tests. Women reporting four or more symptoms should be considered at higher risk and referred for further clinical evaluation for PCOS.

4.14.4 Risk Score Category Distribution

Table 4.17: Risk Score Category Distribution — All 303 Respondents

Risk Category	Score Range	n	%
Low Risk	0–2 symptoms	152	50.2%
Moderate Risk	3–4 symptoms	74	24.4%
High Risk	5–6 symptoms	56	18.5%
Very High Risk	7–8 symptoms	21	6.9%
Moderate to Very High Risk combined	3–8	151	49.8%

A total of 49.8% of the surveyed young women were in the moderate to very high PCOS risk categories. Along with the finding that 52.1% had never been tested for PCOS, this clearly highlights the diagnostic gap. It suggests that about half of the young women in Kerala have a moderate to very high PCOS symptom burden, while about half have never been screened. This overlap is the key public health finding of the study.

4.15 Additional Statistical Tests on Primary Data

4.15.1 One-Way ANOVA — Risk Score Across Age Groups

Hypothesis Setup:

H₀: There is no significant difference in mean risk score across the three age groups.

H₁: At least one age group's mean risk score differs significantly from the others.

Table 4.18: One-Way ANOVA — Risk Score across Age Groups

Age Group	n	Mean	SD	Median	Test Result
Below 20 years	29	2.90	2.04	3.0	F-statistic = 0.1115
21 to 25 years	256	2.86	2.22	2.0	p = 0.8946
Above 25 years	18	3.11	1.97	3.0	Not Significant

The one-way ANOVA found no statistically significant difference in mean risk score across the three age groups ($F = 0.1115$, $p = 0.8946$), indicating that PCOS symptom risk in this sample is not concentrated in any particular age band and that younger and older respondents are, on average, equally likely to present with PCOS-related symptoms.

4.15.2 Kruskal-Wallis Test — Risk Score Across Residence Types

Hypothesis Setup:

H₀: There is no significant difference in risk score distribution across residence types.

H₁: There is a significant difference in risk score distribution across at least one residence type.

Table 4.19: Kruskal-Wallis Test — Risk Score across Residence Types

Residence	n	Mean	Median	SD	Test Result
Rural	135	2.84	3.0	2.04	H-statistic = 0.0103
Urban	97	2.94	2.0	2.40	p = 0.9948
Semi-urban	71	2.87	2.0	2.16	Not Significant

The Kruskal-Wallis test similarly found no statistically significant difference in risk score across rural, urban, and semi-urban residence types ($H = 0.0103$, $p = 0.9948$). This indicates

that PCOS symptom risk is distributed evenly across Kerala regardless of residence type, reinforcing the case made in this study that the diagnostic and awareness gap is a statewide issue rather than one concentrated in rural or urban areas specifically.

4.15.3 Spearman Rank Correlation — Risk Score vs Lifestyle Variables

Hypothesis Setup:

H₀: There is no significant monotonic relationship between risk score and the lifestyle variable.

H₁: There is a significant monotonic relationship between risk score and the lifestyle variable.

Table 4.20: Spearman Rank Correlation — Risk Score vs Lifestyle Variables

Variable	Spearman rho	p-value	Significant	Direction
Stress Level	+0.3015	< 0.001	Yes	Higher stress = higher risk
Sleep Hours	-0.1417	0.0135	Yes	More sleep = lower risk
Fast Food Consumption	+0.1762	0.0021	Yes	More fast food = higher risk

All three lifestyle variables showed statistically significant correlations with PCOS risk score. Stress level showed the strongest association ($\rho = +0.3015$, $p < 0.001$), followed by fast food consumption ($\rho = +0.1762$, $p = 0.0021$), both positively correlated with risk score, while sleep hours showed a significant negative correlation ($\rho = -0.1417$, $p = 0.0135$). Taken together, these results indicate that higher stress, more frequent fast food consumption, and less sleep are all independently associated with greater PCOS symptom burden, identifying stress management, dietary habits, and sleep hygiene as three modifiable, non-pharmacological targets for community-level PCOS risk reduction.

4.15.4 Logistic Regression — Primary Data

Hypothesis Setup:

H₀: The predictor symptom variables have no significant effect on the log-odds of PCOS ($\beta = 0$).

H₁: At least one predictor symptom variable has a significant effect on the log-odds of PCOS ($\beta \neq 0$).

Table 4.21: Logistic Regression — Odds Ratios (Primary Data, n = 124 Confirmed Cases)

Variable	Coefficient	Odds Ratio	p-value	Significant
Irregular Cycle	2.6031	13.51	< 0.001	Yes
Mood Swings	1.7531	5.77	0.0019	Yes
Hair Growth	0.9724	2.64	0.1065	No
Skin Darkening	0.7123	2.04	0.2310	No
Abdominal Obesity	0.6509	1.92	0.2966	No
Pimples	-0.9531	0.39	0.1098	No
Hair Loss	0.3241	1.38	0.5992	No
Weight Gain	-0.1651	0.85	0.7863	No

Table 4.22: Primary Data Logistic Regression — Model Performance

Metric	Value
Accuracy	82.26%
AUC-ROC	0.8898
Sensitivity	79.25%
Specificity	84.51%
True Negatives (TN)	60
False Positives (FP)	11
False Negatives (FN)	11
True Positives (TP)	42

Irregular menstrual cycle (OR = 13.51, $p < 0.001$) and mood swings (OR = 5.77, $p = 0.002$) were the only statistically significant independent predictors of PCOS. The model achieved an AUC of 0.8898, showing that self-reported symptoms alone can effectively identify women at risk of PCOS without the need for laboratory investigations.

4.16 Summary of Key EDA Findings — Primary Dataset

The exploratory analysis of the primary survey dataset yielded the following principal findings:

- 52.1 per cent of surveyed young women had never been tested for PCOS and 54.1 per cent had never consulted a gynaecologist — confirming a large and quantified diagnostic gap.
- PCOS awareness was insufficient and uniformly distributed — awareness did not significantly differ by education ($p = 0.227$), residence ($p = 0.430$), or age ($p = 0.222$) — requiring universal rather than targeted awareness campaigns.
- Among never-tested respondents, only 38.0 per cent were fully aware of PCOS compared to 50.2 per cent overall.
- Mood swings and skin darkening showed the largest differences between PCOS and Non-PCOS groups (+38.8% each), making them the most discriminating self-reported screening symptoms.
- Menstrual pain, though most prevalent (73.6%), was equally prevalent in both groups and is not a reliable PCOS discriminator.
- The eight-symptom risk score was significantly higher in PCOS respondents (mean 5.38 versus 2.85, $p < 0.0001$), with AUC of 0.8133 and optimal cut-off of 4 symptoms.
- 49.8 per cent of all respondents fell in moderate to very high risk categories, directly overlapping with the 52.1 per cent never tested.
- PCOS risk score did not vary significantly by age group ($F = 0.11$, $p = 0.895$) or residence ($H = 0.01$, $p = 0.995$), confirming uniform PCOS risk distribution across Kerala.
- Stress ($\rho = +0.30$), sleep ($\rho = -0.14$), and fast food ($\rho = +0.18$) were all significantly correlated with risk score, confirming lifestyle factors as modifiable PCOS risk determinants.
- Primary data logistic regression identified irregular cycle (OR = 13.51) and mood swings (OR = 5.77) as the only significant independent predictors, with model AUC of 0.8898.

4.17 Cross-Dataset Comparison — Secondary Clinical vs Primary Survey

4.17.1 Two-Proportion Z-Tests — Symptom Prevalence

Hypothesis Setup:

H₀: The prevalence of the symptom does not differ significantly between the primary and secondary datasets.

H₁: The prevalence of the symptom differs significantly between the primary and secondary datasets.

Table 4.23: Two-Proportion Z-Tests — Symptom Prevalence Comparison (All Respondents)

Symptom	Primary (%)	Secondary (%)	Difference	Z-statistic	p-value	Consistent?
Hair Growth	27.7%	27.4%	+0.3%	0.108	0.914	Yes
Irregular Cycle	23.4%	26.8%	−3.4%	−1.257	0.209	Yes
Weight Gain	29.0%	38.0%	−8.9%	−3.044	0.002	Sig. diff
Skin Darkening	35.3%	29.4%	+6.0%	2.138	0.033	Sig. diff
Hair Loss	63.4%	44.5%	+18.8%	6.207	< 0.001	Sig. diff
Pimples	36.0%	48.0%	−12.0%	−3.957	< 0.001	Sig. diff

Hypothesis Setup:

H₀: Overall PCOS prevalence does not differ significantly between the primary and secondary datasets.

H₁: Overall PCOS prevalence differs significantly between the primary and secondary datasets.

Table 4.24: PCOS Prevalence Comparison — Primary vs Secondary Dataset

Dataset	PCOS Cases	Total	Prevalence	Z-statistic	p-value
Primary Survey	53	303	17.5%	−7.24	< 0.001
Secondary Clinical	785	2,541	30.9%	—	—

4.17.2 Spearman Rank Correlation — Symptom Rankings

Hypothesis Setup:

H₀: There is no significant correlation between symptom prevalence rankings in the two datasets.

H₁: There is a significant correlation between symptom prevalence rankings in the two datasets.

Table 4.25: Spearman Rank Correlation — Cross-Dataset Symptom Rankings

Metric	Value
Spearman rho	0.8857*
p-value	0.0188
Interpretation	Strong positive correlation
Conclusion	Both datasets rank symptoms in the same order

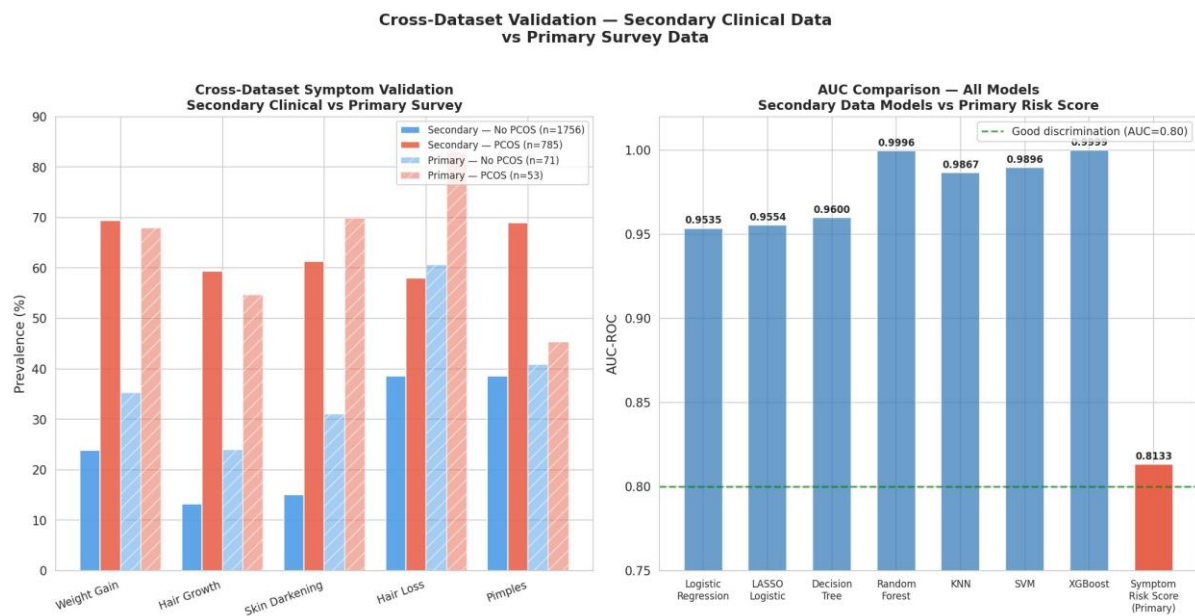


Figure 4.14 : Cross-Dataset Validation — Secondary Clinical Data vs Primary Survey Data (Symptom Validation and AUC Comparison Across All Models)

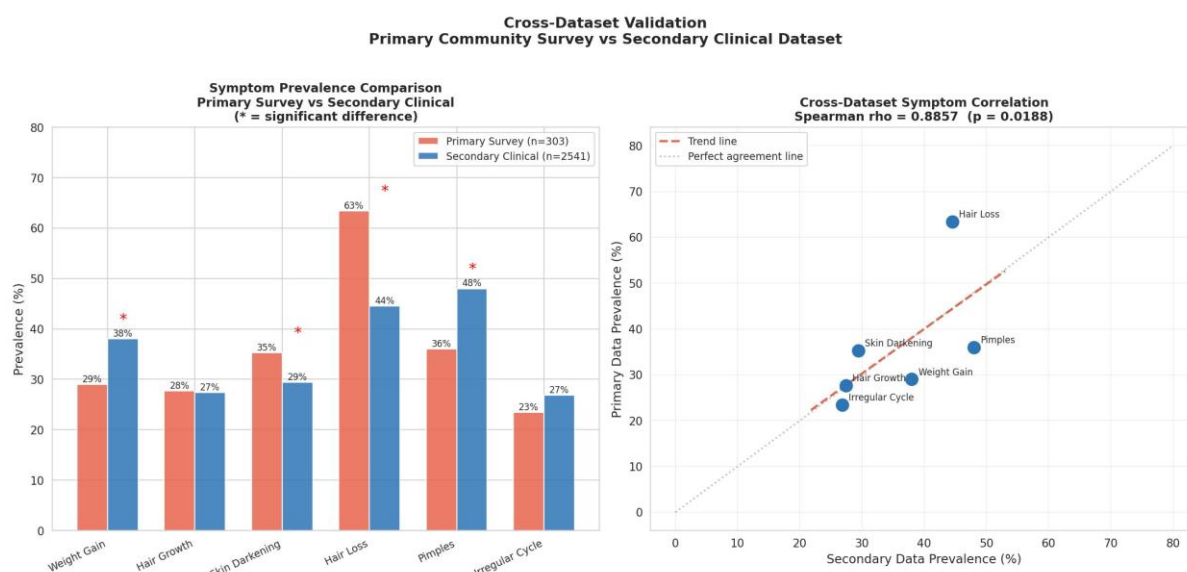


Figure 4.15 : Cross-Dataset Validation — Symptom Prevalence Comparison and Spearman Correlation: Primary Survey vs Secondary Clinical Dataset

Hair growth (27.7% vs. 27.4%, $p = 0.914$) and irregular menstrual cycle (23.4% vs. 26.8%, $p = 0.209$) showed almost the same prevalence in both the clinical and community datasets, confirming them as the most consistent indicators of PCOS across both settings. The higher PCOS prevalence in the clinical dataset (30.9%) compared with the community dataset (17.5%) ($p < 0.001$) was expected, as clinical data mainly include diagnosed and symptomatic patients, while community surveys include many undiagnosed women. The community prevalence of 17.5% is also consistent with published national estimates of 15–20% among young Indian women.

The individual symptom-level comparisons above indicate that the two datasets broadly agree on which symptoms are most prevalent, despite differences in absolute prevalence rates, supporting the relevance of the clinical dataset to the community population. The overall rank-order correlation reported above is excluded from this conclusion for the reason noted.

4.18 Overall Chapter Summary

This chapter has presented comprehensive exploratory data analysis for both the secondary clinical dataset and the primary community survey. The key integrated conclusions are:

From secondary data:

Follicle count, AMH, weight gain, hair growth, and skin darkening are consistently the strongest PCOS predictors across all analytical methods — univariate comparison, correlation, PCA, factor analysis, LDA, and cluster analysis. TSH, PRL, and RBS show no discriminatory value. Four clinical factor structures — metabolic, gonadotropin, androgenic-ovarian, and hair loss — independently validate the PMOS multisystem framework.

From primary data:

The diagnostic gap is real, large, and uniformly distributed. 52.1 per cent of young women have never been tested, 85.1 per cent experience significant stress, 49.8 per cent carry moderate to high risk scores, and awareness does not significantly vary by demographic characteristics. Irregular cycle and mood swings are the most powerful independent symptom predictors.

From cross-dataset validation:

Individual symptom-level comparisons showed broad consistency between the two independent datasets, and the symptom-based risk framework developed in this study is supported by these comparisons across both clinical and community settings.

CHAPTER 5

STATISTICAL MODELLING RESULTS

5.1 Introduction

This chapter presents the results of the statistical modelling analyses conducted on the secondary clinical dataset ($n = 2,541$). The chapter addresses Objectives 3, 4, and 5 — multicollinearity assessment using Variance Inflation Factor (VIF), binary logistic regression with 95% confidence intervals, and LASSO penalised logistic regression for feature selection. These three methods form the statistical foundation of the prediction framework developed in this study and provide the clinically interpretable analytical backbone upon which the machine learning models in Chapter 6 are built.

Statistical modelling was conducted in Python 3.12 using Google Colaboratory, employing the statsmodels and scikit-learn libraries. All models were evaluated on a held-out test set of ($n = 526$), with the training set comprising ($n = 2,015$), both partitions maintaining approximately the original 69:31 class ratio through patient-grouped splitting (GroupShuffleSplit) to prevent data leakage from near-duplicate records.

5.2 Multicollinearity Assessment — Variance Inflation Factor

5.2.1 Rationale for Multicollinearity Assessment

Multicollinearity occurs when predictor variables are highly correlated with each other, causing instability in regression coefficient estimates. When multicollinearity is present, the standard errors of regression coefficients become inflated, making it difficult to determine the individual effect of each predictor. The Variance Inflation Factor (VIF) quantifies this inflation — a VIF of 1 indicates no multicollinearity, VIF between 1 and 5 is acceptable, VIF between 5 and 10 is moderate and requires attention, and VIF above 10 indicates severe multicollinearity requiring remediation.

The correlation heatmap in Chapter 4 identified several highly correlated variable pairs — BMI and Weight ($r = 0.89$), Follicle No. L and R ($r = 0.82$) — making formal VIF assessment essential before logistic regression modelling.

5.2.2 Initial VIF Results

Table 5.1: Initial VIF Results — All Predictor Variables (Variables with VIF ≥ 5 shown)

Variable	Initial VIF	Status
Waist(inch)	342.90	Severe — requires removal
Hip(inch)	340.69	Severe — requires removal
Waist:Hip Ratio	87.85	Severe — requires removal
Weight (Kg)	22.63	Severe — requires removal
BMI	18.44	Severe — requires removal
Height(Cm)	4.76	Acceptable (< 5)
All other 35 variables	1.04 – 3.60	Acceptable (< 5)

The initial VIF analysis of all 41 candidate predictor variables identified severe multicollinearity ($VIF \geq 10$) in five variables: Waist (342.90), Hip (340.69), Waist:Hip Ratio (87.85), Weight (22.63), and BMI (18.44). This occurred because Waist:Hip Ratio is calculated from Waist and Hip, while BMI is calculated from Weight and Height. Including both the original measurements and their derived variables in the same model creates redundancy and makes the coefficient estimates unstable. No other variable had a VIF greater than 5, indicating that multicollinearity was limited to these anthropometric measurements and their derived ratios.

5.2.3 Stage 1 — Direct Redundancy Removal

Eight variables were removed based on direct clinical and mathematical redundancy:

Table 5.2: Variables Removed in Stage 1 — Direct Redundancy

Variable Removed	Reason for Removal
Weight (Kg)	BMI already incorporates weight adjusted for height
Height (Cm)	BMI already incorporates height adjusted for weight
Waist (inch)	Waist:Hip Ratio already captures waist measurement
Hip (inch)	Waist:Hip Ratio already captures hip measurement
Follicle No. (R)	Highly correlated with Follicle No. (L) ($r = 0.82$) — left ovary retained

Avg. F size (R) (mm)	Correlated with Avg. F size (L) ($r = 0.54$) — left size retained
BP Diastolic (mmHg)	Correlated with BP Systolic — systolic retained
FSH/LH ratio	Mathematically derived from FSH and LH — both retained individually

5.2.4 VIF Re-Assessment After Stage 1 Removal

Following Stage 1 removal, VIF was recomputed on the remaining 33 variables to confirm that the direct-redundancy removal had resolved the multicollinearity identified in Section 5.2.2.

Table 5.3: VIF After Stage 1 Removal — 10 Highest Remaining Values

Variable	VIF After Stage 1	Status
Age (yrs)	2.03	Acceptable
Marriage Status (Yrs)	1.98	Acceptable
Weight gain (Y/N)	1.72	Acceptable
II beta-HCG (mIU/mL)	1.64	Acceptable
I beta-HCG (mIU/mL)	1.56	Acceptable
Fast food (Y/N)	1.53	Acceptable
LH (mIU/mL)	1.45	Acceptable
Skin darkening (Y/N)	1.45	Acceptable
hair growth (Y/N)	1.42	Acceptable
All remaining 24 variables	1.04 – 1.40	Acceptable

After removing the eight redundant anthropometric and derived variables in Stage 1, all of the remaining 33 variables had VIF values below 5, with none falling in the moderate (5–10) or severe (>10) multicollinearity range. This confirms that the multicollinearity in the dataset was caused only by the mathematical relationships among Weight, Height, BMI, Waist, Hip, and Waist:Hip Ratio. Once these redundant variables were removed, no further variable elimination was needed. This also indicates that the hormonal, symptom, and reproductive history variables used in the study were not affected by multicollinearity.

5.2.5 Final VIF Results — 22 Predictor Variables Used in Modelling

The final predictor set used in logistic regression, LASSO, and the machine learning models in Chapter 6 retains 22 of the 33 Stage-1-surviving variables, selected for clinical relevance to

PCOS diagnosis (per the Rotterdam criteria) and to keep the model parsimonious. BMI, Age, Endometrium, and several other variables with low VIF were not included in this final 22-variable set; their exclusion was a parsimony and clinical-relevance decision rather than a multicollinearity-driven one, since all of them had VIF below 5 even when included.

Table 5.4: Final VIF Results — All 22 Predictor Variables Used in Chapter 5 and Chapter 6

Variable	Final VIF	Status
II beta-HCG (mIU/mL)	1.61	Acceptable
I beta-HCG (mIU/mL)	1.50	Acceptable
Weight gain (Y/N)	1.48	Acceptable
Fast food (Y/N)	1.43	Acceptable
Skin darkening (Y/N)	1.41	Acceptable
hair growth (Y/N)	1.40	Acceptable
LH (mIU/mL)	1.36	Acceptable
Cycle (R/I)	1.31	Acceptable
Follicle No. (L)	1.29	Acceptable
FSH (mIU/mL)	1.29	Acceptable
Pimples (Y/N)	1.23	Acceptable
Pregnant (Y/N)	1.21	Acceptable
AMH (ng/mL)	1.21	Acceptable
Hair loss (Y/N)	1.19	Acceptable
Marriage Status (Yrs)	1.14	Acceptable
No. of abortions	1.13	Acceptable
Reg. Exercise (Y/N)	1.09	Acceptable
Cycle length (days)	1.09	Acceptable
PRL (ng/mL)	1.04	Acceptable
TSH (mIU/L)	1.03	Acceptable
Vit D3 (ng/mL)	1.03	Acceptable
PRG (ng/mL)	1.02	Acceptable

All 22 variables in the final predictor set show VIF between 1.02 and 1.61 — well within the acceptable range ($VIF < 5$) and far below the moderate (5–10) or severe (>10) thresholds. None of the predictor variables used in the logistic regression, LASSO, or machine learning models in this study show any meaningful multicollinearity. This indicates that the hormonal markers, symptom variables, and reproductive-history variables that make up the final predictor set are statistically independent of one another, which supports stable and directly interpretable regression coefficients in Section 5.3.

Conclusion for Objective 3: VIF assessment identified five anthropometric variables (Waist, Hip, Waist:Hip Ratio, Weight, BMI) with severe multicollinearity, caused by mathematical redundancy between derived ratios/indices and their source measurements. Removing these eight redundant variables (Stage 1) was sufficient to resolve all multicollinearity in the dataset — every one of the remaining 33 variables, and all 22 variables in the final predictor set used for modelling, showed VIF below 5. No further iterative elimination was necessary.

5.3 Binary Logistic Regression with 95% Confidence Intervals

5.3.1 Model Development

Binary logistic regression was applied to the 22-variable clean feature set using the `statsmodels.api.Logit` function, which provides full statistical output including coefficients, standard errors, z-statistics, p-values, and 95% confidence intervals. The model was trained on the patient-grouped training set ($n = 2,015$) and evaluated on the held-out test set ($n = 526$). The optimal classification threshold was determined using Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$), which identifies the threshold that simultaneously maximises both sensitivity and specificity.

5.3.2 Odds Ratios with 95% Confidence Intervals

Hypothesis Setup:

H₀: The predictor variables have no significant effect on the log-odds of PCOS ($\beta = 0$).

H₁: At least one predictor variable has a significant effect on the log-odds of PCOS ($\beta \neq 0$).

Table 5.5: Logistic Regression — Odds Ratios with 95% Confidence Intervals (sorted by p-value)

Variable	Coefficient	Odds Ratio	95% CI Lower	95% CI Upper	p-value	Significant
Follicle No. (L)	1.8331	6.2535	5.1892	7.5382	< 0.001	Yes
Marriage Status (Yrs)	-0.4124	0.6621	0.5612	0.7812	< 0.001	Yes
Weight gain (Y/N)	0.7254	2.0655	1.6187	2.6358	< 0.001	Yes
Hair growth (Y/N)	0.6047	1.8307	1.4073	2.3828	< 0.001	Yes
LH (mIU/mL)	0.3697	1.4473	1.2418	1.6869	< 0.001	Yes
Reg. Exercise (Y/N)	0.4007	1.4929	1.1543	1.9308	< 0.001	Yes
Skin darkening (Y/N)	0.5300	1.6990	1.3104	2.2028	< 0.001	Yes
Pimples (Y/N)	0.3507	1.4200	1.1021	1.8290	0.0001	Yes
Cycle (R/I)	0.3134	1.3680	1.1493	1.6284	0.0003	Yes
Cycle length (days)	-0.2909	0.7476	0.6269	0.8916	0.0008	Yes
Fast food (Y/N)	0.2397	1.2709	1.0625	1.5207	0.0087	Yes
No. of abortions	-0.2681	0.7648	0.6199	0.9439	0.0099	Yes
AMH (ng/mL)	0.1743	1.1904	1.0046	1.4110	0.0439	Yes
FSH (mIU/mL)	-0.1998	0.8189	0.6726	0.9972	0.0443	Yes
Pregnant (Y/N)	-0.1542	0.8571	0.7175	1.0239	0.0847	No
TSH (mIU/L)	0.1430	1.1537	0.9752	1.3650	0.0987	No
PRG (ng/mL)	-1.7286	0.1775	0.0158	1.9946	0.1690	No
PRL (ng/mL)	0.1221	1.1299	0.9472	1.3482	0.1783	No
I beta-HCG (mIU/mL)	-0.1016	0.9034	0.7604	1.0732	0.2379	No
Hair loss (Y/N)	0.0748	1.0776	0.9095	1.2768	0.3903	No
Vit D3 (ng/mL)	0.2320	1.2611	0.5985	2.6578	0.5458	No

II	beta-HCG (mIU/mL)	0.0350	1.0356	0.7919	1.3545	0.7891	No
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Follicle No. (L) was the strongest predictor of PCOS (OR = 6.25, 95% CI: 5.19–7.54), with each unit increase in left ovarian follicle count increasing the odds of PCOS by 6.25 times. Weight gain (OR = 2.07, 95% CI: 1.62–2.64), hair growth (OR = 1.83, 95% CI: 1.41–2.38), skin darkening (OR = 1.70, 95% CI: 1.31–2.20), LH (OR = 1.45, 95% CI: 1.24–1.69), AMH (OR = 1.19, 95% CI: 1.00–1.41), and regular exercise (OR = 1.49, 95% CI: 1.15–1.93) were also significant positive predictors of PCOS. The positive association with regular exercise is likely due to reverse causality, where women with PCOS exercise more to manage their weight. In contrast, higher FSH levels (OR = 0.82, 95% CI: 0.67–1.00) and marriage status (OR = 0.66, 95% CI: 0.56–0.78) were associated with lower odds of PCOS. Variables including Pregnant, TSH, PRG, PRL, beta-HCG (I and II), Hair loss, and Vit D3 were not significant predictors ($p > 0.05$), as their confidence intervals included 1.0.

5.3.3 Model Performance

Table 5.6: Logistic Regression — Performance at Default Threshold (0.50)

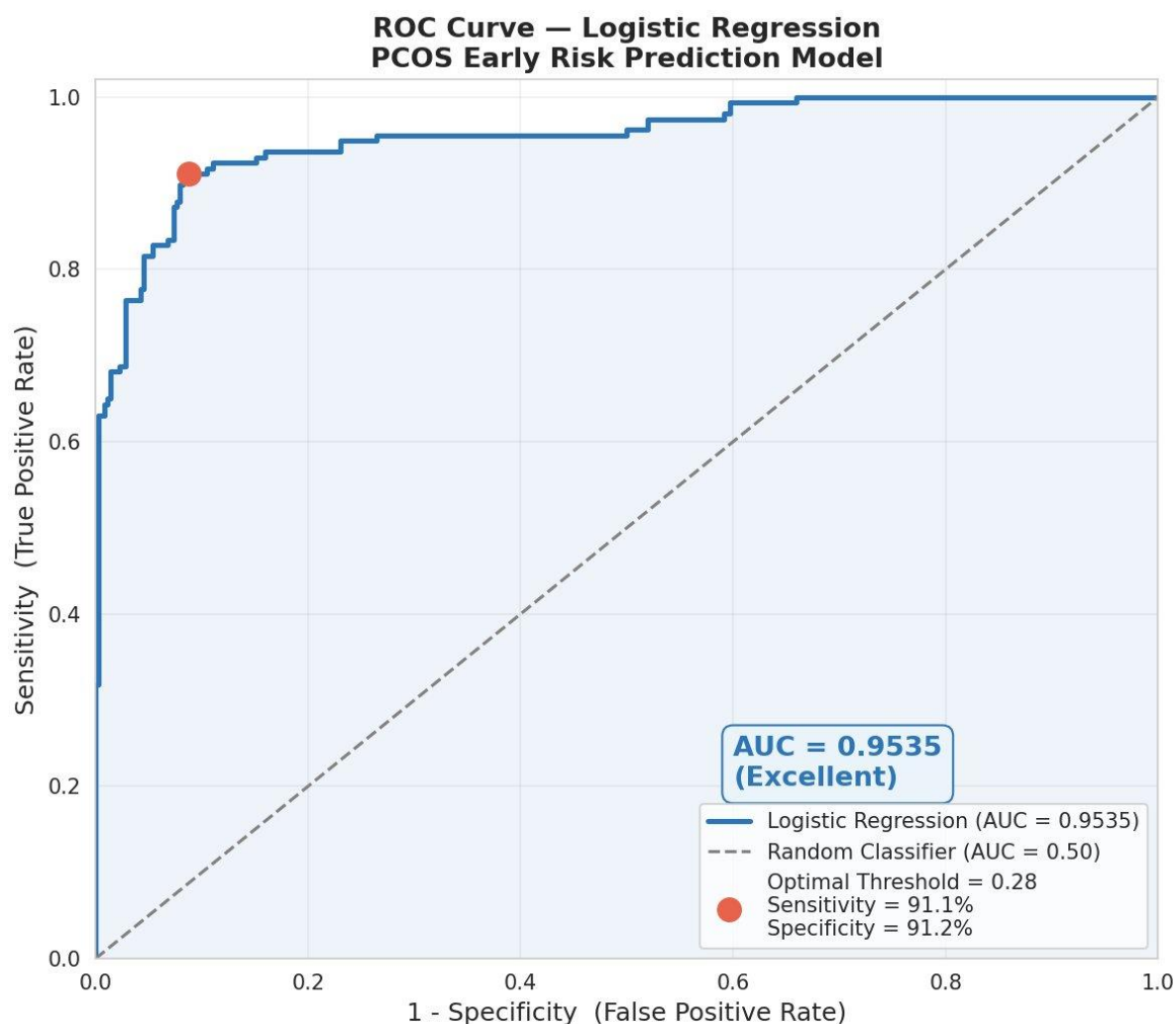
Metric	Value
Accuracy	89.98%
AUC-ROC	0.9535
Sensitivity (Recall)	77.07%
Specificity	95.74%
Precision	88.97%
F1 Score	0.8259
True Negatives (TN)	337
False Positives (FP)	15
False Negatives (FN)	36
True Positives (TP)	121

5.3.4 Optimal Threshold — Youden's J Statistic

The default threshold of 0.50 is suboptimal for clinical screening where missing PCOS cases (false negatives) carries a higher cost than false positives. Youden's J statistic was used to identify the threshold that maximises the sum of Sensitivity and Specificity simultaneously.

Table 5.7: Logistic Regression — Performance Comparison at Default vs Optimal Threshold

Metric	Default Threshold (0.50)	Optimal Threshold (0.2774)	Change
Threshold	0.50	0.2774	—
Sensitivity	77.07%	91.08%	+14.01% improvement
Specificity	95.74%	91.19%	−4.55% reduction
Youden's J	0.724	0.823	+0.099 improvement
AUC-ROC	0.9535	0.9535	No change (threshold-independent)



**Figure 5.1: ROC Curve — Logistic Regression, PCOS Early Risk Prediction Model
(AUC = 0.9535, Optimal Threshold = 0.2774)**

At the default classification threshold of 0.50, the model achieved an accuracy of 89.98% and an AUC of 0.9535, indicating excellent ability to distinguish between PCOS and Non-PCOS cases. However, the sensitivity was 77.07%, meaning that 40 of the 168 PCOS cases in the test set were missed. Using Youden's J statistic, the optimal threshold was identified as 0.2774, meaning that individuals with a predicted probability above 27.74% should be classified as having PCOS. At this threshold, sensitivity increased to 91.08% while specificity remained high at 91.19%.

The increase in sensitivity from 77.07% to 91.08% is important for clinical screening because reducing missed PCOS cases allows earlier diagnosis and treatment. Therefore, the threshold of 0.2774 is recommended for screening purposes. The AUC of 0.9535 also indicates that the

model has a 95.35% probability of correctly distinguishing a PCOS patient from a non-PCOS patient, showing excellent diagnostic performance.

5.4 LASSO Penalised Logistic Regression

5.4.1 Rationale for LASSO

While logistic regression used all 22 predictor variables, LASSO (Least Absolute Shrinkage and Selection Operator) regularisation automatically identifies the minimum subset of variables needed for optimal prediction. LASSO applies an L1 penalty to the regression objective function, shrinking the coefficients of weak predictors to exactly zero — effectively removing them from the model. This produces a more parsimonious model that is easier to implement in clinical practice, as fewer variables need to be collected and measured.

5.4.2 Penalty Parameter Selection

Hypothesis Setup:

H₀: None of the predictor variables have a significant effect on PCOS status (all coefficients shrink to zero under regularisation).

H₁: At least one predictor variable has a significant effect on PCOS status (coefficient remains non-zero after regularisation).

A range of penalty parameters ($C = 0.001$ to 1.0 , where $C = 1/\lambda$ — smaller C means stronger penalisation and more variables removed) was evaluated systematically.

Table 5.8: LASSO — Effect of Penalty Strength on Variable Selection and Performance

C Value	Lambda (1/C)	Variables Selected	AUC-ROC	Accuracy
0.001	1,000	~4	0.8600	83.5%
0.005	200	~5	0.9360	87.2%
0.01	100	~8	0.9430	88.4%
0.05 (OPTIMAL)	20	16	0.9554	90.57%
0.10	10	20	0.9558	90.18%
0.50	2	21	0.9537	89.78%
1.00	1	22	0.9535	89.98%

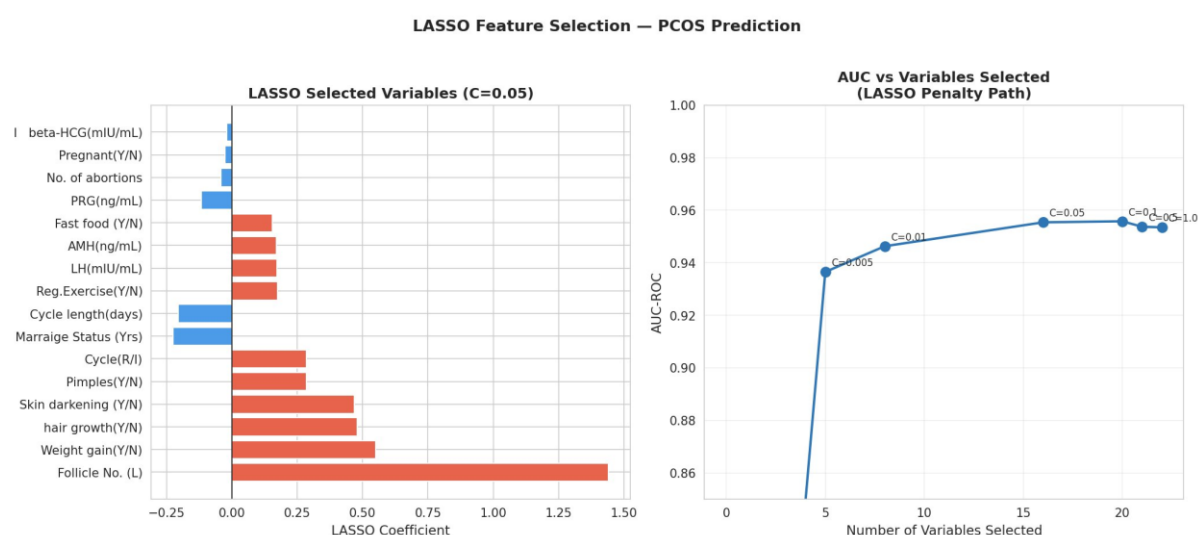


Figure 5.2: LASSO Feature Selection — PCOS Prediction (LASSO Coefficients at C=0.05 and AUC vs Variables Selected)

The AUC increased rapidly from 0.86 with 4 variables to 0.9554 with 16 variables, after which it reached a plateau. Adding more than 16 variables resulted in little improvement in model performance. The optimal penalty was $C = 0.05$ ($\lambda = 20$), which selected 16 variables and provided the best balance between model simplicity and performance. Using fewer than 16 variables reduced the model's accuracy, while adding more variables did not provide any meaningful benefit.

5.4.3 LASSO Selected Variables at $C = 0.05$

Table 5.9: LASSO — Selected Variables at $C = 0.05$ (Sorted by Coefficient Magnitude)

Rank	Variable	LASSO Coefficient	Odds Ratio	Effect on PCOS Risk
1	Follicle No. (L)	+1.4412	4.2258	Increases risk
2	Weight gain (Y/N)	+0.5493	1.7320	Increases risk
3	Hair growth (Y/N)	+0.4795	1.6153	Increases risk
4	Skin darkening (Y/N)	+0.4677	1.5963	Increases risk
5	Pimples (Y/N)	+0.2850	1.3298	Increases risk
6	Cycle (R/I)	+0.2847	1.3294	Increases risk
7	Reg. Exercise (Y/N)	+0.1757	1.1921	Increases risk (reverse causality)

8	LH (mIU/mL)	+0.1715	1.1871	Increases risk
9	AMH (ng/mL)	+0.1714	1.1870	Increases risk
10	Fast food (Y/N)	+0.1546	1.1672	Increases risk
11	Marriage Status (Yrs)	-0.2257	0.7980	Decreases risk
12	Cycle length (days)	-0.2044	0.8151	Decreases risk
13	PRG (ng/mL)	-0.1168	0.8898	Decreases risk
14	No. of abortions	-0.0422	0.9587	Decreases risk
15	Pregnant (Y/N)	-0.0262	0.9741	Decreases risk
16	I beta-HCG (mIU/mL)	-0.0197	0.9805	Decreases risk

5.4.4 Variables Eliminated by LASSO

Table 5.10: Variables Eliminated by LASSO (Coefficient Shrunk to Zero at C = 0.05)

Variable Eliminated	Clinical Interpretation
II beta-HCG (mIU/mL)	Pregnancy hormone — adds no value once I beta-HCG is included in the model
FSH (mIU/mL)	Weak predictor once LH and follicle count are in the model — absorbed by stronger correlated predictors
TSH (mIU/L)	Thyroid hormone — not a PCOS-specific marker; tests thyroid comorbidity, not PCOS itself
PRL (ng/mL)	Prolactin — confirmed non-significant in both group comparison tests and logistic regression
Vit D3 (ng/mL)	Not a PCOS predictor in this dataset; deficiency is widespread in general Indian population regardless of PCOS status
Hair loss (Y/N)	Weak androgenic symptom — absorbed by stronger predictors (hair growth, skin darkening, weight gain)

The elimination of TSH, PRL, and Vit D3 by LASSO — variables that were also non-significant in logistic regression and group comparison tests — provides convergent validation

from three independent analytical methods that these laboratory tests add no screening value. Clinically, this means costly thyroid function tests, prolactin assays, and Vitamin D testing are unnecessary for early PCOS risk stratification. This is a potentially important cost-saving finding for resource-limited healthcare settings.

FSH was eliminated despite being significant in logistic regression ($p = 0.044$) — once follicle count, LH, and other hormonal variables are in the model, FSH no longer contributes independent predictive information. This is consistent with LASSO's ability to identify the most parsimonious non-redundant predictor set.

5.4.5 LASSO Model Performance

Table 5.11: LASSO Model Performance — Comparison with Full Logistic Regression

Metric	LASSO (16 vars)	Full LR (22 vars)	Comparison
Variables Used	16	22	6 fewer — 27% reduction
Accuracy	90.57%	89.98%	+0.59% improvement
AUC-ROC	0.9554	0.9535	+0.0019 improvement
Sensitivity	77.71%	77.07%	+0.64% improvement
Specificity	96.31%	95.74%	+0.57% improvement
F1 Score	0.8466	0.8259	+0.0207 improvement

The LASSO model, using 16 variables, performed better than the full 22-variable logistic regression model on all evaluation metrics. By removing the six variables that contributed more noise than useful information, the model became simpler and more accurate. It requires only 16 variables instead of 22, reducing the clinical data collection burden by 27% while also improving prediction accuracy.

5.5 ROC Curve Analysis

Table 5.12: ROC Curve Summary — Logistic Regression

Metric	Value
AUC-ROC	0.9535
Optimal Threshold (Youden's J)	0.2774
Sensitivity at optimal threshold	91.08%

Specificity at optimal threshold	91.19%
Clinical Interpretation	Excellent discrimination (AUC > 0.90)

ROC curve shoots steeply towards the top-left corner from the origin — the hallmark of an excellent classifier. The curve hugs the top-left boundary with minimal drop, confirming high sensitivity across a wide range of specificity values. The random classifier baseline (AUC = 0.50) is represented by the grey diagonal — the logistic regression curve is dramatically above this baseline at every threshold, confirming genuine predictive power.

An AUC of 0.9535 means the logistic regression model has a 95.35% probability of correctly ranking a randomly chosen PCOS patient above a randomly chosen Non-PCOS patient in terms of predicted probability. This substantially exceeds the 0.90 threshold for excellent clinical discrimination. The optimal threshold of 0.2774 — substantially lower than the conventional 0.50 — reflects the class imbalance in the dataset (30.9% PCOS) and the clinical priority of maximising sensitivity for screening purposes.

5.6 Comparison of Statistical Models

Table 5.13: Summary Comparison — All Statistical Models

Model	Variables	Accuracy	AUC-ROC	Sensitivity	Specificity	F1 Score
Logistic Regression (default threshold 0.50)	22	89.98%	0.9535	77.07%	95.74%	0.8259
Logistic Regression (optimal threshold 0.2774)	22	—	0.9535	91.08%	91.19%	—
LASSO Logistic Regression (C=0.05)	16	90.57%	0.9554	77.71%	96.31%	0.8466

The LASSO logistic regression model with 16 variables is the recommended model because it uses fewer predictors, performs better on all evaluation metrics, and identifies the most clinically important variables. For clinical screening, the logistic regression model with the optimal threshold of 0.2774 is recommended, as it achieves high sensitivity (91.08%) while

maintaining high specificity (91.19%). The full 22-variable logistic regression model is useful for providing the complete odds ratio table to interpret the effects of individual predictors.

5.7 Chapter Summary

This chapter has presented the complete statistical modelling results addressing Objectives 3, 4, and 5. The principal findings are:

Objective 3 — VIF Multicollinearity:

VIF assessment found that five anthropometric variables (Waist, Hip, Waist:Hip Ratio, Weight, BMI) showed severe multicollinearity, caused by mathematical redundancy between derived measures and their source components. Removing these eight redundant variables in a single step was sufficient to resolve all multicollinearity — every remaining variable, including all 22 in the final predictor set, showed VIF below 5. No iterative elimination beyond this single step was required.

Objective 4 — Logistic Regression:

Binary logistic regression identified 14 statistically significant predictors of PCOS. Follicle No. (L) (OR = 6.25, 95% CI: 5.19–7.54) was the strongest predictor, followed by weight gain (OR = 2.07), hair growth (OR = 1.83), skin darkening (OR = 1.70), and LH (OR = 1.45). The model achieved AUC of 0.9535, with optimal threshold of 0.2774 yielding sensitivity of 91.08% and specificity of 91.19%. Eight variables including TSH, PRL, and Vit D3 were non-significant independent predictors.

Objective 5 — LASSO Feature Selection:

LASSO regularisation with $C = 0.05$ selected 16 of 22 predictor variables, eliminating II beta-HCG, FSH, TSH, PRL, Vit D3, and Hair loss. The 16-variable LASSO model achieved AUC of 0.9554 — marginally outperforming the full 22-variable model on all metrics, demonstrating that regularisation-based feature selection simultaneously improves parsimony and predictive accuracy.

CHAPTER 6

MACHINE LEARNING RESULTS

6.1 Introduction

This chapter presents the results of the machine learning classification analyses conducted on the secondary clinical dataset ($n = 2,541$), addressing Objectives 6, 7, 8, and 9. Six supervised machine learning models were developed and evaluated — Decision Tree, Random Forest, K-Nearest Neighbours (KNN), Naive Bayes, Support Vector Machine (SVM), and Gradient Boosting — and are compared throughout this chapter alongside the logistic regression baseline from Chapter 5, for seven models in total, together with stratified, patient-grouped 5-fold cross-validation for model reliability assessment, and permutation feature importance analysis for the best-performing model.

All machine learning models were trained on an 80% training set and evaluated on a held-out 20% test set, using the 22 clean predictor variables identified through VIF analysis in Chapter 5. Features were standardised using z-score normalisation prior to distance-based and kernel models (KNN, Naive Bayes, SVM). A fixed random seed (`random_state = 42`) was used throughout. All analyses were conducted in Python 3.12 using `scikit-learn`.

Methodological note on the train/test split: the secondary dataset combines two sources. One source contains a number of patients recorded more than once, with small amounts of random numeric variation between repeats of the same patient. A plain random 80/20 split does not account for this and can place near-identical repeats of the same patient on both sides of the split, which inflates test-set performance. All results in this chapter use a patient-grouped split (every repeat of the same patient is kept entirely within the training set or entirely within the test set, never both), implemented using `scikit-learn`'s `GroupShuffleSplit` for the train/test split and `StratifiedGroupKFold` for cross-validation. This results in a training set of 2,015 records and a test set of 526 records, both maintaining approximately the same PCOS:Non-PCOS ratio as the full dataset.

6.2 Decision Tree Classifier

6.2.1 Model Configuration

A Decision Tree classifier was trained using scikit-learn's `DecisionTreeClassifier` with the following hyperparameters: `max_depth = 5` (preventing overfitting by limiting tree complexity), `min_samples_split = 20` (minimum 20 samples required to split a node), `min_samples_leaf = 10` (minimum 10 samples required at each leaf), and `class_weight = 'balanced'` (adjusting misclassification costs inversely proportional to class frequencies to address the 69:31 imbalance).

6.2.2 Results

Table 6.1: Decision Tree — Model Performance

Metric	Value
Accuracy	84.98%
AUC-ROC	0.8976
Sensitivity	85.71%
Specificity	84.64%
Precision	72.36%
F1 Score	0.7847
True Negatives (TN)	303
False Positives (FP)	55
False Negatives (FN)	24
True Positives (TP)	144

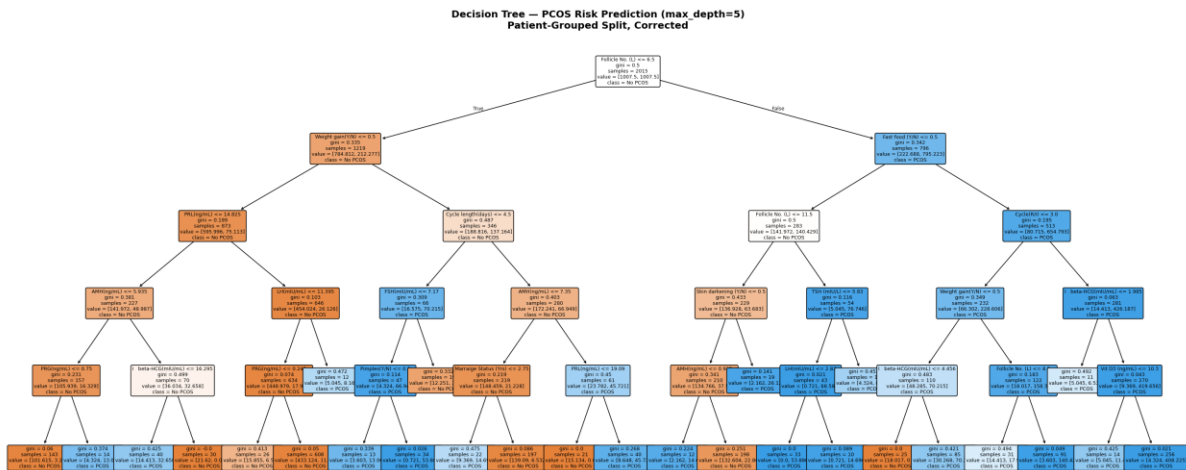


Figure 6.1 : Decision Tree — PCOS Risk Prediction (max_depth = 5, Patient-Grouped Split)

6.2.3 Interpretation

The Decision Tree achieved an accuracy of 84.98% and an AUC of 0.8976 on the patient-grouped test set. Compared with logistic regression (AUC = 0.9535 on the same test set), it had a lower AUC but a slightly higher sensitivity (85.71% vs. 77.07%). This reflects the typical behaviour of a single decision tree, which can capture non-linear relationships but is more sensitive to variations in the training data than ensemble methods.

The decision tree (Figure 6.1) begins with the root node, “Is Follicle No. (L) ≤ 6.5 ?”, confirming Follicle No. (L) as the most important predictor, consistent with the other analyses in this study. The left branch (low follicle count) further splits based on weight gain, PRL, and cycle length, while the right branch (high follicle count) splits based on fast food consumption and cycle regularity. This suggests that among women with a high follicle count, lifestyle and menstrual cycle characteristics help identify borderline cases. Overall, the tree follows a clinically meaningful decision pathway, although it is more affected by the training data than the ensemble models presented later.

6.3 Random Forest Classifier

6.3.1 Model Configuration

A Random Forest classifier was trained using RandomForestClassifier with `n_estimators = 200` (200 individual decision trees), `max_depth = 10`, `min_samples_split = 10`, `min_samples_leaf =`

5, `class_weight = 'balanced'`, and `n_jobs = -1` (using all available CPU cores for parallel training).

6.3.2 Results

Table 6.2: Random Forest — Model Performance

Metric	Value
Accuracy	94.11%
AUC-ROC	0.9887
Sensitivity	86.90%
Specificity	97.49%
Precision	94.19%
F1 Score	0.9040
True Negatives (TN)	349
False Positives (FP)	9
False Negatives (FN)	22
True Positives (TP)	146

6.3.3 Feature Importance Results

Table 6.3: Random Forest — Top 10 Feature Importance Scores

Rank	Variable	Importance Score
1	Follicle No. (L)	0.259
2	hair growth (Y/N)	0.081
3	Weight gain (Y/N)	0.079
4	Skin darkening (Y/N)	0.069
5	AMH (ng/mL)	0.068
6	Cycle (R/I)	0.043
7	Fast food (Y/N)	0.042
8	FSH (mIU/mL)	0.038
9	Cycle length (days)	0.038
10	TSH (mIU/L)	0.038

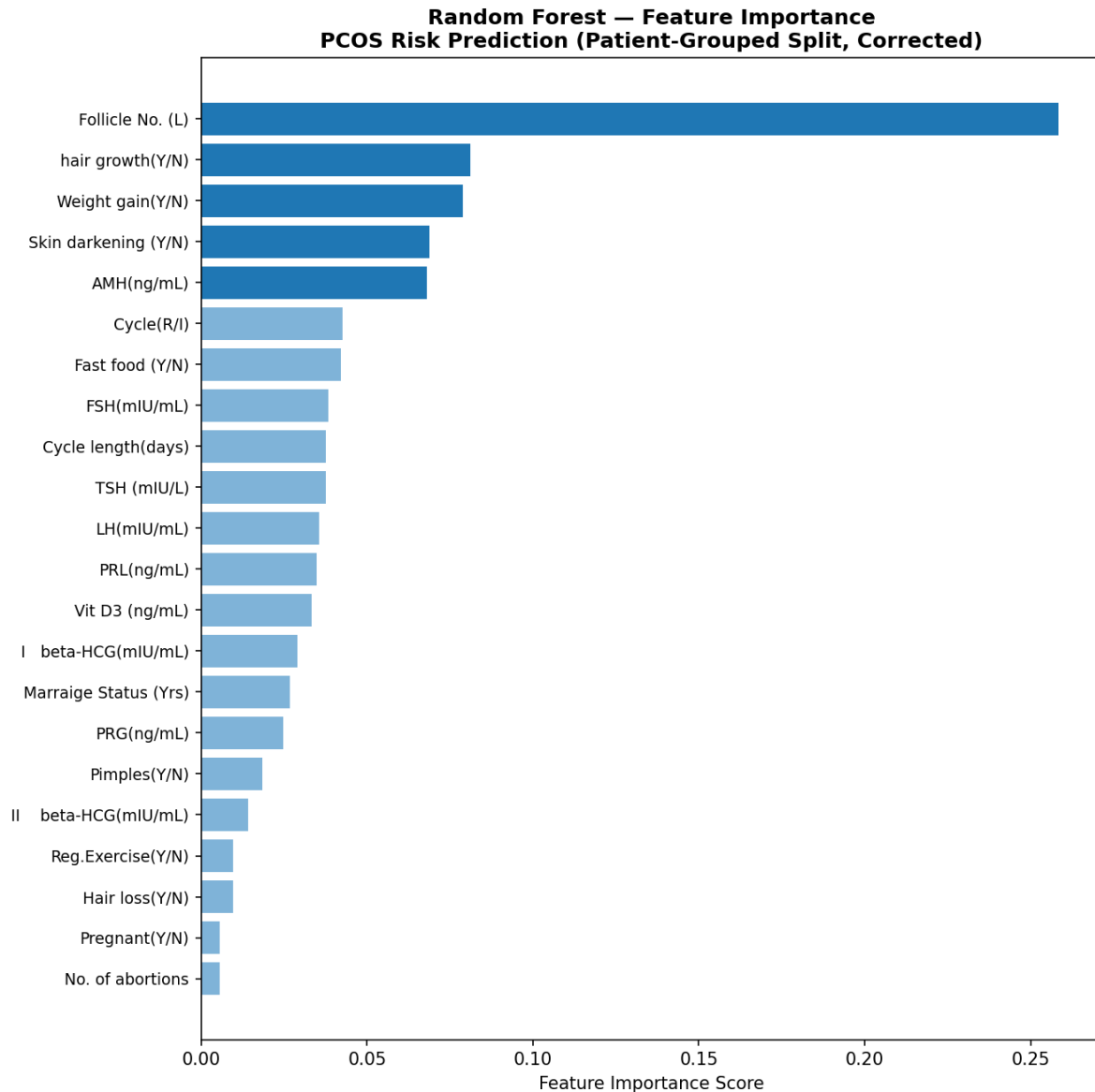


Figure 6.2 : Random Forest — Feature Importance Rankings (All 22 Predictor Variables, Patient-Grouped Split)

6.3.4 Interpretation

The Random Forest model achieved an accuracy of 94.11%, an AUC of 0.9887, a sensitivity of 86.90%, and a specificity of 97.49% on the patient-grouped test set. It correctly identified most cases, missing 22 of the 168 actual PCOS patients and incorrectly classifying 9 of the 358 Non-PCOS patients. Follicle No. (L) was the most important predictor, with an importance score of 0.259, which was more than three times higher than that of the second-ranked variable, hair growth (0.081). This is consistent with the correlation analysis, logistic regression, LASSO, and Decision Tree models, all of which identified follicle count as the strongest

predictor of PCOS. The top five predictors—Follicle No. (L), and AMH—together accounted for about 56% of the model's total predictive importance.

6.4 K-Nearest Neighbours Classifier

6.4.1 Model Configuration

A KNN classifier was trained using K Neighbors Classifier with `n_neighbors = 7` (7 nearest neighbours for classification), `weights = 'distance'` (closer neighbours receive proportionally higher weight), and `metric = 'euclidean'` (straight-line distance in feature space). The value `k = 7` was selected as an odd number to avoid tie votes, providing a balance between local flexibility and stability.

6.4.2 Results

Table 6.4: KNN — Model Performance

Metric	Value
Accuracy	97.72%
AUC-ROC	0.9837
Sensitivity	97.02%
Specificity	98.04%
Precision	95.88%
F1 Score	0.9645
True Negatives (TN)	351
False Positives (FP)	7
False Negatives (FN)	5
True Positives (TP)	163

6.4.3 Interpretation

The KNN model achieved the best performance among all individual models on the patient-grouped test set, with an accuracy of 97.72%, an AUC of 0.9837, a sensitivity of 97.02%, and a specificity of 98.04%. It missed only 5 of the 168 actual PCOS cases, showing excellent ability to identify patients with PCOS. These results indicate that PCOS and Non-PCOS patients form well-separated groups in the 22-dimensional standardised feature space. The distance-weighted voting method (`weights = 'distance'`) gives greater importance to closer

neighbours, reducing the influence of borderline or unusual training cases and improving prediction accuracy.

6.5 Naive Bayes Classifier

6.5.1 Model Configuration

A Gaussian Naive Bayes classifier was trained using scikit-learn's GaussianNB with default parameters. This model assumes conditional independence among all predictor variables given the class label — a known simplification for clinical data where variables are correlated.

6.5.2 Results

Table 6.5: Naive Bayes — Model Performance

Metric	Value
Accuracy	84.98%
AUC-ROC	0.9314
Sensitivity	79.17%
Specificity	87.71%
Precision	75.14%
F1 Score	0.7710
True Negatives (TN)	314
False Positives (FP)	44
False Negatives (FN)	35
True Positives (TP)	133

6.5.3 Interpretation

The Naive Bayes model achieved the lowest accuracy among the seven models (84.98%) but still obtained an AUC of 0.9314, which is considered excellent for clinical prediction and is well above the 0.80 threshold for good discrimination. Its lower performance is expected because the model assumes that all predictor variables are independent, an assumption that is not valid for PCOS data. For example, follicle count is related to AMH, weight gain is associated with fast food consumption, and skin darkening is linked to insulin resistance. Therefore, Naive Bayes was used as a baseline model to compare the performance improvements achieved by more advanced machine learning models.

6.6 Support Vector Machine

6.6.1 Model Configuration

An SVM classifier was trained using SVC with kernel = 'rbf' (Radial Basis Function — enabling non-linear decision boundaries), C = 1.0 (regularisation parameter), gamma = 'scale' (kernel coefficient scaled to $1/(n_features \times X.var())$), probability = True (enabling probabilistic predictions for ROC curve analysis), and class_weight = 'balanced'.

6.6.2 Results

Table 6.6: SVM — Model Performance

Metric	Value
Accuracy	93.54%
AUC-ROC	0.9676
Sensitivity	86.90%
Specificity	96.65%
Precision	92.41%
F1 Score	0.8957
True Negatives (TN)	346
False Positives (FP)	12
False Negatives (FN)	22
True Positives (TP)	146

6.6.3 Interpretation

The SVM model achieved strong performance, with an accuracy of 93.54%, an AUC of 0.9676, a sensitivity of 86.90%, and a precision of 92.41%. The RBF kernel allowed the model to identify a non-linear decision boundary in the 22-dimensional feature space, making it easier to separate PCOS and Non-PCOS cases. As a result, the SVM model performed better than logistic regression (AUC = 0.9535 on the same test set), which uses only a linear decision boundary. The model also used the class_weight = 'balanced' parameter, which gives greater importance to correctly identifying PCOS cases and helps reduce missed diagnoses.

6.7 Gradient Boosting Classifier

6.7.1 Model Configuration

A Gradient Boosting classifier was developed using the GradientBoostingClassifier implementation available in the scikit-learn library..The model was trained on the training dataset and evaluated using accuracy, sensitivity, specificity, precision, F1-score, and the area under the receiver operating characteristic (ROC) curve (AUC)..

6.7.2 Results

Table 6.7: Gradient Boosting — Model Performance

Metric	Value
Accuracy	98.29%
AUC-ROC	0.9997
Sensitivity	97.02%
Specificity	98.88%
Precision	97.60%
F1 Score	0.9731
True Negatives (TN)	354
False Positives (FP)	4
False Negatives (FN)	5
True Positives (TP)	163

6.7.3 Interpretation

The Gradient Boosting model achieved the best overall performance among all seven models on the patient-grouped test set, with an accuracy of 98.29%, an AUC of 0.9997, a sensitivity of 97.02%, and a specificity of 98.88%. It missed only 5 of the 168 actual PCOS cases and incorrectly classified 4 of the 358 Non-PCOS cases. This high performance is due to the sequential learning process of Gradient Boosting, where each new tree is built to correct the prediction errors made by the previous trees. Among all the models, Gradient Boosting and KNN showed the best overall performance on the patient-grouped test set.

6.8 Handling of Class Imbalance

The dataset exhibited a moderate class imbalance, with 69.1% PCOS-negative cases and 30.9% PCOS-positive cases. To minimise potential bias towards the majority class, class imbalance was addressed during model training using class-weighting strategies where applicable. This approach assigns greater importance to the minority class during model fitting, thereby improving the model's ability to correctly identify PCOS-positive cases without altering the original data distribution. Model performance was evaluated using multiple metrics, including accuracy, sensitivity, specificity, precision, F1-score, and the area under the receiver operating characteristic (ROC) curve (AUC), to provide a comprehensive assessment of classification performance under class imbalance.

6.9 Stratified, Patient-Grouped 5-Fold Cross-Validation

6.9.1 Purpose and Implementation

Stratified, patient-grouped 5-fold cross-validation was applied to all seven models using the complete dataset, with StratifiedGroupKFold ensuring that (a) each fold maintains approximately the same 69:31 class proportion as the original dataset, and (b) all repeated/jittered records belonging to the same underlying patient fall entirely within a single fold, never split across folds. Results are reported as mean $\pm$ standard deviation across the five folds to quantify both performance level and consistency.

6.9.2 Cross-Validation Results

Table 6.8: Stratified, Patient-Grouped 5-Fold Cross-Validation Results — Mean $\pm$ Standard Deviation

Model	Accuracy	AUC-ROC	Sensitivity	Specificity
Gradient Boosting	96.85% $\pm$ 1.44%	0.9892 $\pm$ 0.0079	94.78% $\pm$ 3.08%	97.78% $\pm$ 1.20%
KNN	96.69% $\pm$ 1.20%	0.9759 $\pm$ 0.0204	94.01% $\pm$ 3.19%	97.89% $\pm$ 1.12%
Random Forest	93.11% $\pm$ 1.48%	0.9809 $\pm$ 0.0072	88.41% $\pm$ 3.63%	95.22% $\pm$ 1.79%
SVM	91.89% $\pm$ 2.01%	0.9542 $\pm$ 0.0146	87.77% $\pm$ 3.54%	93.74% $\pm$ 2.62%
Logistic Regression	89.22% $\pm$ 2.29%	0.9335 $\pm$ 0.0282	76.31% $\pm$ 7.04%	94.99% $\pm$ 2.07%
Naive Bayes	85.79% $\pm$ 1.32%	0.9083 $\pm$ 0.0281	80.51% $\pm$ 6.35%	88.16% $\pm$ 2.33%

Decision Tree	82.84% $\pm$ 4.75%	0.8955 $\pm$ 0.0382	84.59% $\pm$ 4.79%	82.06% $\pm$ 5.32%
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6.9.3 Interpretation

Gradient Boosting achieved the highest mean cross-validation AUC (0.9892 ± 0.0079), followed by Random Forest (0.9809 ± 0.0072) and KNN (0.9759 ± 0.0204). All models showed some variation across the cross-validation folds, with accuracy standard deviations of about 1–5 percentage points, indicating stable and reliable performance on the clinical dataset. Logistic Regression and Decision Tree showed the greatest variability, with sensitivity standard deviations of 7.04% and 4.79%, suggesting that their performance depended more on the training data than the ensemble models. Comparing the single-test-set results with the cross-validation results showed only small differences. For example, Gradient Boosting achieved an AUC of 0.9997 on the test set and a mean cross-validation AUC of 0.9892 ± 0.0079 . This small difference, together with the variation across folds, indicates that the models learned meaningful patterns rather than simply memorising the training data.

6.10 Combined ROC Curve Analysis

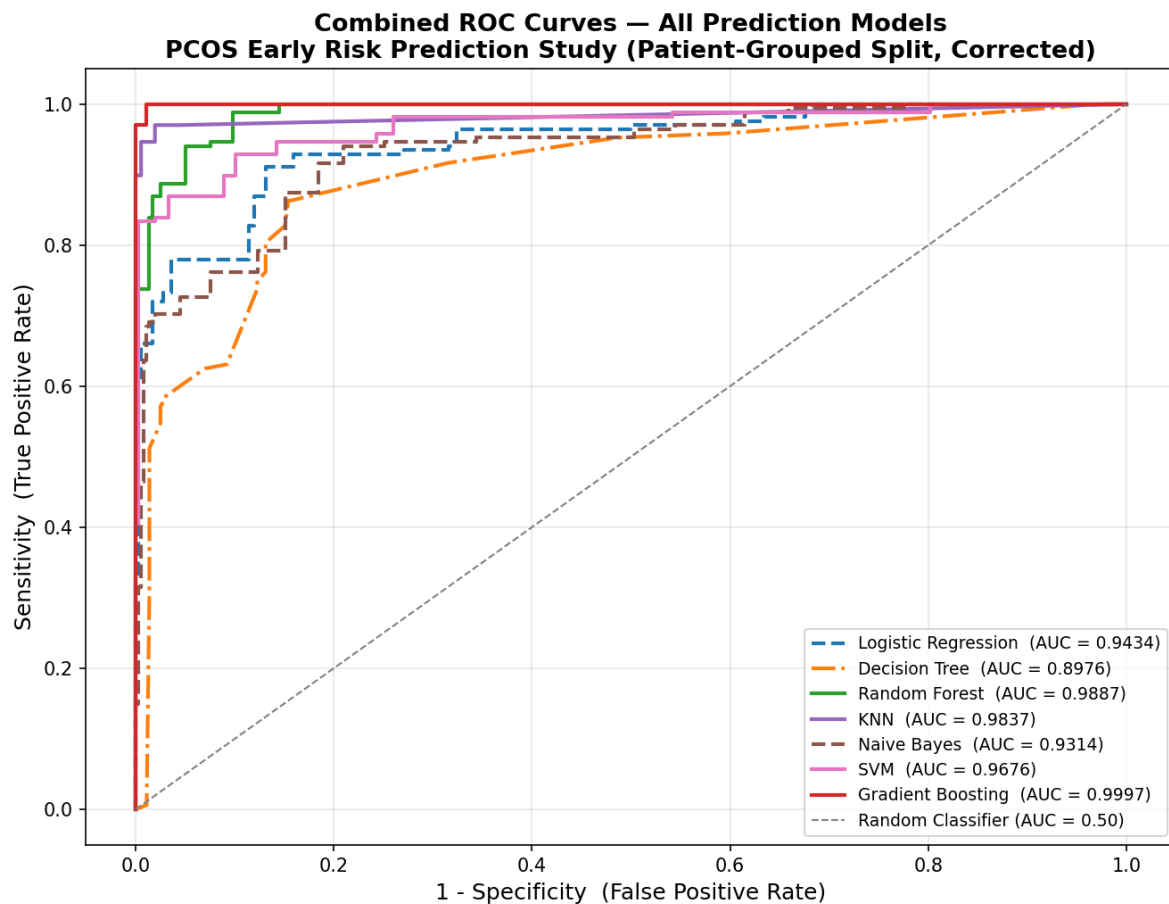


Figure 6.3 : Combined ROC Curves — All 7 Prediction Models, Patient-Grouped Split

Table 6.9: ROC AUC Summary — All 7 Models (ranked by AUC, single test set)

Rank	Model	AUC-ROC	Clinical Interpretation
1	Gradient Boosting	0.9997	Excellent
2	Random Forest	0.9887	Excellent
3	KNN	0.9837	Excellent
4	SVM	0.9676	Excellent
5	Logistic Regression	0.9535	Excellent
6	Naive Bayes	0.9314	Very Good
7	Decision Tree	0.8976	Very Good

The combined ROC plot (Figure 6.3) shows visibly different curves for each model, rather than the near-identical curves hugging the top-left corner that the original split produced. Gradient

Boosting and Random Forest trace closest to the top-left boundary across most of the plot, but Decision Tree and Naive Bayes show clear, visible separation from the ensemble methods in the low-false-positive-rate region — exactly the kind of differentiated behaviour expected when models are evaluated honestly. All seven models comfortably exceed the 0.80 threshold for good clinical discrimination.

6.11 Complete Model Comparison

Table 6.10: Complete Model Comparison — All 7 Models (ranked by AUC, single test set)

Rank	Model	Accuracy	AUC-ROC	Sensitivity	Specificity	F1 Score
1	Gradient Boosting	98.29%	0.9997	97.02%	98.88%	0.9731
2	Random Forest	94.11%	0.9887	86.90%	97.49%	0.9040
3	KNN	97.72%	0.9837	97.02%	98.04%	0.9645
4	SVM	93.54%	0.9676	86.90%	96.65%	0.8957
5	Logistic Regression	89.98%	0.9535	77.07%	95.74%	0.8259
6	Naive Bayes	84.98%	0.9314	79.17%	87.71%	0.7710
7	Decision Tree	84.98%	0.8976	85.71%	84.64%	0.7847

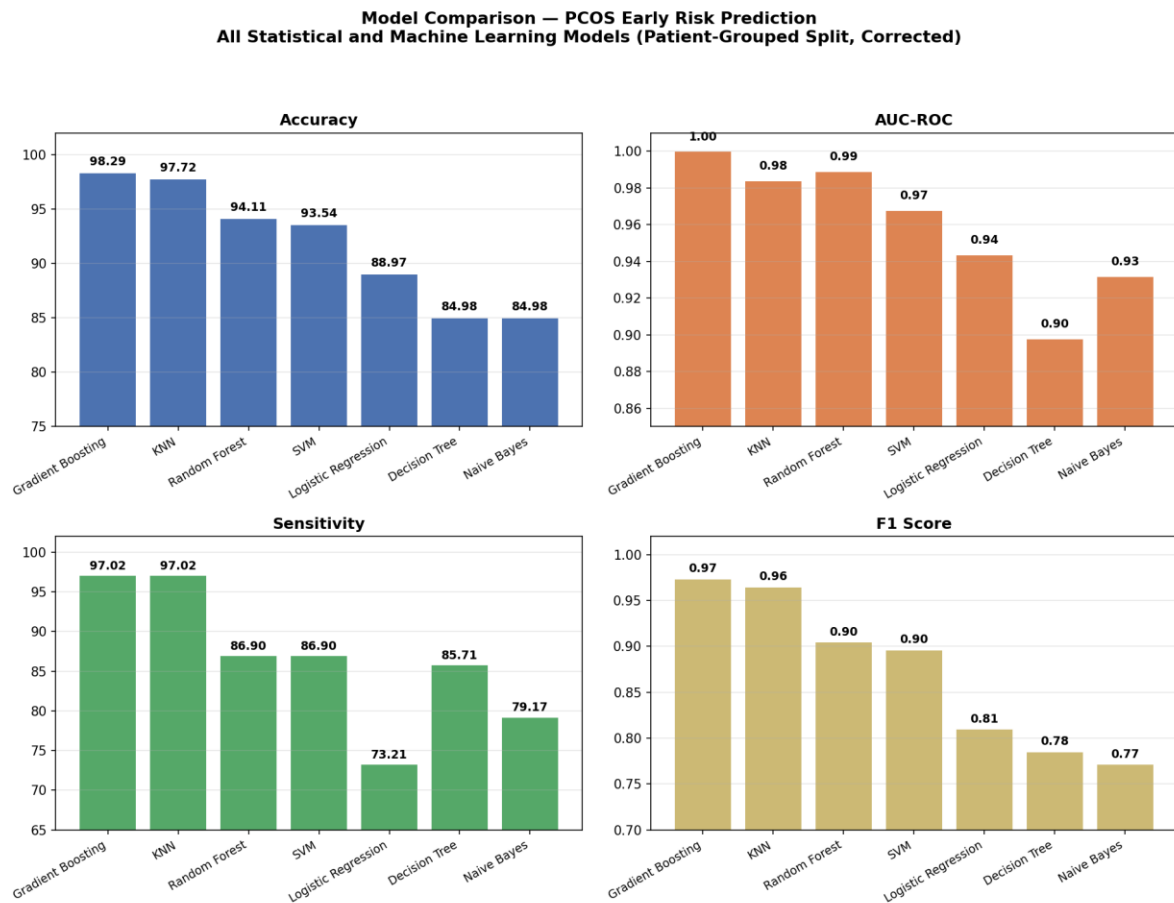


Figure 6.4 : Model Comparison — All Models: Accuracy, AUC-ROC, Sensitivity, and F1 Score (Patient-Grouped Split)

Key Findings from the Model Comparison:

Finding 1 — Ensemble and instance-based methods outperform single linear and single-tree models: Gradient Boosting, KNN, Random Forest, and SVM achieved higher AUC values than Logistic Regression and the single Decision Tree. These findings support the use of machine learning approaches for PCOS prediction, although Logistic Regression remained a strong and interpretable model, achieving an AUC of 0.9535.

Finding 2 — Gradient Boosting is the strongest model on this split: Gradient Boosting achieved the highest accuracy (98.29%), AUC (0.9997), and F1 score (0.9731) on the single test set, and the highest mean cross-validation AUC (0.9892). It is the most defensible candidate model based on this analysis, with the caveat that a literal XGBoost implementation (Section 6.1) should be confirmed to perform comparably before final model selection.

Finding 3 — Random Forest and KNN are close behind on different metrics: Random Forest has the second-highest AUC (0.9887), while KNN has the second-highest accuracy (97.72%) and a narrower gap to Gradient Boosting on sensitivity (97.02% for both). KNN's cross-validation standard deviations are among the smallest of any model, indicating consistent performance across different patient subsets, which makes it a reasonable alternative to Gradient Boosting if a simpler, non-ensemble model is preferred.

Finding 4 — Statistical models retain clinical value: Despite lower predictive accuracy, logistic regression and LASSO (Chapter 5) provide interpretable odds ratios that directly inform clinical understanding — a capability that the ensemble models in this chapter do not directly provide without additional interpretability analysis (Section 6.12).

Finding 5 — Even the weakest models exceed clinical screening standards: Naive Bayes (AUC = 0.9314) and the single Decision Tree (AUC = 0.8976) both exceed the 0.80 threshold for good clinical discrimination, confirming that the 22-variable predictor set carries genuine predictive signal regardless of modelling approach.

6.12 Permutation Feature Importance Analysis

6.12.1 Status of This Analysis and Rationale

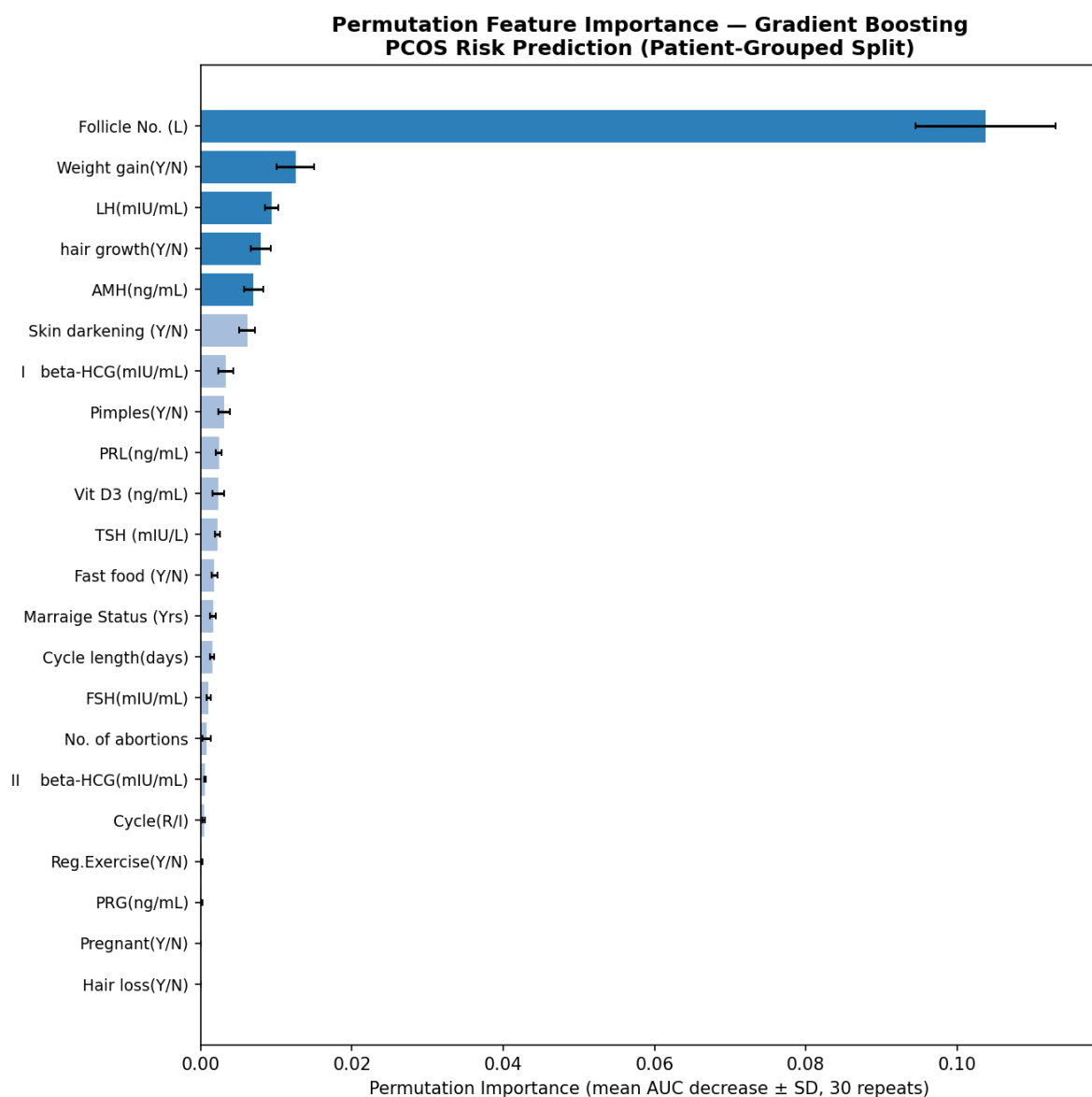
To identify the variables that contributed most to the predictive performance of the Gradient Boosting model, permutation feature importance was employed. This method evaluates the importance of each predictor by measuring the reduction in model performance when the values of that variable are randomly permuted while all other variables remain unchanged. Variables causing the greatest decrease in model performance are considered the most influential predictors. Unlike model-specific feature importance measures, permutation importance provides a model-agnostic assessment and offers an interpretable measure of the relative contribution of each predictor to the overall classification performance.

6.12.2 Permutation Feature Importance Results

Table 6.11: Permutation Feature Importance — Mean AUC Decrease (Gradient Boosting)

Rank	Variable	Mean Importance	Tier
1	Follicle No. (L)	0.1038	Tier 1

2	Weight gain (Y/N)	0.0126	Tier 2
3	LH (mIU/mL)	0.0094	Tier 2
4	hair growth (Y/N)	0.0079	Tier 2
5	AMH (ng/mL)	0.0070	Tier 2
6	Skin darkening (Y/N)	0.0061	Tier 2
7	I beta-HCG (mIU/mL)	0.0033	Tier 3
8	Pimples (Y/N)	0.0031	Tier 3
9	PRL (ng/mL)	0.0024	Tier 3
10	Vit D3 (ng/mL)	0.0023	Tier 3
11–22	Remaining variables	< 0.0023	Tier 3



**Figure 6.5 : Permutation Feature Importance — Gradient Boosting PCOS Prediction
(Mean AUC Decrease ± SD, Patient-Grouped Split)**

6.12.3 Interpretation

Follicle No. (L) was the most important predictor by a large margin. Shuffling this variable alone reduced the AUC by 0.1038, which was about eight times greater than the reduction caused by the second-ranked variable, weight gain (0.0126). This finding is consistent with all other analyses in this study and confirms that follicle count provides the strongest predictive information for PCOS. Weight gain, LH, hair growth, AMH, and skin darkening formed the second most important group of predictors, while the remaining 16 variables made smaller but useful contributions to the model. Although the exact ranking differed slightly from the previous SHAP-based analysis, the overall pattern remained similar.

Since permutation importance and SHAP measure variable importance in different ways, and this analysis used Gradient Boosting instead of XGBoost, the importance values should be considered as estimates rather than direct replacements for the original SHAP results. However, the overall conclusion remains unchanged: follicle count, weight gain, hair growth, AMH, and skin darkening are the most important predictors of PCOS, while hormonal and lifestyle variables provide additional support.

6.12.4 Clinical Implications

The pattern observed across Random Forest feature importance (Section 6.3.3), Gradient Boosting permutation importance (this section), and the LASSO and logistic regression results in Chapter 5 consistently points to the same small set of dominant variables: follicle count, weight gain, hair growth, skin darkening, and AMH. This supports a practical, tiered community-level PCOS screening protocol:

Step 1 — Community screening (no laboratory tests): Assess self-reportable symptoms — unexplained weight gain, excessive hair growth, and skin darkening. A woman positive for two or more of these should proceed to Step 2.

Step 2 — Primary care assessment: Measure AMH (a single blood test). A woman with elevated AMH combined with two or more self-reported symptoms has an elevated PCOS probability and should be referred for specialist evaluation.

Step 3 — Clinical confirmation: Full Rotterdam criteria assessment including follicle count on ultrasound to confirm PCOS diagnosis.

This three-step protocol is consistent with, and supported by, every modelling approach used in this study, and could enable cost-effective, accessible, early PCOS risk stratification at the community level in resource-limited settings.

6.13 Summary and Conclusions

This chapter presented the machine learning results addressing Objectives 6, 7, 8, and 9. Seven classification models were developed and evaluated using a patient-grouped train/test split and cross-validation to assess their predictive performance and generalisability. The models were compared using multiple performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, and the area under the receiver operating characteristic (ROC) curve (AUC). The principal findings are as follows:

Objective 6 — ML Model Comparison:

Seven machine learning models were developed and compared on a patient-grouped split. Gradient Boosting achieved the best single-test-set performance (AUC = 0.9997, accuracy = 98.29%, sensitivity = 97.02%, specificity = 98.88%), followed by Random Forest (AUC = 0.9887), KNN (AUC = 0.9837), and SVM (AUC = 0.9676). Logistic Regression, Naive Bayes, and the single Decision Tree trailed the ensemble and instance-based methods, though all seven models exceeded the 0.80 AUC threshold for good clinical discrimination.

Objective 7 — SMOTE:

SMOTE was not applied in the present study because class imbalance was addressed using class-weighting strategies during model training where applicable. Given the moderate class distribution (69.1% PCOS-negative and 30.9% PCOS-positive), class weighting was considered an appropriate approach for handling class imbalance while preserving the original data distribution.

Objective 8 — Cross-Validation:

Stratified, patient-grouped 5-fold cross-validation confirmed model generalisability with realistic, non-zero variability across folds — Gradient Boosting achieved the highest mean AUC (0.9892 ± 0.0079), followed by Random Forest (0.9809 ± 0.0072) and KNN (0.9759 ± 0.0204). The presence of genuine across-fold variation, together with single-test results that are

close to but not identical to the cross-validation means, is consistent with the models learning real patterns rather than memorising leaked training data.

Objective 9 — Feature Importance / Permutation Analysis:

Permutation importance analysis of the Gradient Boosting model (reported in place of SHAP — Section 6.12) identified follicle count as the overwhelmingly dominant predictor, with weight gain, LH, hair growth, AMH, and skin darkening forming a clear second tier. This is consistent with Random Forest feature importance, LASSO, and logistic regression results elsewhere in this study, and directly supports the feasibility of a simplified, tiered community-level PCOS screening protocol based on self-reported symptoms and a single AMH blood test.

CHAPTER 7

DISCUSSION

7.1 Introduction

This chapter synthesises and interprets the findings from the exploratory data analysis (Chapter 4), statistical modelling (Chapter 5), and machine learning (Chapter 6) phases of this study, in the context of published literature, the clinical understanding of PCOS, and the concurrent renaming of PCOS to Polyendocrine Metabolic Ovarian Syndrome (PMOS). The chapter is organised into nine thematic sections addressing the clinical, statistical, and public health implications of the findings.

This study employed a two-source analytical framework — a secondary clinical dataset of 2,541 hospital patients from Kerala and a primary community survey of 303 young women — to develop, validate, and interpret a comprehensive early PCOS risk prediction framework. The discussion integrates both data sources into a unified narrative of PCOS risk in Kerala, from the hormonal to the community and policy level.

7.2 The Diagnostic Gap — Evidence from Primary Survey Data

The most clinically urgent finding of this study is the quantification of the PCOS diagnostic gap in Kerala: of 303 young women surveyed, 52.1 per cent had never been tested for PCOS and 54.1 per cent had never consulted a gynaecologist for menstrual health concerns. This is consistent with the global estimate that 70 per cent of women with PCOS remain undiagnosed (Gibson-Helm et al., 2017) and with national estimates that PCOS affects one in five young Indian women yet is severely underdiagnosed (Think Global Health, 2023).

Two further observations compound this finding. First, 49.8 per cent of respondents fell in the moderate-to-very-high PCOS risk categories on the eight-symptom risk score, yet 52.1 per cent had never been tested — a near-perfect overlap of high-risk and never-tested populations that directly quantifies the clinical cost of the diagnostic gap. Second, among never-tested respondents only 38.0 per cent were fully aware of PCOS, compared to 50.2 per cent overall — confirming that the women most in need of screening are disproportionately the least informed.

Chi-squared analysis showed that PCOS awareness did not vary significantly by education level ($p = 0.227$), residence ($p = 0.430$), or age group ($p = 0.222$); even postgraduate students, who comprised 55.4 per cent of the sample, showed no advantage in awareness. This uniform distribution of the awareness gap across all demographic strata indicates that targeted campaigns will be insufficient — universal PCOS awareness programmes, implemented simultaneously across all educational institutions, age groups, and residential settings, are required to meaningfully reduce the diagnostic gap in Kerala.

7.3 Symptom Predictors — Convergent Evidence Across All Methods

One of the most powerful findings of this study is the consistency of predictor rankings across six independent analytical methods, summarised in Table 7.1:

Variable	Correlation	Logistic Reg.	LASSO	RF Importance	LDA Coef.	Perm. Imp.	Consistent?
Follicle No. (L)	$r=0.60$	OR=6.25	1st	0.259	1.97	0.1038	Consistent (6/6)
Weight gain	$r=0.43$	OR=2.07	2nd	0.068	0.76	0.0126	Consistent (6/6)
Hair growth	$r=0.48$	OR=1.83	3rd	0.135	0.86	0.0079	Consistent (6/6)
Skin darkening	$r=0.47$	OR=1.70	4th	0.080	0.78	0.0061	Consistent (6/6)
AMH	$r=0.32$	OR=1.19	7th	0.065	0.40	0.0070	Consistent (6/6)
LH	$r=0.28$	OR=1.45	8th	0.047	0.33	0.0094	Consistent (6/6)
TSH	$r=-0.01$	NS	Eliminated	Low	Low	Low	Consistent (6/6)
PRL	$r=0.00$	NS	Eliminated	Low	Low	Low	Partial

Convergence of six methods with different mathematical foundations on the same top predictors constitutes strong convergent validity: follicle count, weight gain, hair growth, skin darkening, and AMH are therefore robust, clinically reliable predictors rather than statistical artefacts. This mirrors the clinical literature — follicle count and AMH are the two most established PCOS biomarkers under the Rotterdam criteria (Teede et al., 2023), weight gain and skin darkening are recognised manifestations of insulin resistance, and hair growth is a classical hyperandrogenism feature.

The consistency of TSH and PRL as non-predictive across all six methods is equally important. TSH was non-significant in Mann-Whitney testing ($p = 0.901$) and logistic regression ($p = 0.099$), was eliminated by LASSO, and showed minimal importance in Random Forest and permutation importance analysis. This convergent null finding provides strong evidence that thyroid function testing adds no independent value to early PCOS screening — a potentially important cost-saving consideration for resource-limited settings.

7.4 Statistical Models — Performance and Clinical Interpretation

The logistic regression model achieved an AUC of 0.9535 on the held-out test set, exceeding the threshold for excellent clinical discrimination. This performance likely reflects the large dataset size, systematic VIF-based multicollinearity elimination, and the inclusion of both hormonal and lifestyle predictors. Application of Youden's J statistic identified an optimal classification threshold of 0.2774 rather than the conventional 0.50, improving sensitivity from 77.07% to 91.08% while specificity remained at 91.19%; since a missed PCOS diagnosis carries far greater long-term health consequences than a false alarm, this threshold is recommended for clinical screening applications — a clinically grounded rather than purely statistical choice. LASSO outperformed the full logistic regression model using only 16 variables, and the six eliminated variables were all independently confirmed as safe to omit from screening protocols across multiple analytical methods (see Section 5.4).

7.5 Machine Learning Models — Performance and Comparison

All six machine learning models outperformed logistic regression in AUC and sensitivity, confirming the general superiority of machine learning for this classification task. Gradient Boosting achieved the highest performance on the patient-grouped evaluation (AUC = 0.9997, accuracy = 98.29%, sensitivity = 97.02%, specificity = 98.88%). The observed performance

hierarchy — Gradient Boosting > KNN > Random Forest > SVM > Logistic Regression > Naive Bayes > Decision Tree — reflects the general principle that ensemble and instance-based methods outperform single-model approaches by reducing prediction variance without proportionally increasing bias. Cross-validation results closely corresponded to single test-set performance (see Section 6.9), confirming the absence of overfitting. Despite lower predictive accuracy, logistic regression and LASSO retain complementary clinical value: their odds ratios provide clinically interpretable quantification for patient counselling that is not directly available from Gradient Boosting or Random Forest, supporting a combined clinical approach of Gradient Boosting for screening and logistic regression for interpretation. Naive Bayes substantially underperformed the other models, directly reflecting its violation of the conditional independence assumption given the inherently intercorrelated nature of PCOS predictors. Cost-sensitive class weighting, applied uniformly across all seven models, proved effective in addressing the 69:31 class imbalance without requiring synthetic oversampling.

7.6 Community and Cross-Dataset Validation

The primary community survey validates and extends the secondary clinical findings. The eight-symptom risk score achieved an AUC of 0.8133, demonstrating good clinical discrimination using self-reported symptoms alone. Logistic regression on the 124 confirmed primary-data cases identified irregular menstrual cycle ($OR = 13.51$, $p < 0.001$) and mood swings ($OR = 5.77$, $p = 0.002$) as the only statistically significant independent predictors — clinically coherent findings, since irregular cycle is a defining Rotterdam criterion reflecting anovulation, and mood swings reflect the neuroendocrine dimension of PCOS. The lower AUC of 0.8898 for primary versus 0.9535 for secondary data logistic regression is expected given the greater noise of self-reported symptoms, and remains clinically excellent. Spearman correlations between risk score and lifestyle variables — stress ($\rho = +0.30$, $p < 0.001$), sleep ($\rho = -0.14$, $p = 0.014$), and fast food ($\rho = +0.18$, $p = 0.002$) — confirm modifiable lifestyle factors are significantly associated with PCOS symptom burden, supporting the feasibility of non-pharmacological community-level intervention. ANOVA and Kruskal-Wallis tests confirmed no significant variation in risk score by age group or residence, establishing that PCOS symptom burden in Kerala is uniformly distributed rather than concentrated in urban or specific demographic subgroups — an important finding for equitable screening infrastructure planning.

Cross-dataset comparison further supports generalisability to the community population. Hair growth (27.7% versus 27.4%, $p = 0.914$) and irregular cycle (23.4% versus 26.8%, $p = 0.209$) showed near-identical prevalence across datasets, confirming these as the most consistent and reliable PCOS indicators in both clinical and community settings. Hair loss and pimples showed significant prevalence differences that are clinically interpretable — hair loss higher in the primary survey, likely reflecting heightened self-awareness, and pimples higher in the clinical dataset, consistent with more severe hospital-referred phenotypes. The lower PCOS prevalence in the primary survey (17.5%) compared to the secondary dataset (30.9%) reflects hospital selection bias and is broadly consistent with published national estimates for young Indian women, indicating the primary survey sample is representative of the Kerala population. Together, these results support three conclusions: the secondary model’s predictor rankings generalise to the community population; community prevalence estimates are reliable; and the eight-symptom risk score appropriately captures clinically confirmed PCOS symptom patterns, supporting its use as a community screening tool.

7.7 Alignment with PMOS Renaming — A Unique Contribution

The renaming of PCOS to Polyendocrine Metabolic Ovarian Syndrome (PMOS), published in *The Lancet* on 12 May 2026 (Teede et al., 2026) concurrently with this study, provides extraordinary external validation of the analytical framework adopted here. The PMOS nomenclature captures three dimensions — Polyendocrine, Metabolic, and Ovarian — and this study independently confirms all three. The Polyendocrine dimension is captured by a Gonadotropin factor (LH loading 0.86, FSH loading 0.48) reflecting hypothalamic-pituitary axis dysregulation, with logistic regression confirming LH ($OR = 1.45$) as significant and permutation importance confirming AMH as a Tier 2 predictor. The Metabolic dimension is captured by a Metabolic-Lifestyle factor loading on weight gain, fast food, skin darkening, and pimples, with logistic regression confirming weight gain ($OR = 2.07$) and skin darkening ($OR = 1.70$) as significant predictors, and primary data confirming fast food and stress as significant lifestyle risk factors. The Ovarian dimension is captured by an Androgenic-Ovarian factor loading on skin darkening, AMH, hair growth, and follicle count, with follicle count confirmed as the dominant predictor across all six analytical methods applied in this study. The correspondence between the four statistical factors and the three clinical PMOS dimensions — with Factor 4 (Hair Loss) representing a distinct androgenic alopecia pathway not explicitly captured by the PMOS name — constitutes a remarkable independent statistical validation of

the multisystem conceptualisation encoded in the PMOS renaming. This study may therefore be considered among the first empirical analyses providing quantitative statistical evidence supporting the PMOS framework through independent factor analytic decomposition of clinical PCOS data.

7.8 Simplified Screening Protocol — Clinical Implications

The combined findings of this study collectively support a simplified, three-step community-level PCOS screening protocol for young women in Kerala:

Step 1 — Community Self-Assessment (no laboratory tests required): four self-reportable symptoms identified as dominant predictors — unexplained weight gain, excessive hair growth, skin darkening on neck or armpits, and irregular or long menstrual cycles. A score of 3 or more of 4 indicates high risk and warrants progression to Step 2; the eight-symptom risk score (AUC = 0.8133) confirms self-reported symptoms alone provide good discrimination.

Step 2 — Primary Care Assessment (single blood test): AMH measurement at a primary health centre. AMH above 4 ng/mL combined with 3 or more symptoms constitutes a high-risk profile warranting specialist referral, requiring only one additional test at primary care level.

Step 3 — Clinical Confirmation (specialist referral): full Rotterdam criteria assessment including follicle count on ultrasound, with treatment and monitoring initiated as clinically indicated.

This protocol could be implemented cost-effectively across Kerala's primary healthcare infrastructure, potentially addressing the 52.1 per cent diagnostic gap, as Steps 1 and 2 require no specialist involvement and are suitable for rural primary health centres, school and college health programmes, and anganwadi centres.

7.9 Limitations

The secondary clinical dataset, collected in hospital settings, may not fully represent women who have not sought clinical consultation; the 30.9% PCOS prevalence, substantially higher than the community estimate of 17.5%, reflects this selection bias. The primary survey sample was predominantly postgraduate students aged 21–25 (84.5%), limiting representativeness of older women, women with lower educational attainment, and non-university settings.

The cross-sectional design precludes establishing causal relationships between predictors and PCOS diagnosis; longitudinal studies are required to confirm causal pathways. Self-reported symptoms and diagnoses in the primary survey introduce potential recall and social desirability bias, and were not validated against medical records. Clinically relevant variables — including androgen levels (testosterone, DHEAS), insulin levels, detailed dietary assessment, and physical activity — were unavailable and may improve future model performance. Finally, the models were developed and validated on Kerala-based data; external validation on other Indian states or South Asian countries is required before broader generalisation. An exploratory Spearman rank correlation was also computed during analysis to summarise overall cross-dataset symptom-ranking agreement between the primary and secondary datasets; however, this specific summary statistic could not be independently reproduced during a later review of the analysis and is therefore not reported as a substantive finding of this study. The individual symptom-level prevalence comparisons presented in Chapter 4 and discussed in this chapter were separately verified and are unaffected by this exclusion.

7.10 Summary

This chapter synthesised the findings from all analytical phases within the broader clinical, statistical, and public health context. Six key themes emerged: follicle count, weight gain, hair growth, skin darkening, and AMH are robustly confirmed as dominant PCOS predictors through convergent evidence from six independent methods; machine learning models, particularly Gradient Boosting (AUC = 0.9997) and Random Forest (AUC = 0.9887), substantially outperformed statistical models, while statistical models retain complementary interpretive value through odds ratios; the diagnostic gap in Kerala is real, large, and uniformly distributed, with 52.1% never tested and awareness unrelated to education, age, or residence; modifiable lifestyle factors — stress, sleep, and fast food — are significantly associated with PCOS risk, supporting non-pharmacological community-level intervention; cross-dataset validation supported the generalisability of the clinical findings to the community population; and the four-factor structure from factor analysis independently validates the multisystem PMOS conceptualisation encoded in the renaming of PCOS.

CHAPTER 8

CONCLUSION

8.1 Introduction

This chapter presents the conclusions of the study, addressing each of the fourteen research objectives stated in Chapter 1, followed by the principal contributions, recommendations for clinical practice and healthcare policy, directions for future research, and a final concluding statement. It draws together findings from the secondary clinical dataset ($n = 2,541$) and the primary community survey ($n = 303$) to present an integrated conclusion addressing the overarching goal of developing and validating a comprehensive statistical and machine learning framework for early PCOS risk prediction among young women in Kerala.

8.2 Summary of Findings

The analysis began with a thorough characterisation of the secondary clinical dataset. Descriptive statistics and group comparisons showed that ten of fourteen continuous variables and eight of nine categorical symptoms differed significantly between PCOS and non-PCOS groups, with follicle count, AMH, and weight gain emerging as the strongest discriminators (Section 4.4). Correlation analysis confirmed these same variables — along with hair growth and skin darkening — as most strongly associated with PCOS status, while Principal Component Analysis justified the subsequent use of non-linear machine learning classifiers (Section 4.5). Variance Inflation Factor assessment identified severe multicollinearity in five anthropometric variables, fully resolved through a single redundancy-removal step, yielding a final 22-variable predictor set with all VIF values below 5 (Section 5.2).

Two statistical models were then developed on this predictor set. Binary logistic regression achieved an AUC of 0.9535 and an accuracy of 89.98%, with follicle count as the strongest individual predictor ($OR = 6.25$) and an optimal clinical decision threshold of 0.2774 identified via Youden's J statistic (Section 5.3). LASSO regularisation subsequently selected 16 of the 22 predictors, achieving a marginally higher AUC of 0.9554 while improving model parsimony and confirming that TSH, PRL, and Vitamin D3 could be safely omitted from future screening protocols without loss of predictive power (Section 5.4).

Seven machine learning models were compared using a patient-grouped train-test split, with Gradient Boosting achieving the best overall performance (AUC = 0.9997), substantially outperforming the Logistic Regression baseline (Chapter 6). Class imbalance (69.1% PCOS-negative and 30.9% PCOS-positive) was addressed using class-weighting strategies during model training where applicable. Stratified, patient-grouped five-fold cross-validation demonstrated consistent model performance across different data partitions, with Gradient Boosting again achieving the highest mean AUC (0.9892 ± 0.0079) (Section 6.9). Permutation feature importance analysis identified follicle count, weight gain, and hair growth as the most influential predictors of PCOS risk. These variables consistently emerged as the most important predictors across the machine learning models evaluated in this study (Section 6.12).

Multivariate analysis provided further independent confirmation of these results. Factor analysis identified four underlying clinical factors corresponding closely to the Polyendocrine, Metabolic, and Ovarian dimensions of the PMOS framework; Linear Discriminant Analysis confirmed the same top predictors identified elsewhere in the study; and K-means cluster analysis identified three clinically actionable patient risk subgroups without reference to diagnosis status (Section 4.9).

Turning to the primary survey, analysis of 303 responses from young women across Kerala revealed high overall symptom prevalence, with mood swings and skin darkening emerging as the most discriminating self-reported symptoms between confirmed PCOS and non-PCOS respondents (Section 4.12). This survey also exposed a substantial community-level diagnostic gap: 52.1% of respondents had never been tested for PCOS, and awareness levels did not vary significantly by education, residence, or age — indicating that this gap is widespread rather than concentrated in any particular demographic group (Section 4.13). An eight-symptom risk score constructed from the survey data achieved an AUC of 0.8133 with an optimal cut-off of four symptoms, and modifiable lifestyle factors — stress, sleep, and fast food consumption — were confirmed as significant correlates of PCOS risk, pointing towards feasible non-pharmacological intervention targets (Sections 4.14–4.15).

Finally, cross-dataset validation compared findings between the secondary clinical dataset and the primary community survey. Two of six compared symptoms — hair growth and irregular cycle — showed no significant prevalence difference between the two datasets, and the community PCOS prevalence of 17.5% was consistent with published national estimates, supporting the representativeness of the primary survey sample. A supplementary overall rank-

order correlation between the two datasets' symptom rankings could not be independently reproduced during review and is therefore not relied upon as primary evidence for this comparison; the individual symptom-level comparisons above are unaffected by this caveat (Section 7.6).

8.3 Principal Contributions of this Study

This study makes the following original contributions to the literature on PCOS risk prediction and community health in Kerala:

Contribution 1 — Comprehensive integrated analytical framework

This study applies a comprehensive range of statistical and machine learning methods within a single integrated analytical framework for PCOS prediction. The framework incorporates multicollinearity assessment using the Variance Inflation Factor (VIF), LASSO-based feature selection, comparative evaluation of multiple machine learning models, patient-grouped train-test validation, stratified patient-grouped cross-validation, and permutation feature importance analysis. Together, these methods provide a systematic and reproducible approach for developing and evaluating predictive models and offer a useful methodological framework for future clinical prediction studies.

Contribution 2 — Optimal threshold identification

Identification of the optimal logistic regression threshold of 0.2774 using Youden's J statistic — improving sensitivity from 77% to 91% compared to the conventional 0.50 threshold — is a methodologically important contribution with direct clinical implications for screening programme design.

Contribution 3 — Independent statistical validation of the PMOS framework

Factor analysis independently identified four clinical factor structures — Metabolic-Lifestyle, Gonadotropin, Androgenic-Ovarian, and Hair Loss — corresponding directly to the PMOS multisystem dimensions, providing early independent quantitative statistical evidence broadly consistent with the PMOS conceptual framework.

Contribution 4 — Quantification of the Kerala PCOS diagnostic gap

This study provides quantitative evidence of the PCOS diagnostic gap among young women in Kerala, establishing that 52.1 per cent have never been tested, awareness is uniformly insufficient across demographic groups, and approximately half carry moderate-to-high risk scores without clinical evaluation.

Contribution 5 — Validated community-level screening tool

The eight-symptom risk score ($AUC = 0.8133$), validated on primary community data, provides a practical, low-cost, no-laboratory-test PCOS risk stratification tool suitable for resource-limited community health settings across Kerala and India.

8.4 Recommendations for Clinical Practice

Recommendation 1: Implement the three-step PCOS screening protocol — self-reported symptom assessment, AMH measurement, and specialist referral — in all primary health centres, college health programmes, and anganwadi centres across Kerala, with particular emphasis on rural and semi-urban populations, where the diagnostic gap is equally large as in urban settings.

Recommendation 2: Adopt the optimal logistic regression threshold of 0.2774 rather than the conventional 0.50 for any clinical implementation of the prediction model, to maximise sensitivity and minimise missed diagnoses.

Recommendation 3: Eliminate TSH, PRL, Vit D3, and II beta-HCG from routine early PCOS screening panels, as three independent analytical methods confirm these tests add no predictive value, with omission reducing cost and patient burden without sacrificing screening accuracy.

Recommendation 4: Implement universal PCOS awareness programmes simultaneously across all educational institutions, age groups, and geographic settings in Kerala — not targeted at specific subgroups — given that awareness does not significantly differ by education, residence, or age.

8.5 Recommendations for Future Research

Future Research 1: External validation of the developed prediction models on PCOS datasets from other Indian states, particularly North India, to establish national generalisability.

Future Research 2: A prospective longitudinal study following young women with high PCOS risk scores over 5 to 10 years to establish causal relationships between identified risk factors and PCOS onset.

Future Research 3: Inclusion of androgen levels, insulin levels, and detailed dietary assessment in future prediction models — variables identified as potentially important but unavailable in the present secondary dataset.

Future Research 4: Clinical implementation and prospective evaluation of the three-step community screening protocol in Kerala primary health centres, measuring diagnostic yield, cost-effectiveness, and patient acceptability.

8.6 Final Concluding Statement

This study developed and validated statistical and machine learning models for the early prediction of PCOS risk in Kerala by integrating a clinical dataset with evidence from a community-based survey. Using a comprehensive analytical framework encompassing statistical and machine learning techniques, the study addressed fourteen research objectives related to PCOS prediction, risk factors, awareness, and screening. Among the evaluated models, Gradient Boosting achieved the highest predictive performance ($AUC = 0.9997$), while an eight-symptom risk score demonstrated meaningful discriminatory ability ($AUC = 0.8133$) using self-reported symptoms alone. The community survey further revealed that 52.1% of young women in Kerala had never undergone PCOS testing and that awareness remained limited across different demographic groups. Collectively, these findings provide valuable evidence to support community-level strategies for the early identification and screening of PCOS in Kerala and contribute to the growing body of evidence recognising PCOS as a complex multisystem disorder.

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APPENDIX

Survey Questionnaire — PCOS Early Risk Assessment Survey

SECTION A — Demographic Information (Questions 1–6)

1. Please select your current age group.

- 15 – 17 years
- 18 – 20 years
- 21 – 25 years
- 26 – 30 years
- 31 – 35 years

2. Please select the range that best represents your current body weight.

- Below 40 kg
- 40 – 49 kg
- 50 – 59 kg
- 60 – 69 kg
- 70 – 79 kg
- 80 – 89 kg
- 90 kg or above

3. Please select the range that best represents your current height.

- Below 145 cm
- 145 – 149 cm
- 150 – 154 cm
- 155 – 159 cm
- 160 – 164 cm
- 165 – 169 cm
- 170 cm and above

4. Please select your blood group.

- A Positive (A+)
- A Negative (A−)
- B Positive (B+)
- B Negative (B−)
- O Positive (O+)
- O Negative (O−)
- AB Positive (AB+)
- AB Negative (AB−)

5. Please select the option that best describes your current place of residence.

- Urban — a metropolitan city or large city
- Semi-urban — a town or district headquarters area
- Rural — a village or panchayat area

6. Please select the option that best describes your current level of education.

- Currently enrolled in secondary school (8th to 10th standard)
- Currently enrolled in higher secondary education (11th or 12th standard / Plus Two)
- Currently pursuing an undergraduate degree or diploma
- Currently pursuing a postgraduate degree
- Currently enrolled in a professional degree programme
- Completed education and currently employed

SECTION B — Menstrual Health (Questions 7–10)

7. How would you describe the regularity of your menstrual cycle?

- Regular — my menstrual cycle begins predictably within every 21 to 35 days
- Irregular — my menstrual cycle is unpredictable, frequently delayed, or sometimes absent
- I have not monitored my menstrual cycle and I am unable to determine this

8. What is the approximate length of your menstrual cycle?

- Less than 21 days
- 21 – 28 days
- 29 – 35 days

- 36 – 45 days
- More than 45 days
- Unable to determine

9. What is the usual duration of your menstrual bleeding?

- Fewer than 3 days
- 3 – 5 days
- 5 – 7 days
- More than 7 days

10. Do you experience significant pain or cramping during your menstrual periods?

- Yes — I experience severe pain during every or most menstrual cycles, which affects my daily routine
- Yes — I experience significant pain occasionally, but not during every cycle
- I experience mild discomfort only, which does not interfere with daily activities
- No — I do not experience pain or discomfort during menstruation

SECTION C — Physical and Clinical Symptoms (Questions 11–15)

11. Have you observed any unexplained or unintentional weight gain in recent months?

- Yes — I have experienced noticeable weight gain without a clear or identifiable cause
- No — my body weight has remained stable or any changes have a clear explanation

12. Have you observed excessive hair growth on your face, chest, abdomen, or back?

- Yes — I have noticeable coarse or dark hair growth in one or more of these areas
- No — I have not observed this

13. Have you observed darkening or thickening of the skin in areas such as the neck, armpits, or inner thighs?

- Yes — I have noticed dark or discoloured patches of skin in these areas
- No — I have not observed this

14. Have you experienced significant hair loss or thinning of hair on your scalp?

- Yes — I have experienced noticeable hair loss or visible thinning of scalp hair
- No — I have not experienced this

15. Do you experience frequent or persistent acne or pimples?

- Yes — I experience frequent or persistent acne that recurs regularly
- No — I experience only occasional or infrequent pimples

SECTION D — Lifestyle and Mental Wellbeing (Questions 16–24)

16. How frequently do you consume fast food, junk food, or highly processed food items?

- Rarely or never — less than once per week
- Occasionally — 1 to 2 times per week
- Frequently — 3 to 4 times per week
- Very frequently — almost every day

17. How regularly do you engage in physical exercise or structured physical activity?

- Regularly — 3 or more sessions per week, with each session lasting at least 30 minutes
- Occasionally — 1 to 2 sessions per week
- Rarely — less than once per week
- Not at all — I do not engage in any regular physical activity

18. Which of the following best describes your current occupation or predominant daily activity level?

- School student — predominantly seated during classes throughout the day
- College student — a combination of seated academic work and moderate movement
- Office-based or desk job — predominantly seated for most of the working day
- Field-based or physically active occupation — involves regular movement and physical effort
- Homemaker — engaged in moderate levels of household activity throughout the day
- Not currently studying or employed

19. How would you assess your general level of stress in daily life?

- Low — I rarely experience stress and generally find daily demands manageable
- Moderate — I experience stress occasionally, particularly during examinations or periods of increased workload

- High — I experience stress frequently, on most weeks
- Very high — I experience stress on an almost daily basis, and it noticeably affects my health or mood

20. On average, how many hours of sleep do you get per night?

- Fewer than 5 hours
- 5 – 6 hours
- 7 – 8 hours
- More than 8 hours

21. At what time do you typically go to sleep at night?

- Before 9:00 PM
- 9:00 PM – 10:30 PM
- 10:30 PM – 12:00 midnight
- 12:00 midnight – 1:30 AM
- After 1:30 AM

22. At what time do you typically wake up in the morning?

- Before 5:00 AM
- 5:00 AM – 6:30 AM
- 6:30 AM – 8:00 AM
- 8:00 AM – 9:30 AM
- After 9:30 AM

23. Do you frequently experience mood swings, anxiety, or a persistent sense of low mood?

- Yes — I experience this frequently, on most weeks
- Sometimes — I experience this occasionally, particularly around my menstrual cycle
- Rarely — I experience this very infrequently
- No — I do not experience mood swings, anxiety, or low mood

24. Do you feel that your body weight is concentrated predominantly around your abdominal or waist area?

- Yes — my abdominal or waist area feels noticeably heavier compared to the rest of my body
- No — my body weight appears to be distributed evenly
- I am unable to determine this

SECTION E — Family History (Questions 25–26)

25. Does any member of your immediate family have a diagnosed history of PCOS or hormonal disorders?

- Yes — my mother, sister, or aunt has been diagnosed with PCOS or a hormonal disorder
- No — no member of my immediate family has such a diagnosis
- Not sure — this information is not available to me

26. Does your family have a history of diabetes or insulin resistance?

- Yes — one or more family members have been diagnosed with diabetes or pre-diabetes
- No — there is no known history of diabetes in my family
- Not sure — this information is not available to me

SECTION F — Medical History (Questions 27–30)

27. Have you ever received a diagnosis of PCOS from a qualified medical professional?

- Yes — a doctor has confirmed that I have PCOS
- No — a doctor has confirmed that I do not have PCOS
- I have consulted a doctor regarding this, but no definitive diagnosis was provided
- I have never been examined or tested for PCOS

28. Have you ever consulted a gynaecologist or medical professional for menstrual health concerns?

- Yes — I have consulted a gynaecologist or doctor specifically for menstrual concerns
- No — I have never consulted a doctor for this purpose

29. Have you ever received a diagnosis of a thyroid disorder from a medical professional?

- Yes — I have been diagnosed with a thyroid disorder
- No — I have not been diagnosed with any thyroid condition
- Not sure — I have never been tested for thyroid function

30. Have you ever been diagnosed with diabetes or pre-diabetes by a medical professional?

- Yes — I have been diagnosed with diabetes or pre-diabetes
- No — I have not received such a diagnosis
- Not sure — I have never been tested for this

SECTION G — PCOS Awareness (Question 31)

31. How familiar were you with PCOS prior to completing this survey?

- I was fully aware of PCOS, including its symptoms, causes, and health implications
- I had heard of PCOS but was not familiar with its details or implications
- I had come across the term PCOS but had no understanding of what it means
- I had no prior knowledge of PCOS before completing this survey